Compressive Covariance Sensing via Quadratic Sampling based on Non-convex Learning

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***Abstract*—The covariance matrix is fundamental in many signal and information processing tasks as it quantifies the linear relationships and variability among multiple variables. When data rapidly evolves or sampling acquisition devices have limited processing power and storage, it is desirable to efficiently estimate the covariance matrix from a single pass over the data and limited stored measurements, particularly given the large dimensionality of modern datasets. Quadratic samplers have emerged as a promising approach in Compressive Covariance Sensing. In this paper, we focus on the quadratic measurement model, assuming that the covariance matrix exhibits sparse structure in high-dimensional settings. We propose a least-squares-based framework regularized by non-convex sparsity-inducing penalties. To efficiently compute this estimator, we develop a multistage convex relaxation algorithm based on the majorization-minimization (MM) framework. This algorithm provides strong computational and statistical guarantees by leveraging local restricted strong convexity and Hessian smoothness. We demonstrated that the proposed algorithm achieves quadratic convergence to approximate solutions for each convex relaxation subproblem. Furthermore, we establish the statistical property of all sequential approximate solutions generated by the MM-based algorithm, showing that the optimal solution reached after sufficient iterations possesses oracle statistical properties in the Frobenius norm. Extensive numerical simulations validate our theoretical results, confirming the effectiveness of our approach.**

***Index Terms*—Compressive covariance sensing, compressive covariance sketching, quadratic measurements model, rank-one measurements model, majorization-minimization, positive definiteness, sparsity, non-convex statistical optimization.**

I. Introduction

In recent decades, Compressed Sensing (CS) has garnered significant attention due to its remarkable compression efficiency and low computational complexity in large-scale dataset processing [1]–[6]. The core principle of CS lies in exploiting the sparsity inherent in many real-world signals when represented in appropriate bases. The sparsity enables the original high-dimensional signals to be efficiently subsampled using random linear projections and accurately reconstructed from their low-dimensional measurements [7]–[12]. In many practical applications, however, random processes are involved, rendering the reconstruction of original signals itself less meaningful [13]–[16]. Instead, the focus shifts towards extracting critical quantities from signals (such as second-order or even higher-order statistics), which contain valuable features and provide reliable information about the random processes [17]–[19].

The covariance matrix $\mathbf{\Sigma} = \mathbb{E}\left[\mathbf{x}\mathbf{x}^\top\right] \in \mathbb{R}^{d\times d}$ is a fundamental second-order statistical characterization of the (zero-mean) random signals $\mathbf{x} \in \mathbb{R}^d$, which plays a critical role in various estimation or detection tasks within signal processing and information theory [20]–[24]. However, due to its large dimensionality, it is impractical to store or communicate $\mathbf{\Sigma_x}$ (or the sample covariance matrix (SCM)). As a result, it is infeasible to adopt the SCM estimator directly. Instead, one can acquire compressive linear measurements of $\mathbf{x}$ through observation of the form $y_i = \boldsymbol{a}_i^\top \mathbf{x}$, where $\boldsymbol{a}_i \in \mathbb{R}^d$ is the measurement vector. The variance Σ_{y_i} of y_i serves as a compressive sketch of $\mathbf{\Sigma_x}$, which can be effectively stored and processed. The relationship between the high-dimensional covariance matrix $\mathbf{\Sigma_x}$ and its compressive sketch Σ_{y_i} is given by

$$\Sigma_{y_i} = \boldsymbol{a}_i^\top \mathbf{\Sigma_x} \boldsymbol{a}_i.$$

Notably, Σ_{y_i} and $\mathbf{\Sigma_x}$ are still linearly related via the Kronecker product $\boldsymbol{a}_i \otimes \boldsymbol{a}_i$, which is seen more clearly when expressed in the following vectorized form:

$$\Sigma_{y_i} = \left(\boldsymbol{a}_i \otimes \boldsymbol{a}_i\right)^\top \operatorname{vec}\left(\mathbf{\Sigma_x}\right). \tag{1}$$

It is obvious that Σ_{y_i} is a quadratic function of $\boldsymbol{a}_i$ and the noise-free measurements $\boldsymbol{a}_i^\top \mathbf{\Sigma_x} \boldsymbol{a}_i$ are referred to as quadratic measurements (or rank-one measurements).

The central question in Compressive Covariance Sensing (CCS) is whether it is possible to design the sensing vector $\boldsymbol{a}_i$ in such a way that the resulting compressive sketch exhibits statistically desirable properties, which can then be leveraged to accurately reconstruct $\mathbf{\Sigma_x}$ from a minimal number of measurements. Although $\mathbf{\Sigma_x}$ may generally not be identifiable under arbitrary sensing designs, reliable recovery is possible via clever design of $\boldsymbol{a}_i$ when $\mathbf{\Sigma_x}$ possesses certain latent structures, such as sparsity, which is consistent with the principles in CS [2]–[4], [8], [9], [25]. The sparsity assumption is widely recognized in high-dimensional covariance matrix estimation problems, where the sample size is small relative to the ambient dimension of the underlying parameter to be estimated [22], [26]–[30]. This assumption is justified by practical considerations in real-world applications [31]. Furthermore, careful exploitation of the positive semidefinite property of $\mathbf{\Sigma_x}$ can potentially lead to more efficient compression.

A number of works have investigated the problem of CCS, which involves compressively sampling the underlying random process to obtain sketches and subsequently recovering the covariance matrix from these sketches [14], [32]–[36]. In [37], [38], the authors explored the recovery of second-order statistics of a cyclostationary signal from random linear measurements, assuming that the covariance matrix is approximately sparse. They proposed an approach based on ℓ_1-minimization, although the analysis did not provide explicit performance guarantees. The authors in [34] presented a unified framework for compressing and recovering sparse covariance matrix using the ℓ_1-norm heuristic. They demonstrate that $\mathcal{O}\left(s\log\left(d^2/s\right)\right)$ measurements are sufficient for the compression of d-dimensional sparse covariance matrix with s-sparse. In a concurrent work by *Dasarathy et al.* [33], the covariance matrix $\boldsymbol{\Sigma}_{\mathbf{x}}$ is sketched through quadratic measurements of the form $\boldsymbol{\Sigma}_{\mathbf{y}} = \mathbf{A}\boldsymbol{\Sigma}_{\mathbf{x}}\mathbf{A}^\top$, where $\boldsymbol{\Sigma}_{\mathbf{x}}$ is assumed to exhibit distributed sparsity and the sketching matrix $\mathbf{A}$ is constructed from expander graphs. And the sample complexity required to compress distributed sparse covariance matrices is proved to be $\mathcal{O}\left(\sqrt{d}\log d\right)$.

The ℓ_1-minimization method [33], [34], [37], [38] has been extensively studied for the CCS problem. However, it is now well-known that methods based on the ℓ_1-norm, e.g., in Lasso [39], introduces a non-negligible bias into the resulting estimator [40], which compromises the estimation accuracy. To alleviate this bias effect, non-convex penalties such as smoothly clipped absolute deviation (SCAD) penalty [40] and minimax concave penalty (MCP) [41] have been proposed as alternatives. It has been demonstrated in [40]–[42] that the non-convex penalized regression can effectively eliminate the estimation bias in Lasso and achieve improved statistical convergence rates. The advantages of non-convex penalties have been further highlighted in many other contexts, such as high-dimensional graph models [43], [44], high-dimensional covariance estimation [30], graph trend filtering [45], etc. However, the theory of non-convex penalty in the context of CCS problem remains underexplored.

In this paper, we investigate a quadratic measurement model for CCS and estimate the covariance matrix from compressive sketches using a non-convex penalty. A significant challenge in this model is the design of effective sensing vectors that ensure successful recovery from compressive sketches. We show that for a broad class of sub-Gaussian sensing vectors, we can derive the covariance matrix exactly and provide theoretical performance guarantees. Additionally, the analysis of the theoretical performance guarantee for the proposed estimator is complicated due to the non-convexity of the objective function. Since non-convex problems generally lack global convergence guarantees, directly application of iterative algorithms such as stochastic gradient descent may fail to reach the global optimum. Moreover, the statistical properties of local optima are generally difficult to characterize. To address this, we introduce an efficient multistage convex relaxation algorithm based on the majorization-minimization (MM) framework, which solves a sequence of convex subproblems. We demonstrate that our proposed estimator achieves faster statistical convergence rates compared to the conventional estimator using the ℓ_1-norm penalty. Furthermore, under mild assumptions, we rigorously prove that the proposed estimator exhibits oracle (or optimal) statistical properties. Numerical simulations are provided to support the theoretical findings and validate the superior performance of the proposed estimator.

A. Organization

The remainder of this paper is organized as follows. In Section II, we introduce the quadratic measurement model for CCS and present the proposed estimator with a non-convex penalty. Section IV details the design of the algorithm, focusing on the development of a proximal Newton method within the MM framework. In Section V, we provide an asymptotic statistical analysis that characterizes the reconstruction performance of (3), specifically addressing the convergence rate of the estimator and the iteration complexity of the algorithm. Numerical experiments are subsequently presented in Section VI. Finally, Section VIII concludes the paper with a discussion. The appendices provide proofs for all theoretical claims made in the main text.

B. Notation

This subsection provides a concise overview of the essential notations employed throughout the paper. Scalar values are denoted by conventional lower-case or upper-case letters, while boldface lower-case letters signify vectors and upper-case letters represent matrices. The symbols $\mathbf{0}$ and $\mathbf{I}$ are used to denote the all-zero matrix and the identity matrix, respectively, with dimensions that are contextually appropriate. The set of all $m\times n$ matrices with real entries is represented by $\mathbb{R}^{m\times n}$.

Let $\mathbf{X}\in\mathbb{R}^{m\times n}$ be a matrix. The elements of the matrix are denoted by X_{ij} and $[\mathbf{X}]_{ij}$, both referring to the entry in the (i,j)-th position. We denote $\mathbf{X}\succeq\mathbf{0}$ (resp. $\mathbf{X}\succ\mathbf{0}$) to indicate that $\mathbf{X}$ is positive semidefinite (resp. definite). The transpose, inverse, and determinant of the matrix $\mathbf{X}$ are represented by $\mathbf{X}^\top$, $\mathbf{X}^{-1}$, and $\det(\mathbf{X})$, respectively. The smallest and largest eigenvalues of $\mathbf{X}$ are denoted by $\lambda_{\min}(\mathbf{X})$ and $\lambda_{\max}(\mathbf{X})$. The p-norm of $\mathbf{X}$ is defined as $\|\mathbf{X}\|_p = \left(\sum_{i=1}^m\sum_{j=1}^n |X_{ij}|^p\right)^{\frac{1}{p}}$ for a real $p>0$. For instance, the Frobenius norm and spectral norm are indicated by $\|\mathbf{X}\|_{\mathrm{F}}$ and $\|\mathbf{X}\|_2$, respectively, while $\|\mathbf{X}\|_1$ represents the sum of the absolute values of all entries of $\mathbf{X}$. Specifically, the maximum-absolute-value norm of $\mathbf{X}$ is expressed as $\|\mathbf{X}\|_{\max}$, and the minimum-absolute-value norm as $\|\mathbf{X}\|_{\min}$. The notation $\mathbf{X}_{\cdot j}$ ($\mathbf{X}_{k\cdot}$) is used to denote the j-th column (k-th row) of $\mathbf{X}$. The vectorization of $\mathbf{X}$, achieved by stacking its columns, is denoted as $\operatorname{vec}(\mathbf{X})$, while $\operatorname{mat}(\cdot)$ represents the inverse operation. Furthermore, the Kronecker product and Hadamard product (also referred to as the entry-wise product) of matrices $\mathbf{X}$ and $\mathbf{Y}$ are denoted by $\mathbf{X}\otimes\mathbf{Y}$ and $\mathbf{X}\odot\mathbf{Y}$, respectively. The Euclidean inner product is defined as $\langle\mathbf{X},\mathbf{Y}\rangle := \operatorname{tr}\left(\mathbf{X}\mathbf{Y}^\top\right)$, where $\operatorname{tr}(\cdot)$ denotes the trace of a matrix. For a real-valued function $f(\mathbf{X})$, the gradient $\nabla f(\mathbf{X})$ is a $d\times d$ matrix with the (i,j)-th element given by $\frac{\partial}{\partial X_{ij}} f(\mathbf{X})$, denoted by $\nabla_{ij} f(\mathbf{X})$, while $\nabla^2 f(\mathbf{X})$ represents the $d^2\times d^2$ Hessian matrix. Additionally, we define $\|\mathbf{X}\|_{\mathbf{H}}^2 := \operatorname{vec}^\top(\mathbf{X})\,\mathbf{H}\operatorname{vec}(\mathbf{X})$ for the purpose of simplified representation.

For an index set $\mathcal{E}$, its cardinality is represented as $|\mathcal{E}|$, while its complement is denoted by $\overline{\mathcal{E}}$. The notation $\mathbf{X}_{\mathcal{E}}$ refers to the matrix comprising entries X_{ij} for $(i,j)\in\mathcal{E}$ and zero elsewhere. The function $\operatorname{sign}(x)$ is defined as $\operatorname{sign} = x/|x|$ when $x\neq 0$ and is equal to zero otherwise. For the functionals $f(n)$ and $g(n)$, we express $f(n)\gtrsim g(n)$ if $f(n)\geq cg(n)$, $f(n)\lesssim g(n)$ if $f(n)\leq Cg(n)$, and $f(n)\asymp g(n)$ if $cg(n)\leq f(n)\leq Cg(n)$ for some positive constants c and

C. Additionally, $\mathcal{O}_p(\cdot)$ is utilized to indicate boundedness in probability.

II. Preliminaries

In this section, we introduce a comprehensive framework for CCS, followed by the proposed estimator with a non-convex penalty.

A. The Quadratic Measurement Model

Consider n independent observations $\{\mathbf{x}_i\}_{i=1}^n$, each drawn from a zero-mean random vector $\mathbf{x}$. The quadratic measurement model is given by

$$y_i = \frac{1}{n}\sum_{t=1}^{n}\left|\boldsymbol{a}_i^\top \mathbf{x}_t\right|^2 + \eta_i = \boldsymbol{a}_i^\top \mathbf{S}\boldsymbol{a}_i + \eta_i,\ i = 1,\ldots,m, \quad (2)$$

where $\{y_i\}_{i=1}^m$ are a sequence of measurements, $\{\boldsymbol{a}_i\}_{i=1}^m$ are the sensing vectors, and $\{\eta_i\}_{i=1}^m$ are additive noise components assumed to be drawn from a sub-exponential distribution with mean 0 and variance proxy σ^2. $\mathbf{S} = \frac{1}{n}\sum_{i=1}^n \mathbf{x}_i\mathbf{x}_i^\top$ is the SCM, which can be decomposed as $\mathbf{S} = \mathbf{\Sigma}^* + \mathbf{E}$, where $\mathbf{\Sigma}^*$ is the true covariance matrix, and $\mathbf{E}$ is the bias term.

For convenience, we define $\mathbf{A}_i := \boldsymbol{a}_i\boldsymbol{a}_i^\top$ as the corresponding sensing matrix, so that the measurement process can be written as $\{\langle \mathbf{A}_i, \mathbf{S}\rangle\}_{i=1}^m$. Thus, the measurements $\{y_i\}_{i=1}^m$ obeys $y_i := \langle \mathbf{A}_i, \mathbf{S}\rangle + \eta_i$. Let the measurement vector be $\boldsymbol{y} := [y_1, \cdots, y_m]^\top$ and the noise vector be $\boldsymbol{\eta} := [\eta_1, \cdots, \eta_m]^\top$. Additionally, we introduce the linear operator $\mathcal{A}(\mathbf{S}) : \mathbb{R}^{d\times d} \mapsto \mathbb{R}^m$, which maps a matrix $\mathbf{S} \in \mathbb{R}^{d\times d}$ to the set $\{\langle \mathbf{A}_i, \mathbf{S}\rangle\}_{i=1}^m$. Hence, the quadratic measurement model can be expressed as $\boldsymbol{y} = \mathcal{A}(\mathbf{S}) + \boldsymbol{\eta}$. It is noteworthy that the measurement model introduced in (2) has been previously explored in [34], [46], [47]. Importantly, the model presented in (2) is significantly simpler to implement and entails lower computational complexity when compared to full-rank measurement matrices with independent and identically distributed (i.i.d.) entries.

B. Practical Examples

The quadratic measurement model in (2) has numerous practical applications across various domains. We briefly review two examples: Spectrum estimation of stochastic processes from energy measurements and Direction-Of-Arrival (DOA) estimation. For further examples and detailed discussions, readers are referred to the works [33], [34], [47].

Example 1 (Spectrum Estimation)**.** *In the context of spectrum estimation for stochastic processes based on energy measurements, one typically employs a sensing vector $\boldsymbol{a}_i$, which can be implemented using random demodulators [48]. The energy measurements are averaged over N instances $\{\mathbf{x}_t\}_{1\le t\le N}$, leading to observations in the form of quadratic forms.*

Example 2 (DOA)**.** *In the context of DOA, the objective is to determine the angles of arrival for a finite number of signal sources. This problem can be framed as a sparse power spectrum estimation task, which naturally leads to a quadratic measurement model. A wide range of techniques have been developed to address this challenge [49]–[52].*

C. Proposed Estimator

We propose a novel estimator tailored to the CCS framework. Given a collection of m samples $\{(y_i, \boldsymbol{a}_i)\}_{i=1}^m$, which are assumed to be generated according to the quadratic measurement model in (2), we estimate the unknown covariance matrix $\mathbf{\Sigma}$ by solving the following optimization problem

$$\min_{\mathbf{\Sigma}\succ\mathbf{0}}\left\{\frac{1}{2m}\left\|\boldsymbol{y} - \mathcal{A}(\mathbf{\Sigma})\right\|_2^2 - \tau\log\det\mathbf{\Sigma} + \sum_{i,j} p_\lambda\left(|\Sigma_{ij}|\right)\right\}, \quad (3)$$

where the first term aims to minimize the empirical errors, the second term incorporates a log-determinant barrier function to enforce positive definiteness of $\mathbf{\Sigma}$, with a barrier parameter $\tau > 0$ as introduced by Rothman [53]. While covariance matrices are typically semi-positive definite, practical applications often require that estimators be strictly positive definite, even for finite sample sizes. This property is particularly important in supervised learning contexts, such as in quadratic discriminant analysis and covariance regularized regression [54]. The third term $p_\lambda(\cdot)$ represents a non-convex penalty function governed by a regularization parameter $\lambda > 0$. We consider a class of non-convex penalty functions $p_\lambda(\cdot)$ satisfying the following assumptions.

Assumption 3. *The function $p_\lambda(t) : \mathbb{R} \to \mathbb{R}$ satisfies:*

- $p_\lambda(t)$ is symmetric around zero with $p_\lambda(0) = 0$, nondecreasing on the nonnegative, differentiable almost everywhere on $(0, +\infty)$, and subdifferentiable at $t = 0$;
- $0 \le p'_\lambda(t_1) \le p'_\lambda(t_2) \le \lambda$ for all $t_1 \ge t_2 \ge 0$ and $\lim_{t\to 0^+} p'_\lambda(t) = \lambda$;
- There exists an $\alpha > 0$ such that $p'_\lambda(t) = 0$ for $t \ge \alpha\lambda$;

It is suggested that a well-designed penalty function should yield an estimator with three properties: unbiasedness, sparsity, continuity [40], which coincides with the three conditions in Assumption 3. A variety of functions have been explored through past research on non-convex regularization, of which we present a few representative examples here:

- Smooth clipped absolute deviation penalty: This penalty, due to [40], is given by
$$p'_\lambda(t) := \begin{cases}\lambda, & \text{for } 0 < t \le \lambda,\\ \frac{b\lambda - t}{b-1}, & \text{for } \lambda \le t \le b\lambda,\\ 0, & \text{for } t \ge b\lambda,\end{cases}$$
where $b > 2$ is an additional tuning parameter. The authors in [40] proposed $b = 3.7$ through a Bayesian rationale, applicable when the variable dimension is less than 100.
- Minimax concave penalty regularizer: This penalty, due to [42], is defined as follows:
$$p_\lambda(t) := \operatorname{sign}(t)\,\lambda\cdot\int_0^{|t|}\left(1 - \frac{z}{\lambda b}\right)_+ dz$$
for some $b > 0$.

Note that the capped ℓ_1-penalty, takes the form $p_\lambda(t) = \lambda\cdot\min\{|t|, b\lambda\}$, where $b > 0$ is a fixed parameter. It is a simpler but less smooth version of the smooth clipped absolute deviation penalty [41].

III. Sparse Eigenvalue

In this section, we investigate the sparse eigenvalue condition, which is crucial for establishing the following theorem on the universal recovery of CCS.

It is well-established in CS that perfect recovery from a minimal number of samples is achievable if the dimensionality reduction projection preserves the signal strength within the class of matrices under consideration [2], [4]. Several incoherence measures, such as the Uniform Uncertainty Principle (UUP) and the Restricted Isometry Property (RIP), have been proposed to characterize this [55]. These concepts similarly facilitate the adaptive recovery of CCS. In [34], the mixed-norm RIP-ℓ_2/ℓ_1 was introduced, which combines the Frobenius norm and the ℓ_1 norm to assess input and output strengths, respectively. An analogous criterion, RIP-ℓ_1/ℓ_1, has also been applied to CCS recovery [46].

A. The Design Matrix $\widetilde{\mathbf{A}}$

Recall (1), we have

$$\boldsymbol{y} = \widetilde{\mathbf{A}}\,\mathrm{vec}\left(\mathbf{S}\right) + \boldsymbol{\eta},$$

where $\widetilde{\mathbf{A}} = \left[\begin{array}{ccc} (\boldsymbol{a}_1 \otimes \boldsymbol{a}_1) & \cdots & (\boldsymbol{a}_m \otimes \boldsymbol{a}_m) \end{array}\right]^\top \in \mathbb{R}^{m\times d^2}$ is the equivalent design sensing matrix. We present several fundamental assumptions on the sensing vectors $\boldsymbol{a}_i$.

Assumption 4. *We assume that the sensing vectors $\boldsymbol{a}_i$'s $(1 \le i \le m)$ are i.i.d. sub-Gaussian random variables. Each vector $\boldsymbol{a}_i$ contains d elements, denoted as $(\boldsymbol{a}_i)_j$ $(1 \le j \le d)$, which are characterized by the following conditions:*

$$\mathbb{E}\left[(\boldsymbol{a}_i)_j\right] = 0,\ \mathbb{E}\left[(\boldsymbol{a}_i)_j^2\right] = 1. \tag{4}$$

Assumption 4 asserts that each row of the design matrix $\widetilde{\mathbf{A}}$ is composed of i.i.d. sub-exponential random variables.

B. Restricted Eigenvalue and Sparse Eigenvalue

Bickel et al. [56] introduced the Restricted Eigenvalue (RE) condition, demonstrating that it is one of the weakest and most general conditions in sparse reconstruction problems. This condition provides a relaxation of the UUP [41], [57]. Here, we present one version of the RE condition, as described by Bickel et al. [56].

Definition 5 (Restricted Eigenvalue)**.** For some integer $0 < s_0 < d$ and a positive constant c_0, the RE$(s_0, c_0, \mathbf{X})$ for the matrix $\mathbf{X} \in \mathbb{R}^{d\times d}$ necessitates that the following inequality holds:

$$\forall\, \mathbf{u} \neq 0,\ \min_{\substack{J\subset\{1,\dots,d\},\\ |J|\le s_0}}\ \min_{\left\|\mathbf{u}_{\overline{J}}\right\|_1 \le c_0 \left\|\mathbf{u}_J\right\|_1} \frac{\left\|\mathbf{X}\mathbf{u}\right\|_2}{\left\|\mathbf{u}_J\right\|_2} > 0, \tag{5}$$

where $\mathbf{v}_J$ denotes the subvector of $\mathbf{v} \in \mathbb{R}^d$ restricted to the subset J of $\{1,\dots,d\}$.

In the following, we introduce the Sparse Eigenvalues (SE) condition as delineated by Fan et al. [58], which exhibits a significant correlation with the RE properties. For the detailed relationships between RE and SE, please refer to Appendix A.

Definition 6 (Sparse Eigenvalue)**.** For a positive integer $0 < s_0 < d$, the localized sparse eigenvalues are defined as

$$\rho^+_{s_0,r} = \sup\left\{\frac{\mathbf{u}^\top \nabla^2 f\left(\boldsymbol{\Sigma}\right)\mathbf{u}}{\mathbf{u}^\top\mathbf{u}} \mid \left\|\mathbf{u}\right\|_0 \le s_0, \left\|\boldsymbol{\Sigma} - \boldsymbol{\Sigma}^*\right\|_{\mathrm{F}} \le r\right\};$$

$$\rho^-_{s_0,r} = \inf\left\{\frac{\mathbf{u}^\top \nabla^2 f\left(\boldsymbol{\Sigma}\right)\mathbf{u}}{\mathbf{u}^\top\mathbf{u}} \mid \left\|\mathbf{u}\right\|_0 \le s_0, \left\|\boldsymbol{\Sigma} - \boldsymbol{\Sigma}^*\right\|_{\mathrm{F}} \le r\right\}.$$

The SE condition has been widely studied within the domain of high-dimensional sparse recovery and compressed sensing [56], [59]–[62]. The SE condition with parameter s_0 holds for i.i.d. sub-exponential random matrices, provided that $m = \mathcal{O}\left(s_0 \log^2 d/s_0\right)$ [63]. This condition is satisfied with probability at least $1 - c_1 \exp\left(-c_2\sqrt{d}\right)$, where $c_1, c_2 > 0$ are universal numerical constants.

Hence, we present the following assumption:

Assumption 7. *There exists a universal constant c_3 such that for $\widetilde{s} \ge c_3 s^*$, the SE property is guaranteed with parameters $\rho^-_{2s^*+2\widetilde{s},r}$ and $\rho^+_{2s^*+2\widetilde{s},r}$ satisfying*

$$0 < \rho^-_{2s^*+2\widetilde{s},r} < \rho^+_{2s^*+2\widetilde{s},r} < +\infty$$

with probability at least $1 - c_4 \exp\left(-c_5\sqrt{d}\right)$ for $c_4, c_5 > 0$.

Assumption 7 asserts that the sparse eigenvalues of the Hessian matrix $\nabla^2 f\left(\boldsymbol{\Sigma}\right)$ are lower and upper bounded when $\boldsymbol{\Sigma}$ is adequate sparsity and in close proximity to $\boldsymbol{\Sigma}^*$ with high probability (w.h.p). In other words, $\nabla^2 f\left(\boldsymbol{\Sigma}\right)$ possesses both bounded maximum and non-zero minimum sparse eigenvalues over a local cone.

IV. Optimization Algorithm

In this section, we first revisit the MM framework, which plays a crucial role in the forthcoming theoretical analysis. We then introduce an MM-based multistage convex relaxation algorithm to solve (3), and provide the corresponding pseudocode. Furthermore, we describe the solution method employed at each stage of the MM-based algorithm—proximal Newton homotopy algorithm.

A. The MM Framework and Multistage Convex Relaxation Algorithm

The MM algorithm framework is an iterative process that encompasses two main steps: the Majorization step and the Minimization step. Suppose we want to minimize a real-valued function $F\left(\mathbf{x}\right)$, below is the basic procedure of this algorithm framework:

- Majorization Step: In this step, the goal is to select or construct a surrogate function $\overline{F}\left(\mathbf{x} \mid \mathbf{x}^{(k-1)}\right)$ which equals the value of the original objective function $F\left(\mathbf{x}\right)$ at the current iteration point $\mathbf{x}^{(k-1)}$, but is an upper bound for $F\left(\mathbf{x}\right)$ at all other points. Specifically, $\overline{F}\left(\mathbf{x} \mid \mathbf{x}^{(k-1)}\right) \ge F\left(\mathbf{x}\right)$ for all $\mathbf{x}$ and $\overline{F}\left(\mathbf{x}^{(k-1)} \mid \mathbf{x}^{(k-1)}\right) = F\left(\mathbf{x}^{(k-1)}\right)$.
- Minimization Step: After the surrogate function is determined, the next step is to find the point $\mathbf{x}^{(k)}$ that minimizes $\overline{F}\left(\mathbf{x} \mid \mathbf{x}^{(k-1)}\right)$, such that $\mathbf{x}^{(k)} \in \arg\min \overline{F}\left(\mathbf{x} \mid \mathbf{x}^{(k-1)}\right)$.

Algorithm 1: The MM-Based Multistage Convex Relaxation Algorithm for Solving (3).

Input: $\{y_i, \boldsymbol{a}_i\}_{i=1}^m$, τ, λ;

Initialize $\widetilde{\boldsymbol{\Sigma}}^{(0)} = \mathbf{I}$

for $k = 1, 2, \ldots, K$ **do**

  $\Lambda_{ij}^{(k-1)} = p'_\lambda\left(\left|\widetilde{\Sigma}_{ij}^{(k-1)}\right|\right)$;

  $\widetilde{\boldsymbol{\Sigma}}^{(k)} = \arg\min F(\boldsymbol{\Sigma})$;

  $k = k + 1$;

end

Output: $\widetilde{\boldsymbol{\Sigma}}^{(K)}$

These two steps are executed in an alternating fashion, with each iteration aimed at reducing or at least not increasing the value of the objective function $F(\mathbf{x})$, ensuring $F\left(\mathbf{x}^{(k)}\right) \leq F\left(\mathbf{x}^{(k-1)}\right)$. Instead of directly minimizing $F(\mathbf{x})$, the algorithm focuses on solving a series of simple optimization problems sequentially. The process begins with a feasible initial point $\mathbf{x}^{(0)}$ and continues iteratively through $\mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \ldots$ until a specified convergence criterion is satisfied.

We now introduce a multistage convex relaxation algorithm based on the MM approach to solve the problem (3). The algorithm is outlined in Algorithm 1, with the initialization of $\boldsymbol{\Sigma}^{(0)} = \mathbf{I}$ as a trivial start. To simplify, we define

$$f(\boldsymbol{\Sigma}) = \frac{1}{2m}\|\boldsymbol{y} - \mathcal{A}(\boldsymbol{\Sigma})\|_2^2 - \tau \log\det \boldsymbol{\Sigma}.$$

In each iteration of the algorithm, a weighted ℓ_1-norm serves as the surrogate function for $\sum_{i,j} p_\lambda(\Sigma_{ij})$. Specifically, for each $1 \leq k \leq K$, we minimize the following convex relaxation subproblem:

$$F(\boldsymbol{\Sigma}) = f(\boldsymbol{\Sigma}) + \sum_{i,j} p'_\lambda\left(\left|\widehat{\Sigma}_{ij}^{(k-1)}\right|\right)|\Sigma_{ij}|, \tag{6}$$

where $\widehat{\boldsymbol{\Sigma}}^{(k)}$ represents the optimal solution to the k-th subproblem. Each subproblem can further be reformulated as follows:

$$\min_{\boldsymbol{\Sigma} \succ \mathbf{0}} \left\{f(\boldsymbol{\Sigma}) + \|\boldsymbol{\Lambda} \odot \boldsymbol{\Sigma}\|_1\right\}, \tag{7}$$

where $\boldsymbol{\Lambda}$ is the regularized parameter matrix with entries $\Lambda_{ij} = p'_\lambda(|\widehat{\Sigma}_{ij}|) \in [0, \lambda]$. According to the Karush-Kuhn-Tucker (KKT) conditions, the optimal solution $\widehat{\boldsymbol{\Sigma}}$ for each subproblem satisfies the first-order optimal condition:

$$\nabla f\left(\widehat{\boldsymbol{\Sigma}}\right) + \boldsymbol{\Lambda} \odot \widehat{\boldsymbol{\Xi}} = 0,$$

where $\widehat{\boldsymbol{\Xi}} \in \partial\left\|\widehat{\boldsymbol{\Sigma}}\right\|_1$ and the gradient $\nabla f(\boldsymbol{\Sigma}) = -\frac{1}{m}\mathcal{A}^*(\boldsymbol{y} - \mathcal{A}(\boldsymbol{\Sigma})) - \tau\boldsymbol{\Sigma}^{-1}$. Here, $\mathcal{A}^*(\cdot)$ is the conjugate operator of $\mathcal{A}(\cdot)$. Given that the problem (7) lacks an analytical solution, we seek a sub-optimal solution within a pre-specified tolerance level, as defined in Definition 8, and terminate the iterations when the approximate KKT conditions are satisfied.

Definition 8. Let ε denote the tolerance level. The solution $\widetilde{\boldsymbol{\Sigma}}^{(k)}$ is considered ε-optimal for the k-th subproblem (7) if the condition $\omega_{\boldsymbol{\Lambda}}(\widetilde{\boldsymbol{\Sigma}}^{(k)}) \leq \varepsilon$ is satisfied, where

$$\omega_{\boldsymbol{\Lambda}^{(k-1)}}\left(\widetilde{\boldsymbol{\Sigma}}^{(k)}\right) = \min_{\boldsymbol{\Xi} \in \partial\left\|\boldsymbol{\Sigma}^{(k)}\right\|_1}\left\|\nabla f\left(\boldsymbol{\Sigma}^{(k)}\right) + \boldsymbol{\Lambda}^{(k-1)} \odot \boldsymbol{\Xi}\right\|_{\max}.$$

In order to guarantee that the solution $\widetilde{\boldsymbol{\Sigma}}^{(k)}$ derived at each stage attains sufficient precision for all $k \geq 1$, we present the following assumption, which is essential for the convergence analysis of multistage convex relaxation.

Assumption 9. *For each iteration $k \geq 1$ of the convex relaxation process* (7)*, we assume the accuracy parameter ε as follows:*

$$\varepsilon = \frac{c_6}{\sqrt{mn}} \leq \frac{\lambda}{8},$$

where c_6 is a predetermined small constant.

B. Proximal Newton Algorithm

At each stage, the subproblem is obviously convex, enabling the application of various solution techniques. Among these, first-order methods like the proximal gradient algorithm and second-order methods like the proximal Newton algorithm are particularly noteworthy. In this study, we adopt the proximal Newton method to enhance computational efficiency. The proximal Newton algorithm solves a proximal subproblem in each iteration, which involves updating the current estimate using both gradient and Hessian information. Specifically, the algorithm computes a Newton direction to serve as the descent direction, followed by determining an appropriate step size that ensures a sufficient decrease in the objective function, thereby facilitating convergence.

For the sake of notational simplicity, we denote the iteration index within the k-th stage as t and omit the stage index k. We build a quadratic approximation by considering the second-order Taylor expansion of $f(\boldsymbol{\Sigma})$:

$$\begin{aligned}\overline{f}(\boldsymbol{\Sigma} \mid \boldsymbol{\Sigma}_t) =& f(\boldsymbol{\Sigma}_t) + \langle \nabla f(\boldsymbol{\Sigma}_t), \boldsymbol{\Sigma} - \boldsymbol{\Sigma}_t\rangle \\ &+ \frac{1}{2}\|\boldsymbol{\Sigma} - \boldsymbol{\Sigma}_t\|_{\nabla^2 f(\boldsymbol{\Sigma}_t)}^2.\end{aligned} \tag{8}$$

We then solve

$$\boldsymbol{\Sigma}_{t+\frac{1}{2}} = \underset{\boldsymbol{\Sigma} \succ \mathbf{0}}{\arg\min}\, \overline{F}(\boldsymbol{\Sigma} \mid \boldsymbol{\Sigma}_t, \boldsymbol{\Lambda}), \tag{9}$$

where $\overline{F}(\boldsymbol{\Sigma} \mid \boldsymbol{\Sigma}_t, \boldsymbol{\Lambda}) = \overline{f}(\boldsymbol{\Sigma} \mid \boldsymbol{\Sigma}_t) + \|\boldsymbol{\Lambda} \odot \boldsymbol{\Sigma}\|_1$. Denote $\Delta\boldsymbol{\Sigma}_t = \boldsymbol{\Sigma}_{t+\frac{1}{2}} - \boldsymbol{\Sigma}_t$ as the Newton direction for the function $F(\boldsymbol{\Sigma})$. Particularly, we adopt coordinate descent algorithms combined with an active set strategy to get the Newton direction, which can improve the time complexity. The detailed computation will be introduced in the Appendix B.

An additional backtracking line searching procedure is included to guarantee the descent of the objective value. We try to find a step size $\beta \in (0, 1]$ such that the Armijo condition [64] holds. Specially, we start with a fixed constant $\mu \in (0.5, 1)$ and update $\beta = \mu^q$ from $q = 0$ with a constant decrease rate until we find the smallest $q \in \mathbb{N}$ such that

$$F(\boldsymbol{\Sigma}_t + \beta\Delta\boldsymbol{\Sigma}_t) \leq F(\boldsymbol{\Sigma}_t) + \alpha\beta\delta_t,$$

Algorithm 2: Proximal Newton Algorithm (ProxNewton) With Back-tracking Line Search for Solving (7).

Input: $\widetilde{\boldsymbol{\Sigma}}^{(k-1)}$, $\boldsymbol{\Lambda}^{(k-1)}$, ε;

Initialize $t=0$, $\boldsymbol{\Sigma}_t^{(k)}=\widetilde{\boldsymbol{\Sigma}}^{(k-1)}$, $\mu=0.8$, $\alpha=0.3$

repeat

  $\boldsymbol{\Sigma}_{t+\frac{1}{2}}^{(k)}=\arg\min_{\boldsymbol{\Sigma}\succ\mathbf{0}}\overline{F}\left(\boldsymbol{\Sigma}\mid\boldsymbol{\Sigma}_t^{(k)},\boldsymbol{\Lambda}^{(k-1)}\right)$

  $\Delta\boldsymbol{\Sigma}_t^{(k)}=\boldsymbol{\Sigma}_{t+\frac{1}{2}}^{(k)}-\boldsymbol{\Sigma}_t^{(k)}$

  $\delta_t=\left\langle\nabla f\left(\boldsymbol{\Sigma}_t^{(k)}\right),\Delta\boldsymbol{\Sigma}_t^{(k)}\right\rangle-\left\|\boldsymbol{\Lambda}^{(k-1)}\odot\boldsymbol{\Sigma}_t^{(k)}\right\|_1$
  $\quad+\left\|\boldsymbol{\Lambda}^{(k-1)}\odot\left(\boldsymbol{\Sigma}_t^{(k)}+\Delta\boldsymbol{\Sigma}_t^{(k)}\right)\right\|_1$

  $\beta=1$, $q=0$

  repeat

    $\beta=\mu^q$

    $q=q+1$

    if $\boldsymbol{\Sigma}_t^{(k)}+\beta\Delta\boldsymbol{\Sigma}_t^{(k)}\preceq\mathbf{0}$ **then**

      continue;

    end

  until $F\left(\boldsymbol{\Sigma}_t^{(k)}+\beta\Delta\boldsymbol{\Sigma}_t^{(k)}\right)\leq F\left(\boldsymbol{\Sigma}_t^{(k)}\right)+\alpha\beta\delta_t$;

  $\boldsymbol{\Sigma}_{t+1}^{(k)}=\boldsymbol{\Sigma}_t^{(k)}+\beta\Delta\boldsymbol{\Sigma}_t^{(k)}$

  $t=t+1$

until $\omega_{\boldsymbol{\Lambda}^{(k-1)}}\left(\boldsymbol{\Sigma}_t^{(k)}\right)\leq\varepsilon$;

Output: $\widetilde{\boldsymbol{\Sigma}}^{(k)}=\boldsymbol{\Sigma}_t^{(k)}$

where $\alpha\in(0,0.5)$ and

$$\begin{aligned}\delta_t=&\left\langle\nabla f\left(\boldsymbol{\Sigma}_t\right),\Delta\boldsymbol{\Sigma}_t\right\rangle-\left\|\boldsymbol{\Lambda}\odot\boldsymbol{\Sigma}_t\right\|_1\\&+\left\|\boldsymbol{\Lambda}\odot\left(\boldsymbol{\Sigma}_t+\Delta\boldsymbol{\Sigma}_t\right)\right\|_1.\end{aligned}\tag{10}$$

Then $\boldsymbol{\Sigma}_{t+1}$ is set as $\boldsymbol{\Sigma}_{t+1}=\boldsymbol{\Sigma}_t+\beta\Delta\boldsymbol{\Sigma}_t$. The whole proximal Newton algorithm is summarized in Algorithm 2.

V. Statistical and Computational Theories

In this section, we first introduce some technical assumptions and important definitions. Following these, we establish the statistical convergence rate of our proposed covariance estimator and the iteration complexity of the proposed algorithm. The proofs are provided in the supplementary material.

A. Assumptions

We present several assumptions about the true covariance matrix $\boldsymbol{\Sigma}^*$. Define the support set of $\boldsymbol{\Sigma}^*$ as $\mathcal{S}^*=\left\{(i,j)\mid\boldsymbol{\Sigma}_{ij}^*\neq 0\right\}$, with s^* representing its size, i.e, $s^*=|\mathcal{S}^*|$. In the following, we impose some mild conditions on the $\boldsymbol{\Sigma}^*$.

Assumption 10. *For the ground truth covariance matrix $\boldsymbol{\Sigma}^*$, there exist finite upper and positive lower bounds on its eigenvalues. Specifically, there exists a constant $\kappa\geq 1$ that satisfies*

$$0<\frac{1}{\kappa}\leq\lambda_{\mathsf{min}}\left(\boldsymbol{\Sigma}^*\right)\leq\lambda_{\mathsf{max}}\left(\boldsymbol{\Sigma}^*\right)\leq\kappa<\infty.$$

Assumption 10 is a common premise in the literature concerning the estimation of sparse covariance matrices [28], [30], [53], [65]. This assumption offers several advantages, and readers are encouraged to consult [53] for further details.

Assumption 11. *Given the true covariance matrix $\boldsymbol{\Sigma}^*$, there exist universal constants α, c_7 such that*

$$\left\|\boldsymbol{\Sigma}_{\mathcal{S}^*}^*\right\|_{\min}=\min_{(i,j)\in\mathcal{S}^*}\left|\boldsymbol{\Sigma}_{ij}^*\right|\geq(\alpha+c_7)\,\lambda,$$

where α is a constant introduced in Assumption 3, and $c_7\in(0,\alpha)$ satisfies $p_\lambda'\left(c_7\lambda\right)\geq\frac{\lambda}{2}$.

Assumption 11 is referred to as the minimum signal strength condition, commonly applied in the analysis of non-convex penalized regression problems [40], [42], [58]. Additionally, it is essential to select λ appropriately to ensure that the regularization is sufficiently large to eliminate irrelevant coordinates, thereby yielding that the solution is adequately sparse.

Assumption 12. *There exists a generic constant c_8 such that*

$$\lambda=c_8\sqrt{\frac{\log d}{m}}\geq 4\left(\left\|\nabla f\left(\boldsymbol{\Sigma}^*\right)\right\|_{\max}+\varepsilon\right).$$

Assumption 12 sets the tuning parameter λ in the order of $\sqrt{\frac{\log d}{mn}}$. It prevents λ from becoming overly large as the number of measurements m or the number of data n increases, thereby ensuring that the estimators maintain a close correspondence with the true model parameters.

Definition 13. Define a local region around $\boldsymbol{\Sigma}^*$ as

$$\mathcal{B}\left(\boldsymbol{\Sigma}^*,r\right)=\left\{\boldsymbol{\Sigma}\succ 0\mid\left\|\boldsymbol{\Sigma}-\boldsymbol{\Sigma}^*\right\|_{\mathrm{F}}\leq r\right\}.$$

In our analysis, we set the radius r as $\frac{\rho_{2s^*+2\widetilde{s}}^-}{4\tau\kappa}$. Under Assumption 12, we prove that the solution produced by our algorithm will fall within the local region $\mathcal{B}\left(\boldsymbol{\Sigma}^*,\frac{\rho_{2s^*+2\widetilde{s}}^-}{4\tau\kappa}\right)$ after a finite number of iterations.

B. Statistical Guarantees and Consequences

The following presents the main result, which demonstrates the contraction property of the solution path $\{\widetilde{\boldsymbol{\Sigma}}^{(k)}\}_{k\geq 1}$.

Theorem 14 (Contraction Property)**.** *Consider the estimator in* (3) *and suppose the Assumptions 4, 7, 10, and 11 hold. Then with probability exceeding $1-c_4\exp\left(-c_5\sqrt{d}\right)$ for some $c_4,c_5>0$, the ε-optimal solution $\widetilde{\boldsymbol{\Sigma}}^{(k)}$ is bounded by:*

$$\begin{aligned}\left\|\widetilde{\boldsymbol{\Sigma}}^{(k)}-\boldsymbol{\Sigma}^*\right\|_{\mathrm{F}}\leq\frac{1}{\rho_{2s^*+2\widetilde{s}}^-}&\Bigg(\underbrace{\left\|\left(\nabla f\left(\boldsymbol{\Sigma}^*\right)\right)_{\mathcal{S}^*}\right\|_{\mathrm{F}}}_{\textit{oracle rate}}+\underbrace{\varepsilon\sqrt{s^*}}_{\textit{optimization error}}\\&+\delta\underbrace{\left\|\widetilde{\boldsymbol{\Sigma}}^{(k-1)}-\boldsymbol{\Sigma}^*\right\|_F}_{\textit{contraction}}\Bigg),\end{aligned}$$

for $1\leq k\leq K$, where $\delta\in(0,1)$ is the contraction factor, provided that $m=\mathcal{O}\left(\left(s^*+\widetilde{s}\right)\log^2\left(d/\left(s^*+\widetilde{s}\right)\right)\right)$.

Remark 15. The oracle estimator $\widehat{\boldsymbol{\Sigma}}^O$ is defined with prior knowledge of the true support set $\mathcal{S}^*$, and is given by

$$\widehat{\boldsymbol{\Sigma}}^O=\arg\min_{\boldsymbol{\Sigma}}f\left(\boldsymbol{\Sigma}\right)\qquad\text{s.t. }\boldsymbol{\Sigma}_{\overline{\mathcal{S}^*}}=\mathbf{0}.$$

The statistical convergence rate of this oracle estimator, commonly referred to as the oracle rate, provides a benchmark for evaluating the theoretical upper limit of estimator performance when the true support set is known. Specifically, the distance between $\widehat{\boldsymbol{\Sigma}}^{O}$ and $\boldsymbol{\Sigma}^*$ is characterized by $\left\|\widehat{\boldsymbol{\Sigma}}^{O}-\boldsymbol{\Sigma}^*\right\|_{\mathrm{F}} \lesssim \|(\nabla f(\boldsymbol{\Sigma}^*))_{\mathcal{S}^*}\|_{\mathrm{F}}$. This inequality indicates that the oracle estimator can theoretically approximate the true parameter matrix $\boldsymbol{\Sigma}^*$ very closely when the true support sets are known.

Theorem 14 elaborates the estimation error between the ε-optimal solution $\widetilde{\boldsymbol{\Sigma}}^{(k)}$ and the ground truth $\boldsymbol{\Sigma}^*$ is constrained by three primary factors: the oracle rate, the optimization error, and a contraction term. The oracle rate represents the error bound achievable under ideal conditions using the optimal strategy; the optimization error reflects the performance gap between the actual optimization algorithm and the optimal strategy; the contraction term indicates the rate at which the error converges to its minimum value as the number of iterations increases. The ensuing results specify the exact statistical convergence rate.

Corollary 16. *Let $\mathbf{x}$ be a sub-Gaussian random vector with zero mean and covariance $\boldsymbol{\Sigma}^*$ and $\{\mathbf{x}_i\}_{i=1}^n$ be a collection of i.i.d. samples from $\mathbf{x}$. Suppose that Assumptions 4, 7, 9, 10, and 11 hold, that is, $\lambda \asymp \sqrt{\frac{\log d}{mn}}$ and $\varepsilon \lesssim \sqrt{\frac{1}{mn}}$. If $\tau \lesssim \sqrt{\frac{1}{mn}}\left\|(\boldsymbol{\Sigma}^*)^{-1}\right\|_{\max}^{-1}$, then the ε-optimal solution $\widetilde{\boldsymbol{\Sigma}}^{(1)}$ satisfies*

$$\left\|\widetilde{\boldsymbol{\Sigma}}^{(1)}-\boldsymbol{\Sigma}^*\right\|_{\mathrm{F}} \lesssim \sqrt{\frac{s^*\log d}{mn}}$$

with high probability (w.h.p.).

Corollary 17. *Let $\mathbf{x}$ be a sub-Gaussian random vector with zero mean and covariance $\boldsymbol{\Sigma}^*$ and $\{\mathbf{x}_i\}_{i=1}^n$ be a collection of i.i.d. samples from $\mathbf{x}$. Suppose that Assumptions 4, 7, 9, 10, and 11 hold, that is, $\lambda \asymp \sqrt{\frac{\log d}{mn}}$ and $\varepsilon \lesssim \sqrt{\frac{1}{mn}}$. If $\tau \lesssim \sqrt{\frac{1}{mn}}\left\|(\boldsymbol{\Sigma}^*)^{-1}\right\|_{\max}^{-1}$, and $K \gtrsim \log(\lambda\sqrt{mn}) \gtrsim \log\log d$, then the ε-optimal solution $\widetilde{\boldsymbol{\Sigma}}^{(K)}$ satisfies*

$$\left\|\widetilde{\boldsymbol{\Sigma}}^{(K)}-\boldsymbol{\Sigma}^*\right\|_{\mathrm{F}} = \mathcal{O}_p\left(\sqrt{\frac{s^*}{mn}}\right),$$

w.h.p.

Corollary 16 and Corollary 17 are direct consequences of Theorem 14. The former addresses the scenario where $k=1$ and describes a contraction property resulting from the MM-based multistage convex relaxation algorithm. It demonstrates that to achieve the oracle rate, the optimization error ε must be chosen such that $\varepsilon \leq \frac{\|(\nabla f(\boldsymbol{\Sigma}^*))_{\mathcal{S}^*}\|_{\mathrm{F}}}{\sqrt{s^*}}$, and the parameter K must be sufficiently large. The latter implies that, with minimal assumptions, solving no more than approximately $\log\log d$ convex problems is enough to achieve the oracle rate $\sqrt{\frac{s^*}{mn}}$.

C. Computational Theory

Lemma 18. *Under Assumptions 7 and 10, $f(\boldsymbol{\Sigma})$ is $(\rho^-_{2s^*+2\widetilde{s}}+\frac{16\tau^3}{\kappa^2(4\tau+\rho^-_{2s^*+2\widetilde{s}})^2})$-strongly convex and $(\rho^+_{2s^*+2\widetilde{s}}+\frac{16\tau^3\kappa^2}{(4\tau-\rho^-_{2s^*+2\widetilde{s}})^2})$-smooth in the region $\mathcal{B}(\boldsymbol{\Sigma}^*,\frac{\rho^-_{2s^*+2\widetilde{s}}}{4\tau\kappa})$, where κ is defined in Assumption 10, $\rho^-_{2s^*+2\widetilde{s}}$, $\rho^+_{2s^*+2\widetilde{s}}$ is defined in Assumption 7 and τ is the positive tuning hyper-parameter.*

Lemma 18 ensures that within the specified region, the function exhibits strong convexity and smoothness properties, which are essential for stability and convergence analysis. Next, we present the explicit iteration complexity of the proposed algorithm. We begin by characterizing the convergence for the first stage.

Theorem 19. *For $k=1$, suppose Assumption 4, 7, 9, 10, and 11 hold. After a sufficient number of iterations $T<\infty$, for all $t \geq T$, the following conditions are satisfied: $\left\|\boldsymbol{\Sigma}_t^{(1)}-\boldsymbol{\Sigma}^*\right\|_{\mathrm{F}} \leq r$, $\left\|\left(\boldsymbol{\Sigma}_t^{(1)}\right)_{\overline{\mathcal{S}^*}}\right\|_0 \leq \widetilde{s}$ and*

$$\left\|\boldsymbol{\Sigma}_{t+1}^{(1)}-\widehat{\boldsymbol{\Sigma}}^{(1)}\right\|_{\mathrm{F}} \leq \frac{\tau\kappa^3}{\rho^-_{2s^*+2\widetilde{s}}}\left\|\boldsymbol{\Sigma}_t^{(1)}-\widehat{\boldsymbol{\Sigma}}^{(1)}\right\|_{\mathrm{F}}^2.$$

Moreover, we need at most $T+\log\log(\frac{3\rho^+_{2s^*+2\widetilde{s}}}{\varepsilon})$ iterations to terminate the proximal Newton algorithm.

Theorem 20. *For $k \geq 2$, suppose 4, 7, 9, 10, and 11 hold. For all iterations t, we have $\left\|\boldsymbol{\Sigma}_t^{(k)}-\boldsymbol{\Sigma}^*\right\|_{\mathrm{F}} \leq r$, $\left\|\left(\boldsymbol{\Sigma}_t^{(k)}\right)_{\overline{\mathcal{S}^*}}\right\| \leq \widetilde{s}$, which guarantees $\beta=1$, and*

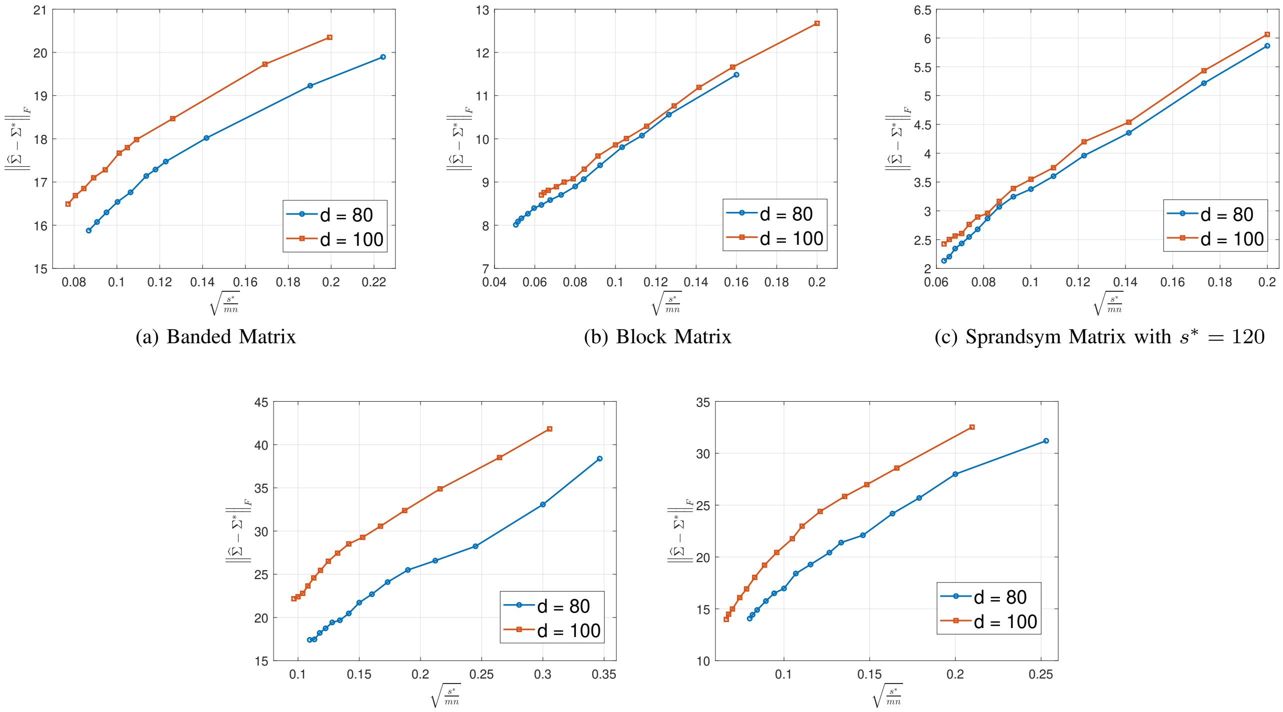
$$\left\|\boldsymbol{\Sigma}_{t+1}^{(k)}-\widehat{\boldsymbol{\Sigma}}^{(k)}\right\|_{\mathrm{F}} \leq \frac{\tau\kappa^3}{\rho^-_{2s^*+2\widetilde{s}}}\left\|\boldsymbol{\Sigma}_t^{(k)}-\widehat{\boldsymbol{\Sigma}}^{(k)}\right\|_{\mathrm{F}}^2.$$

Moreover, the proximal Newton algorithm requires at most $\log\log(\frac{3\rho^+_{2s^*+2\widetilde{s}}}{\varepsilon})$ iterations to terminate.

Theorem 19 demonstrates that the solution enters the ball $\mathcal{B}(\boldsymbol{\Sigma}^*,\frac{\rho^-_{2s^*+2\widetilde{s}}}{4\tau\kappa})$ following the initial stage. To satisfy the approximate KKT conditions, the number of iterations is capped at $\mathcal{O}(T+\log\log(\frac{3\rho^+_{2s^*+2\widetilde{s}}}{\varepsilon}))$. The resulting solution $\widetilde{\boldsymbol{\Sigma}}^{(1)}$ exhibits notable qualities, laying the foundation for improved computational efficiency in the subsequent stages of our proximal Newton algorithm. Theorem 20 states that the sparsity of the solution is preserved throughout the iterations, and the solution remains within $\mathcal{B}(\boldsymbol{\Sigma}^*,\frac{\rho^-_{2s^*+2\widetilde{s}}}{4\tau\kappa})$. Due to the sparsity of the solution, the algorithm exhibits quadratic convergence. Moreover, the algorithm achieves a logarithmic number of iterations, $\mathcal{O}(\log\log(\frac{3\rho^+_{2s^*+2\widetilde{s}}}{\varepsilon}))$, to satisfy the approximate KKT condition.

VI. Numerical Experiments

In this section, we examine the practical recovery performance of the proposed estimator and provide numerical results for the proposed algorithm for sparse CCS. Section VI-A describes the generation of synthetic data and the measurement criteria used in the paper. In Section VI-B and Section VI-C, the statistical theoretical results of Section V-B and the computational theoretical results of Section V-C are demonstrated, respectively. Section VI-D compares the relative estimation error of our proposed estimator and existing ℓ_1 estimator [34]. The non-convex penalty function chosen for these experiments is the MCP, with a constant setting of $b=2$



(a) Banded Matrix (b) Block Matrix (c) Sprandsym Matrix with $s^* = 120$

(d) Probability Matrix with $\mathbf{P} = 0.97$ (e) Probability Matrix with $\mathbf{P} = 0.85$

Fig. 1. The oracle rate for different dimensions of different covariance models[1].

across all trials. The selection of the tuning parameters λ and τ is determined by applying five-fold cross-validation. All methods are implemented in MATLAB and run on an Intel $i7-10700$ 2.90 GHz $\times 8$ with 16 GB of RAM. All results are averaged on 100 Monte Carlo realizations.

A. Synthetic Datasets and Measurement Criteria

For different fixed d, we generate n independent data points by sampling i.i.d. from a multivariate normal distribution $\mathcal{N}(0, \boldsymbol{\Sigma}^*)$. In this paper, we consider three typical covariance models and two additional models with random sparsity. The details of each model are as follows:

- Banded matrix:

$$\Sigma_{ij}^* = \begin{cases} 1 - \frac{|i-j|}{10}, & |i-j| \leq 10, \\ 0, & \text{otherwise}. \end{cases}$$

- Block matrix: The indices $1, 2, \ldots, d$ are partitioned into 10 ordered groups of equal size with

$$\Sigma_{ij}^* = \begin{cases} 1, & i = j, \\ 0.5, & i \text{ and } j \ (i \neq j) \text{ belong to the same group}, \\ 0, & \text{otherwise}. \end{cases}$$

- Toeplitz matrix: $\Sigma_{ij}^* = 0.75^{|i-j|}$.
- Sprandsym matrix: $\boldsymbol{\Sigma}^*$ is generated using the "*sprandsym*" built-in function in MATLAB with a sparsity controlled by the parameter s^*.
- Probability matrix: $\boldsymbol{\Sigma}^*$ is generated with off-diagonal entries drawn $i.i.d.$ from a uniform $(-1, 1)$ distribution. These entries are set to zero with a fixed probability $\mathbf{P}$. Finally, a multiple of the identity was added to the resulting matrix.

The first two models are sparse, while the third one is approximately sparse. The fourth model is positive, symmetric, and sparse when s^* is small. The fifth model is positive definite, well-conditioned, and can be sparse when $\mathbf{P}$ is appropriate (in this case, either $\mathbf{P} = 0.97$ or $\mathbf{P} = 0.85$ to simulate a very sparse and a somewhat sparse model).

The estimation performance is measured by the absolute error and relative error under both the Frobenius norm and the spectral norm, respectively. Specifically, the Frobenius absolute error (FAE) is defined as $\|\widehat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}^*\|_{\mathrm{F}}$, where $\widehat{\boldsymbol{\Sigma}}$[2] is the estimated covariance matrix and $\boldsymbol{\Sigma}^*$ is the true value. Similarly, the Frobenius relative error (FRE) is defined as $\frac{\|\widehat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}^*\|_{\mathrm{F}}}{\|\boldsymbol{\Sigma}^*\|_{\mathrm{F}}}$. The selection performance is examined by the false positive rate (FPR) and the true positive rate (TPR), defined as, respectively

$$\mathrm{FPR} = \frac{\#\left\{(i,j) : \widehat{\Sigma}_{ij} \neq 0 \ \& \ \Sigma_{ij}^* = 0\right\}}{\#\left\{(i,j) : \Sigma_{ij}^* = 0\right\}},$$

$$\mathrm{TPR} = \frac{\#\left\{(i,j) : \widehat{\Sigma}_{ij} \neq 0 \ \& \ \Sigma_{ij}^* \neq 0\right\}}{\#\left\{(i,j) : \Sigma_{ij}^* \neq 0\right\}}.$$

[1]The Toeplitz model is not reported due to the fact that it is not sparse but approximately sparse.

[2]In the whole Section VI, we use $\widehat{\boldsymbol{\Sigma}}$ to denote a generic covariance estimator, which can be the estimation results obtained by various algorithms and estimators.

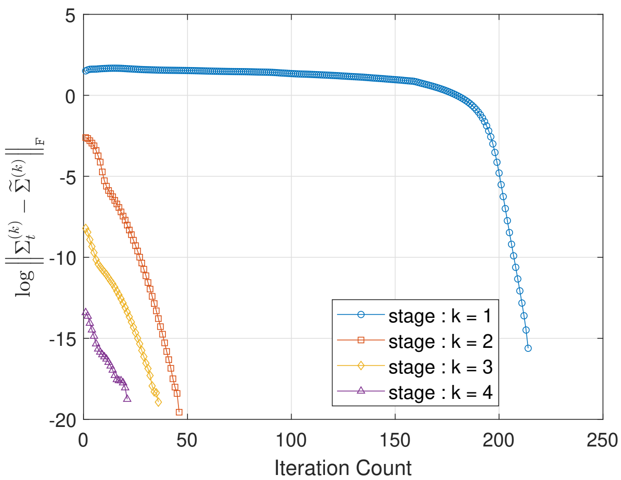


Fig. 2. Computational rate of convergence in each stage for simulation experiment. The $\mathbf{x}$-axis is the iteration count t within the k-th subproblem. A distinct phase transition in the algorithmic convergence rate from sub-linear to super-linear is observable as the iterations progress into the sparse region.

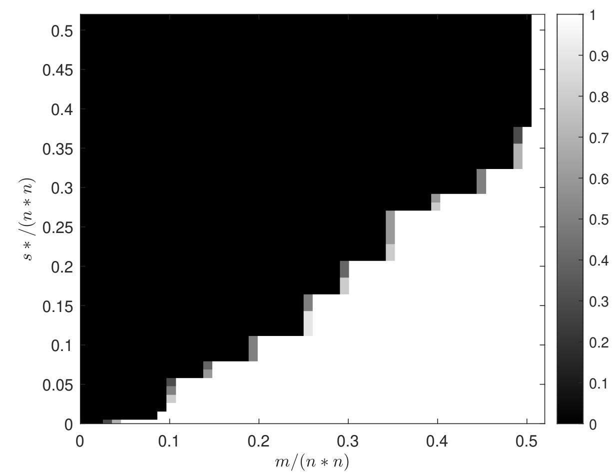


Fig. 3. The rate of successful covariance reconstruction when $d = 100$.

B. Demonstration of Oracle Rate

The oracle rate derived for the proposed estimator and algorithm in Section V-B is shown to be dependent on the sparsity level s^*, the number of measurements m, and the number of samples n. To demonstrate this theoretical result, we apply our proposed algorithm to synthetic datasets, as described in Section VI-A. Figure 1 illustrates the estimation performance across various matrix configurations differentiated by dimensions d, sparsity level s^*, measurement count m, and number of samples n. The x-axis is $\sqrt{\frac{s^*}{mn}}$, and the y-axis represents FAE. Two distinct lines depict the FAE for dimensions $d = \{80, 100\}$. We observe that an increase in measurement size m and sample count n corresponds to a reduction in FAE, visually demonstrating how the estimation quality of the covariance matrix varies with the oracle rate. According to Corollary 17, the Frobenius norm difference $\|\widehat{\mathbf{\Sigma}} - \mathbf{\Sigma}^*\|_F$ converges to the statistical errors as the stage index k increases, following the order of $\mathcal{O}\left(\sqrt{\frac{s^*}{mn}}\right)$. We can see that the estimation errors grow approximately linearly with the theoretical rate, which validates our theoretical guarantee.

C. Demonstration of Computational Rate

Figure 2 illustrates a detailed analysis of the convergence rates observed at each stage of the computational process. In the initial phase, the algorithm exhibits sub-linear convergence, attributed to the absence of strong convexity in the optimization problem. As the computation progresses and the solution approaches the sparse region, the convergence dynamics undergo a significant transformation. Within this region, feasible solutions become increasingly sparse, and the objective function effectively reduces to a low-dimensional form. This reduced form is characterized by strong convexity and smoothness properties that are crucial for accelerated convergence. Consequently, the algorithm achieves a quadratic convergence rate for $k \geq 2$. Notably, even for $k = 1$, where the convergence rate is initially sub-linear, the algorithm transitions into the contraction region and subsequently attains a quadratic convergence rate.

D. Simulation Experiment Comparisons

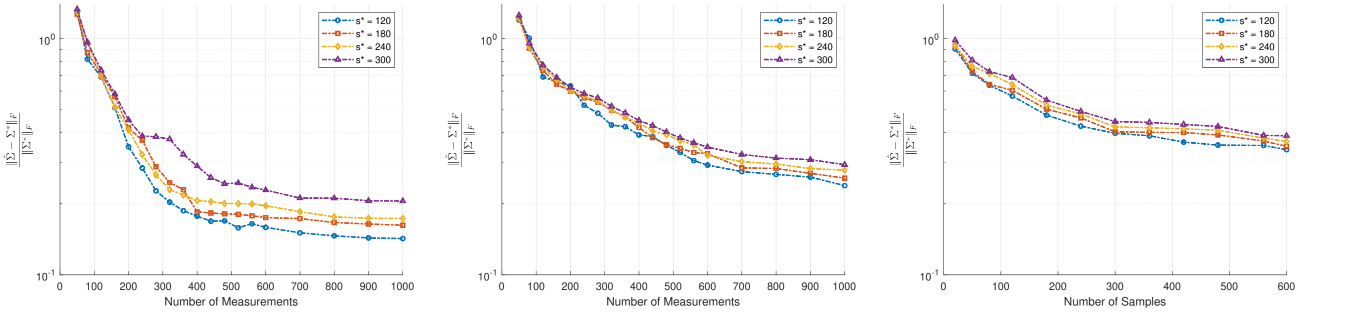
We conduct a series of Monte Carlo trials to evaluate the performance of our proposed CCS algorithm using "Sprandsym matrix model" with $d = 100$ and $n = 50$. The efficacy of the algorithm is analyzed under varying sparsity levels and noise conditions, and comparisons are made against existing estimators.

To evaluate the empirical probability of successful recovery, we use a color-coded matrix representation for each cell in Figure 3. To mitigate the influence of sample size n, we directly sense the true covariance matrix $\mathbf{\Sigma}^*$. A recovery is considered successful if the solution $\widehat{\mathbf{\Sigma}}$ satisfies $\frac{\|\widehat{\mathbf{\Sigma}} - \mathbf{\Sigma}^*\|_F}{\|\mathbf{\Sigma}^*\|_F} \leq 10^{-3}$.

Next, we examine the FRE of the estimated covariance matrices across different sparsity levels ($s^* \in \{120, 180, 240, 300\}$). Results are depicted in Figure 4a, obtained in the absence of noise and while sensing the true covariance $\mathbf{\Sigma}^*$ with a fixed sample count $n = 50$. These results illustrate an inverse relationship between FRE and the number of measurements m, where an increase in m correlates with enhanced estimation accuracy. Additionally, for a fixed m, a lower s^* value, which corresponds to higher sparsity, results in a reduced FRE, highlighting the influence of sparsity on estimation precision.

To simulate more realistic conditions, we introduce additive noise into each measurement. The noise η_i is modeled as a sub-exponential random variable scaled by γ, specifically $\gamma \cdot \mathcal{M}(0, 1)$. The influence of noise, particularly at $\gamma = 10^{-1}$, on the average relative error is illustrated in Figure 4b. Despite the presence of noise, the proposed estimator demonstrates resilience, maintaining consistent trends in FRE across varying measurements and sparsity levels.

Further, we examine the variations in FRE with respect to the sample size, as shown in Figure 4c, with the number of measurements fixed at $m = 300$. To investigate the impact



(a) (b) (c)

Fig. 4. The FRE of the estimated covariance matrices is examined in two contexts: (a) the true covariance under varying levels of sparsity in the absence of noise; (b) the sample covariance under different sparsity levels when subjected to a noise parameter of $\gamma = 0.1$ and $n = 50$; (c) the sample covariance under different sparsity levels when subjected to a noise parameter of $\gamma = 0.1$ and $m = 300$;

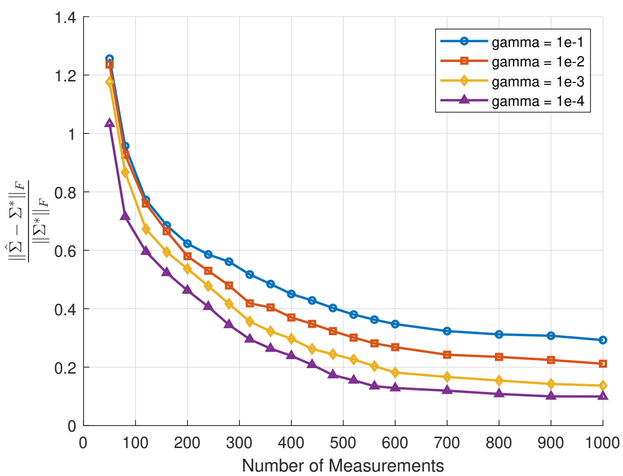


Fig. 5. The FRE of the estimated covariance matrices for different noise levels when $s^* = 300$.

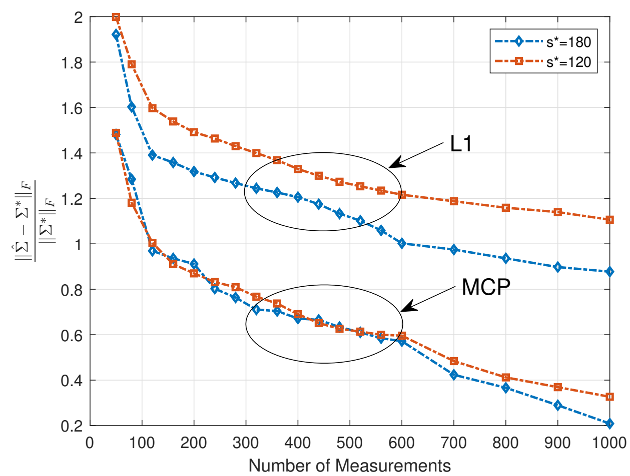


Fig. 6. The FRE of the estimated covariance matrices for different sparsity levels with noise (ℓ_1 v.s. MCP).

of the noise intensity, we analyze the performance of the estimator for $s^* = 300$ under different noise levels, as depicted in Figure 5. The results reveal that at lower m, noise intensity marginally affects recovery accuracy. However, as m increases, the impact of noise diminishes, leading to improved recovery precision. This underscores the importance of sufficient data in mitigating noise effects and enhancing estimation reliability.

Finally, we perform a comparative analysis between our proposed estimator and an existing ℓ_1-norm-based estimator [34] under a consistent noise level $\left(\gamma = 10^{-1}\right)$. The comparative results, shown in Figure 6, demonstrate that the proposed estimator, leveraging a non-convex penalty, consistently outperforms the ℓ_1-norm-based estimator in terms of Frobenius norm error. These findings validate our theoretical claim that employing a non-convex penalty can significantly enhance accuracy in CCS.

VII. Discussion

A. Bilinear Rank-one Measurement Model

Our research provides compelling evidence that covariance matrices can be effectively reconstructed from a limited number of quadratic measurements, provided the underlying structure exhibits sparsity. Furthermore, we extend our analysis to the bilinear rank-one measurement model, where measurements are expressed as $y_i = \boldsymbol{a}_i^\top \boldsymbol{\Sigma} \boldsymbol{b}_i$, with the sensing vectors $\boldsymbol{a}_i$ and $\boldsymbol{b}_i$ being independently generated. This extension validates our findings in the context of an asymmetric sensing framework, which holds significant potential for applications in diverse fields, including communication [66], imaging [67], and machine learning [68].

The bilinear model introduces additional complexity due to the interaction between two independent sensing vectors, $\boldsymbol{a}_i$ and $\boldsymbol{b}_i$. Nonetheless, it is straightforward to demonstrate that this configuration satisfies Assumption 5, a critical criterion for ensuring a successful recovery. The derivation closely follows that of the quadratic model, with appropriate adaptations to address the bilinear structure. Notably, we establish the Hessian matrix associated with the bilinear model retains symmetry, thereby preserving the computational efficiency and efficacy of our proposed estimator. Consequently, the bilinear measurement model inherits the oracle property, further underscoring the robustness and broad applicability of our approach.

B. CCS for variable with non-zero mean

In the preceding discussion, we consider the scenario in which the mean signal vector is assumed to be zero. However, this assumption frequently does not hold true in numerous practical applications. When the mean vector is unknown, conventional methods often involve first estimating the mean vector using robust techniques, followed by a subsequent estimation of the covariance matrix based on this mean estimate. This two-step procedure introduces additional tuning parameters and increases both statistical variability and computational complexity.

To address these challenges, we propose an end-to-end methodology that leverages a pairwise difference approach, thereby eliminating the need for direct mean estimation. Specifically, we define $N := n(n-1)/2$ and consider the pairwise differences

$$\{\widetilde{\mathbf{x}}_1, \widetilde{\mathbf{x}}_2, \dots, \widetilde{\mathbf{x}}_N\} = \{\mathbf{x}_1 - \mathbf{x}_2, \mathbf{x}_2 - \mathbf{x}_3, \dots, \mathbf{x}_{n-1} - \mathbf{x}_n\},$$

which are identically distributed with zero mean. The covariance matrix of these differences is given by $\operatorname{cov}(\widetilde{\mathbf{x}}) = 2\mathbf{\Sigma}$, which is a U-statistic. While the original sample covariance is calculated as $\mathbf{\Sigma}_{\text{sam}} = \frac{1}{n}\sum_{i=1}^{n}(\mathbf{x}_i - \overline{\mathbf{x}})(\mathbf{x}_i - \overline{\mathbf{x}})^\top$, where $\overline{\mathbf{x}} = \frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i$, the pairwise difference approach allows us to equivalently express it as

$$\mathbf{\Sigma}_{\text{sam}} = \frac{1}{N}\sum_{i=1}^{N}\widetilde{\mathbf{x}}_i\widetilde{\mathbf{x}}_i^\top/2.$$

This reformulation simplifies the estimation process by obviating the direct calculation of the mean, thereby reducing computational complexity and mitigating statistical errors associated with separate mean and covariance estimations.

VIII. Conclusion

In this paper, we conduct a detailed investigation into high-dimensional sparse covariance matrix estimation within the framework of the quadratic measurements model. This model is relevant for scenarios with stringent processing and memory constraints, such as real-time data acquisition systems. Our results establish that sparse covariance matrices can be reconstructed with high accuracy using a minimal number of quadratic measurements, requiring significantly reduced storage requirements. We provide rigorous theoretical analysis and empirical validations to support these conclusions. Notably, our proposed estimators exhibit superior statistical convergence rates compared to existing methods, highlighting their practical efficacy and potential in real-world applications.

Appendix A
Restricted Eigenvalue & Sparse Eigenvalue

We first introduce an equivalent definition of definition 5.

Definition 21. For three positive integers s_0, ϑ, r, the largest and smallest localized restricted eigenvalues are defined as

$$\psi^+_{s_0,\vartheta,r} = \sup\left\{\frac{\mathbf{u}^\top\nabla^2 f(\mathbf{\Sigma})\,\mathbf{u}}{\mathbf{u}^\top\mathbf{u}} \mid (\mathbf{u},\mathbf{\Sigma}) \in \mathcal{I}(s_0,\vartheta,r)\right\};$$
$$\psi^-_{s_0,\vartheta,r} = \inf\left\{\frac{\mathbf{u}^\top\nabla^2 f(\mathbf{\Sigma})\,\mathbf{u}}{\mathbf{u}^\top\mathbf{u}} \mid (\mathbf{u},\mathbf{\Sigma}) \in \mathcal{I}(s_0,\vartheta,r)\right\},$$

where $\mathcal{I}(s_0,\vartheta,r)$ is defined as

$$\left\{(\mathbf{u},\mathbf{\Sigma})\,\middle|\,\mathcal{S}^* \subseteq \mathcal{J}, |\mathcal{J}| \le s_0, \left\|\mathbf{u}_{\overline{\mathcal{J}}}\right\|_1 \le \vartheta\left\|\mathbf{u}_{\mathcal{J}}\right\|_1, \|\mathbf{\Sigma}-\mathbf{\Sigma}^*\|_{\mathrm{F}} \le r\right\},$$

which represents a local ℓ_1 cone.

We now proceed to introduce the relationships between SE and RE.

Proposition 22. *For any $\mathbf{\Sigma} \in \mathcal{I}(s,\vartheta,r) \cap \mathcal{B}(\mathbf{\Sigma}^*, r)$, the following inequalities hold:*

$$c_1\psi^-_{s,\vartheta,r} \le \rho^-_s \le c_2\psi^-_{s,\vartheta,r},$$
$$c_3\psi^+_{s,\vartheta,r} \le \rho^+_s \le c_4\psi^+_{s,\vartheta,r},$$

where c_1, c_2, c_3 and c_4 are constant.

For a detailed proof of Proposition 22, readers are referred to the foundational work in [55], which provides comprehensive mathematical derivations and discussions. This reference is omitted here for brevity.

Appendix B
Computing the Newton Direction With Active Set Strategy

In this section, we detail the computation of the Newton Direction as applied within our optimization framework. For clarity, we omit the stage index k and focus on a single Newton iteration. The gradient and Hessian for $f(\mathbf{\Sigma})$ are given by

$$\nabla f(\mathbf{\Sigma}) = -\frac{1}{m}\mathcal{A}^*(\boldsymbol{y} - \mathcal{A}(\mathbf{\Sigma})) - \tau\mathbf{\Sigma}^{-1},$$

and

$$\nabla^2 f(\mathbf{\Sigma}) = \frac{1}{m}\sum_{i=1}^{m}(\mathbf{A}_i \otimes \mathbf{A}_i) + \tau\left(\mathbf{\Sigma}^{-1} \otimes \mathbf{\Sigma}^{-1}\right).$$

The direct computation of the Hessian matrix involves significant computational overhead, with a complexity of $\mathcal{O}\left(md^4\right)$, making it impractical for high-dimensional data. However, the inherent symmetric structure of the Hessian enables certain computational optimizations. By utilizing a coordinate descent approach, it is possible to update all variables in $\mathcal{O}\left(md^3\right)$ time. This methodology has demonstrated efficacy in addressing Lasso-type problems [69]–[71] with theoretical justifications [72].

In order to provide a more precise articulation of the issue, we reformulate (8) as follows:

$$\begin{aligned}
&\overline{f}(\mathbf{\Delta}) \\
=&f(\mathbf{\Sigma}_t) + \operatorname{tr}\left((\mathbf{Q} - \tau\mathbf{W})^\top\mathbf{\Delta}\right) \\
&+ \frac{1}{2}\left(\frac{1}{m}\sum_{i=1}^{m}\operatorname{tr}(\mathbf{A}_i\mathbf{\Delta}\mathbf{A}_i\mathbf{\Delta}) + \tau\operatorname{tr}(\mathbf{W}\mathbf{\Delta}\mathbf{W}\mathbf{\Delta})\right),
\end{aligned} \tag{11}$$

where $\mathbf{Q} = -\frac{1}{m}\mathcal{A}^*(\boldsymbol{y} - \mathcal{A}(\mathbf{\Sigma}))$ and $\mathbf{W} = \mathbf{\Sigma}^{-1}$.

In refining the Newton direction, we represent the current approximation as $\mathbf{D}$ and the updated direction as $\mathbf{D}'$. For the coordinate descent update of the variable Σ_{ij} (assuming $i < j$), the update is constructed to preserve symmetry as follows:

$$\mathbf{D}' = \mathbf{D} + \mu\left(\boldsymbol{e}_i\boldsymbol{e}_j^\top + \boldsymbol{e}_j\boldsymbol{e}_i^\top\right),$$

where μ is the scalar update magnitude, and $\boldsymbol{e}_i$, $\boldsymbol{e}_j$ are the standard basis vectors. The corresponding one-dimensional optimization problem associated with the equation for (9) is expressed as follows:

$$\underset{\mu}{\operatorname{argmin}}\ \overline{f}\left(\mathbf{D}+\mu\left(\boldsymbol{e}_i\boldsymbol{e}_j^\top+\boldsymbol{e}_j\boldsymbol{e}_o^\top\right)\right)+2\Lambda_{ij}\left|\Sigma_{ij}+D_{ij}+\mu\right|.$$

After substituting $\mathbf{D}'=\mathbf{D}+\mu\left(\boldsymbol{e}_i\boldsymbol{e}_j^\top+\boldsymbol{e}_j\boldsymbol{e}_i^\top\right)$ for $\boldsymbol{\Delta}$ in (11), the function $\overline{f}$ is refined to focus exclusively on terms dependent on μ, with other terms omitted. The contributions to the function from three terms are:

- The linear term $\operatorname{tr}\left(\left(\mathbf{Q}-\tau\mathbf{W}\right)^\top\boldsymbol{\Delta}\right)$ contributes $2\mu\left(Q_{ij}-\tau W_{ij}\right)$,
- The penalty term gives $2\Lambda_{ij}\left|\Sigma_{ij}+D_{ij}+\mu\right|$,
- The quadratic term yields

$$\frac{1}{m}\sum_{l=1}^{m}4\mu\left(\mathbf{A}_l\right)_{\cdot i}^\top\mathbf{D}\left(\mathbf{A}_l\right)_{\cdot j}+2\mu^2\tau\left(W_{ij}^2+W_{ii}W_{jj}\right)$$
$$+\frac{1}{m}\sum_{l=1}^{m}2\mu^2\left(\left(\mathbf{A}_l\right)_{ij}^2+\left(\mathbf{A}_l\right)_{ii}\left(\mathbf{A}_l\right)_{jj}\right)+4\mu\tau\mathbf{W}_{\cdot i}^\top\mathbf{D}\mathbf{W}_{\cdot j}.$$

Combining all contributions yields a quadratic function of μ, expressed as (12) and the closed-form solution for μ is given by

$$\mu=-e+\mathcal{T}\left(e-g/h,\Lambda_{ij}/h\right),\tag{13}$$

where $\mathcal{T}(u,v)=\operatorname{sign}(u)\cdot\max\{|u|-v,0\}$ is the soft-thresholding function, and

$$h=\tau\left(W_{ij}^2+W_{ii}W_{jj}\right)+\frac{\sum_{l=1}^{m}\left(\left(\mathbf{A}_l\right)_{ij}^2+\left(\mathbf{A}_l\right)_{ii}\left(\mathbf{A}_l\right)_{jj}\right)}{m},$$
$$g=Q_{ij}-\tau W_{ij}+\tau\mathbf{W}_{\cdot i}^\top\mathbf{D}\mathbf{W}_{\cdot j}+\frac{\sum_{l=1}^{m}\left(\mathbf{A}_l\right)_{\cdot i}^\top\mathbf{D}\left(\mathbf{A}_l\right)_{\cdot j}}{m},$$
$$e=\Sigma_{ij}+D_{ij}.$$

When $i=j$, the substitution of $\mathbf{D}'=\mathbf{D}+\mu\boldsymbol{e}_i\boldsymbol{e}_i^\top$ yields the update formula for D_{ii} as derived from (13). The components of the update formula are defined as follows:

$$h=\tau D_{ii}^2+\frac{\sum_{l=1}^{m}\left(\mathbf{A}_l\right)_{ii}^2}{m},$$
$$g=Q_{ii}-\tau W_{ii}+\tau\mathbf{W}_{\cdot i}^\top\mathbf{D}\mathbf{W}_{\cdot i}+\frac{\sum_{l=1}^{m}\left(\mathbf{A}_l\right)_{\cdot i}^\top\mathbf{D}\left(\mathbf{A}_l\right)_{\cdot i}}{m},$$
$$e=\Sigma_{ii}+D_{ii}.$$

The primary computational challenge arises in evaluating the expressions $\mathbf{W}_{\cdot i}^\top\mathbf{D}\mathbf{W}_{\cdot j}$ and $\sum_{l=1}^{m}\left(\mathbf{A}_l\right)_{\cdot i}^\top\mathbf{D}\left(\mathbf{A}_l\right)_{\cdot j}$. To alleviate the computational burden, which typically necessitates $\mathcal{O}\left(md^2\right)$ time, we introduce intermediary $d\times d$ matrices defined as $\mathbf{U}_l=\mathbf{D}\mathbf{A}_l$, $\mathbf{V}=\mathbf{D}\mathbf{W}$. This strategy facilitates the computation of $\mathbf{W}_{\cdot i}^\top\mathbf{V}_{\cdot j}$ and $\mathbf{W}_{\cdot j}^\top\left(\mathbf{A}_l\right)_{\cdot i}$ using $\mathcal{O}(d)$ floating-point operations. Furthermore, the maintenance $\mathbf{U}_l$ and $\mathbf{V}$ requires the adjustment of $\mathcal{O}(d)$ elements. For a comprehensive description of the methodology, please consult Algorithm 3.

Algorithm 3: Coordinate Descent Algorithms in Conjunction with the active set strategy.

Input: $\nabla f(\boldsymbol{\Sigma})$, $\boldsymbol{\Lambda}$, $\boldsymbol{\Sigma}$, $\{\boldsymbol{a}_l\}_{l=1}^m$

Initialize $\mathbf{W}=\boldsymbol{\Sigma}^{-1}$, $\mathbf{A}_l=\boldsymbol{a}_l\boldsymbol{a}_l^\top$, $\mathbf{D}=\mathbf{0}$, $\mathbf{U}_l=\mathbf{0}$, $\mathbf{V}=\mathbf{0}$

while *not converged* **do**

```
// Partition the variables into
   fixed and free sets
```

$\mathcal{C}_{\text{fixed}}=\{(i,j)\mid|\nabla f_{ij}(\boldsymbol{\Sigma})|\le\Lambda_{ij}\text{ and }\Sigma_{ij}=0\}$

$\mathcal{C}_{\text{free}}=\{(i,j)\mid|\nabla f_{ij}(\boldsymbol{\Sigma})|\le\Lambda_{ij}\text{ and }\Sigma_{ij}\neq0\}$

for $(i,j)\in\mathcal{A}_{\text{free}}$ **do**

$h=\tau\left(W_{ij}^2+W_{ii}W_{jj}\right)+\frac{\sum_{l=1}^{m}\left(\left(\mathbf{A}_l\right)_{ij}^2+\left(\mathbf{A}_l\right)_{ii}\left(\mathbf{A}_l\right)_{jj}\right)}{m}$

$g=\nabla_{ij}f(\boldsymbol{\Sigma})+\tau\mathbf{W}_{\cdot i}^\top\mathbf{V}_{\cdot j}+\frac{\sum_{l=1}^{m}\left(\mathbf{A}_l\right)_{\cdot i}^\top\left(\mathbf{U}_l\right)_{\cdot j}}{m}$

$e=\Sigma_{ij}+D_{ij}$

$\mu=-e+\mathcal{T}\left(e-g/h,\boldsymbol{\Lambda}_{jk}/h\right)$

$D_{ij}=D_{ij}+\mu$

$\left(\mathbf{U}_l\right)_{i\cdot}=\left(\mathbf{U}_l\right)_{i\cdot}+\mu\left(\mathbf{A}_l\right)_{j\cdot}$

$\left(\mathbf{U}_l\right)_{j\cdot}=\left(\mathbf{U}_l\right)_{j\cdot}+\mu\left(\mathbf{A}_l\right)_{i\cdot}$

$\mathbf{V}_{i\cdot}=\mathbf{V}_{i\cdot}+\mu\mathbf{W}_{j\cdot}$, $\mathbf{V}_{j\cdot}=\mathbf{V}_{j\cdot}+\mu\mathbf{W}_{i\cdot}$

end

end

Output: $\mathbf{D}$

A. Updating Only a Subset of Variables

We introduce an algorithm that leverages the active set update strategy to exploit the solution sparsity, thereby accelerating computation. The strategy involves two consecutive nested loops. In the outer loop, all coordinates are categorized into two sets based on a heuristic coordinate gradient thresholding rule (strong rule, [73]): the free set (active coordinates) and the fixed set (inactive coordinates). Subsequently, each iteration of the outer loop initiates an inner loop that performs coordinate optimization exclusively on the active set until convergence, while the inactive coordinates remain fixed at zero. Post each inner loop iteration, the algorithm employs a heuristic rule to reassess and update the active set, aiming to further reduce the objective value. This iterative process continues until the

$$\frac{1}{2}\left(\frac{\sum_{l=1}^{m}\left(\left(\mathbf{A}_l\right)_{ij}^2+\left(\mathbf{A}_l\right)_{ii}\left(\mathbf{A}_l\right)_{jj}\right)}{m}+\tau\left(W_{ij}^2+W_{ii}W_{jj}\right)\right)\mu^2+$$
$$\left(Q_{ij}-\tau W_{ij}+\tau\mathbf{W}_{\cdot i}^\top\mathbf{D}\mathbf{W}_{\cdot j}+\frac{\sum_{l=1}^{m}\left(\mathbf{A}_l\right)_{\cdot i}^\top\mathbf{D}\left(\mathbf{A}_l\right)_{\cdot j}}{m}\right)\mu+2\Lambda_{ij}\left|\Sigma_{ij}+D_{ij}+\mu\right|\tag{12}$$

$$\boxed{\widehat{D}_{ij} = \begin{cases} \mathcal{T}\left(-\frac{Q_{ij}}{\tau W_{ii}W_{jj}+\sum_{l=1}^{m}\left((\mathbf{A}_l)_{ij}^2+(\mathbf{A}_l)_{ii}(\mathbf{A}_l)_{jj}\right)}, \frac{\Lambda_{ij}}{\tau W_{ii}W_{jj}+\sum_{l=1}^{m}\left((\mathbf{A}_l)_{ij}^2+(\mathbf{A}_l)_{ii}(\mathbf{A}_l)_{jj}\right)}\right) & \text{if } i \neq j, \\ -\Sigma_{ii} + \mathcal{T}\left(\Sigma_{ii} - \frac{Q_{ii}-\tau W_{ii}}{\tau W_{ii}^2+\sum_{l=1}^{m}(\mathbf{A}_l)_{ii}^2}, \frac{\Lambda_{ii}}{\tau W_{ii}^2+\sum_{l=1}^{m}(\mathbf{A}_l)_{ii}^2}\right) & \text{if } i = j. \end{cases}} \tag{14}$$

composition of the active set stabilizes across subsequent outer loop iterations. The efficacy of the active set strategy is well-documented in practical applications [70] and is underpinned by strong theoretical foundations [74].

The free set and fixed set are respectively defined as follows:

$$\begin{aligned} &\Sigma_{ij} \in \mathcal{C}_{\text{fixed}} \text{ if } \left|\nabla f_{ij}(\mathbf{\Sigma})\right| \le \Lambda_{ij} \text{ and } \Sigma_{ij} = 0, \\ &\Sigma_{ij} \in \mathcal{C}_{\text{free}} \text{ if } \left|\nabla f_{ij}(\mathbf{\Sigma})\right| > \Lambda_{ij} \text{ and } \Sigma_{ij} \neq 0. \end{aligned}$$

Further, we ascertain that a Newton update confined to the variables within the fixed set does not alter any coordinates in that set. For a detailed discussion on this analysis, we refer readers to the work by [70], [75].

Remark 23. The computation of the Newton direction is streamlined when $\mathbf{\Sigma}$ is diagonal, which is encountered in the initial Newton iteration, wherein the Algorithm is started with an identity matrix. This simple scenario has a succinct closed-form optimal solution presented in (14). On this occasion, it only takes $\mathcal{O}(1)$ time to determine each variable, so the time complexity for solving the first Newton direction drops sharply to $\mathcal{O}\left(md^2\right)$. The core cause of this occurrence lies in the fact that, in this case, the Hessian matrix is diagonal, indicating that each one-variable sub-problem is detached from the others. This means that updating each variable individually once suffices to achieve the optimal solution.

Appendix C
Proof of Statistical Theory

In this appendix, we provide the proofs of all the statistical theoretical results in Section V-B. In Subsection C-A, we begin by Lemma 24 that connects the SE condition to the localized version of the sparse strong convexity/sparse strong smoothness. Subsequently, we introduce several preliminary lemmas that support Lemma 26. Following this, we present Lemma 27, which sets bounds on the estimation error for the general problem (7), followed by Lemma 28 that outlines the estimation error limited for approximate solutions derived using the MM-based algorithm. Subsection C-B elaborates on Theorem 14, leveraging Lemma 27 and Lemma 28 to illustrate the solution path's contraction characteristic. Subsection C-C focuses on key concentration inequalities, which are vital for deducing the explicit statistical convergence rate. In Subsections C-D and C-E, the document presents the proofs for Corollary 16 and Corollary 17, respectively.

A. Technical Lemmata

Lemma 24. *Let $f(\cdot)$ be a convex differentiable function, and let $G_f(\mathbf{\Sigma}_1, \mathbf{\Sigma}_2)$ be defined as*

$$G_f(\mathbf{\Sigma}_1, \mathbf{\Sigma}_2) = f(\mathbf{\Sigma}_1) - f(\mathbf{\Sigma}_2) - \langle \nabla f(\mathbf{\Sigma}_2), \mathbf{\Sigma}_1 - \mathbf{\Sigma}_2 \rangle .$$

Define the symmetrized Bregman divergence for $f(\cdot)$ as

$$\begin{aligned} G_f^s(\mathbf{\Sigma}_1, \mathbf{\Sigma}_2) &= G_f(\mathbf{\Sigma}_1, \mathbf{\Sigma}_2) + G_f(\mathbf{\Sigma}_2, \mathbf{\Sigma}_1) \\ &= \langle \nabla f(\mathbf{\Sigma}_1) - \nabla f(\mathbf{\Sigma}_2), \mathbf{\Sigma}_1 - \mathbf{\Sigma}_2 \rangle . \end{aligned}$$

For any $\mathbf{\Sigma}_1, \mathbf{\Sigma}_2 \in \mathcal{B}(\mathbf{\Sigma}^*, r)$ such that $\max\left\{\|(\mathbf{\Sigma}_1)_{\overline{\mathcal{S}^*}}\|_0, \|(\mathbf{\Sigma}_2)_{\overline{\mathcal{S}^*}}\|_0\right\} \le \widetilde{s}$, the following holds:

$$\begin{aligned} \frac{1}{2}\rho_{2s^*+2\widetilde{s}}^{-}\|\mathbf{\Sigma}_1 - \mathbf{\Sigma}_2\|_2^2 &\le G_f(\mathbf{\Sigma}_1, \mathbf{\Sigma}_2) \\ &\le \frac{1}{2}\rho_{2s^*+2\widetilde{s}}^{+}\|\mathbf{\Sigma}_1 - \mathbf{\Sigma}_2\|_2^2 , \end{aligned}$$

$$\begin{aligned} \rho_{2s^*+2\widetilde{s}}^{-}\|\mathbf{\Sigma}_1 - \mathbf{\Sigma}_2\|_2^2 &\le G_f^s(\mathbf{\Sigma}_1, \mathbf{\Sigma}_2) \\ &\le \rho_{2s^*+2\widetilde{s}}^{+}\|\mathbf{\Sigma}_1 - \mathbf{\Sigma}_2\|_2^2 . \end{aligned}$$

Proof. According to the mean value theorem and the convexity of the local region $\mathcal{B}(\mathbf{\Sigma}^*, r)$, there exist parameters $\theta_1, \theta_2, \theta_3 \in [0,1]$ such that

$$\begin{aligned} \widetilde{\mathbf{\Sigma}} &= \theta_1\mathbf{\Sigma}_1 + (1-\theta_1)\mathbf{\Sigma}_2 \in \mathcal{B}(\mathbf{\Sigma}^*, r), \\ \widetilde{\mathbf{\Sigma}}' &= \theta_2\widetilde{\mathbf{\Sigma}} + (1-\theta_2)\mathbf{\Sigma}_2 \in \mathcal{B}(\mathbf{\Sigma}^*, r), \\ \widetilde{\mathbf{\Sigma}}'' &= \theta_3\mathbf{\Sigma}_1 + (1-\theta_3)\mathbf{\Sigma}_2 \in \mathcal{B}(\mathbf{\Sigma}^*, r), \end{aligned}$$

and satisfies $\max\left\{\left\|\widetilde{\mathbf{\Sigma}}\right\|_0, \left\|\widetilde{\mathbf{\Sigma}}'\right\|_0\right\} \le 2\widetilde{s}$. The function $G_f(\mathbf{\Sigma}_1, \mathbf{\Sigma}_2)$ and the symmetrized Bregman divergence can be articulated as follows:

$$\begin{aligned} &f(\mathbf{\Sigma}_1) - f(\mathbf{\Sigma}_2) - \langle \nabla f(\mathbf{\Sigma}_2), \mathbf{\Sigma}_1 - \mathbf{\Sigma}_2 \rangle \\ =& \left\langle \nabla f\left(\widetilde{\mathbf{\Sigma}}\right) - \nabla f(\mathbf{\Sigma}_2), \mathbf{\Sigma}_1 - \mathbf{\Sigma}_2 \right\rangle = \|\mathbf{\Sigma}_1 - \mathbf{\Sigma}_2\|_{\nabla^2 f(\widetilde{\mathbf{\Sigma}}')}^2 , \end{aligned}$$

and

$$\langle \nabla f(\mathbf{\Sigma}_1) - \nabla f(\mathbf{\Sigma}_2), \mathbf{\Sigma}_1 - \mathbf{\Sigma}_2 \rangle = \|\mathbf{\Sigma}_1 - \mathbf{\Sigma}_2\|_{\nabla^2 f(\widetilde{\mathbf{\Sigma}}'')}^2 .$$

By applying the definition of the sparse eigenvalue, we arrive at the desired conclusion. □

Lemma 25. *Consider $\mathbf{\Sigma}(\theta) = \mathbf{\Sigma}^* + \theta(\mathbf{\Sigma} - \mathbf{\Sigma}^*)$ for $\theta \in (0,1]$. Then, we have*

$$G_f^s(\mathbf{\Sigma}(\theta), \mathbf{\Sigma}^*) \le \theta G_f^s(\mathbf{\Sigma}, \mathbf{\Sigma}^*) .$$

Proof. Define $\varphi(\theta)$ as follows:

$$\begin{aligned} \varphi(\theta) &= G_f(\mathbf{\Sigma}(\theta), \mathbf{\Sigma}^*) \\ &= f(\mathbf{\Sigma}(\theta)) - f(\mathbf{\Sigma}^*) - \langle \nabla f(\mathbf{\Sigma}^*), \mathbf{\Sigma}(\theta) - \mathbf{\Sigma}^* \rangle . \end{aligned}$$

The derivative of $f(\mathbf{\Sigma}(\theta))$ with respect to θ is $\langle \nabla f(\mathbf{\Sigma}(\theta)), \mathbf{\Sigma} - \mathbf{\Sigma}^* \rangle$. Thus

$$\varphi'(\theta) = \langle \nabla f(\mathbf{\Sigma}(\theta)) - \nabla f(\mathbf{\Sigma}^*), \mathbf{\Sigma} - \mathbf{\Sigma}^* \rangle .$$

Consequently, the symmetric Bregman divergence $G_f^s\left(\mathbf{\Sigma}\left(\theta\right),\mathbf{\Sigma}^*\right)$ is

$$\begin{aligned}&G_f^s\left(\mathbf{\Sigma}\left(\theta\right),\mathbf{\Sigma}^*\right)\\=&\left\langle\nabla f\left(\mathbf{\Sigma}\left(\theta\right)\right)-\nabla f\left(\mathbf{\Sigma}^*\right),\theta\left(\mathbf{\Sigma}-\mathbf{\Sigma}^*\right)\right\rangle\\=&\theta\varphi'\left(\theta\right)\end{aligned}$$

for $0<\theta\leq 1$. Given $\varphi'\left(1\right)=G_f^s\left(\mathbf{\Sigma},\mathbf{\Sigma}^*\right)$, and assuming the convexity of $\varphi\left(\theta\right)$ based on the convexity of $f\left(\cdot\right)$ and the linearity of $\mathbf{\Sigma}\left(\theta\right)$, $\varphi'\left(\theta\right)$ is non-decreasing. Thus

$$G_f^s\left(\mathbf{\Sigma}\left(\theta\right),\mathbf{\Sigma}^*\right)=\theta\varphi'\left(\theta\right)\leq\theta\varphi'\left(1\right)=\theta G_f^s\left(\mathbf{\Sigma},\mathbf{\Sigma}^*\right),$$

completing the proof. For a more detailed proof of the convexity of $\varphi\left(\theta\right)$, see [43], [58]. □

Lemma 26. *Consider a set $\mathcal{E}$ such that $\mathcal{S}^*\subseteq\mathcal{E}$. Given the condition $\|\nabla f\left(\mathbf{\Sigma}^*\right)\|_{\max}+\varepsilon\leq\|\mathbf{\Lambda}_{\mathcal{E}}\|_{\min}$, we have*

$$\left\|\left(\widetilde{\mathbf{\Sigma}}-\mathbf{\Sigma}^*\right)_{\overline{\mathcal{E}}}\right\|_1\leq 5\left\|\left(\widetilde{\mathbf{\Sigma}}-\mathbf{\Sigma}^*\right)_{\mathcal{E}}\right\|_1.$$

Proof. Define $\mathbf{\Omega}=\nabla f\left(\widetilde{\mathbf{\Sigma}}\right)+\mathbf{\Lambda}\odot\widetilde{\mathbf{\Xi}}$ and $\widetilde{\mathbf{\Delta}}=\widetilde{\mathbf{\Sigma}}-\mathbf{\Sigma}^*$. By the mean value theorem, there exists a $\theta\in[0,1]$, such that

$$\nabla f\left(\widetilde{\mathbf{\Sigma}}\right)-\nabla f\left(\mathbf{\Sigma}^*\right)=\operatorname{mat}\left(\mathbf{H}\left(\theta\right)\operatorname{vec}\left(\widetilde{\mathbf{\Delta}}\right)\right),$$

where $\mathbf{H}\left(\theta\right)=\nabla^2 f\left(\theta\mathbf{\Sigma}^*+\left(1-\theta\right)\widetilde{\mathbf{\Sigma}}\right)$. Then we have

$$\begin{aligned}&\left\langle\nabla f\left(\widetilde{\mathbf{\Sigma}}\right)+\mathbf{\Lambda}\odot\widetilde{\mathbf{\Xi}},\widetilde{\mathbf{\Delta}}\right\rangle\\=&\left\langle\nabla f\left(\mathbf{\Sigma}^*\right)+\operatorname{mat}\left(\mathbf{H}\left(\theta\right)\operatorname{vec}\left(\widetilde{\mathbf{\Delta}}\right)\right)+\mathbf{\Lambda}\odot\widetilde{\mathbf{\Xi}},\widetilde{\mathbf{\Delta}}\right\rangle\\\leq&\|\mathbf{\Omega}\|_{\max}\left\|\widetilde{\mathbf{\Delta}}\right\|_1.\end{aligned}$$

Given $\left\|\widetilde{\mathbf{\Delta}}\right\|_{H(\theta)}^2\geq 0$, it follows that:

$$0\leq\|\mathbf{\Omega}\|_{\max}\left\|\widetilde{\mathbf{\Delta}}\right\|_1-\underbrace{\left\langle\nabla f\left(\mathbf{\Sigma}^*\right),\widetilde{\mathbf{\Delta}}\right\rangle}_{\mathbf{I}}-\underbrace{\left\langle\mathbf{\Lambda}\odot\widetilde{\mathbf{\Xi}},\widetilde{\mathbf{\Delta}}\right\rangle}_{\mathbf{II}}.\tag{15}$$

For term $\mathbf{I}$, separating the support of $\nabla f\left(\mathbf{\Sigma}^*\right)$ and $\widetilde{\mathbf{\Delta}}$ to $\mathcal{E}$ and $\overline{\mathcal{E}}$, and then using the matrix Hölder's inequality, we obtain

$$\begin{aligned}\mathbf{I}=&\left\langle\left(\nabla f\left(\mathbf{\Sigma}^*\right)\right)_{\mathcal{E}},\widetilde{\mathbf{\Delta}}_{\mathcal{E}}\right\rangle+\left\langle\left(\nabla f\left(\mathbf{\Sigma}^*\right)\right)_{\overline{\mathcal{E}}},\widetilde{\mathbf{\Delta}}_{\overline{\mathcal{E}}}\right\rangle\\\geq&-\|\left(\nabla f\left(\mathbf{\Sigma}^*\right)\right)_{\mathcal{E}}\|_{\mathrm{F}}\left\|\widetilde{\mathbf{\Delta}}_{\mathcal{E}}\right\|_{\mathrm{F}}-\|\left(\nabla f\left(\mathbf{\Sigma}^*\right)\right)_{\overline{\mathcal{E}}}\|_{\max}\left\|\widetilde{\mathbf{\Delta}}_{\overline{\mathcal{E}}}\right\|_1\\\geq&-\|\left(\nabla f\left(\mathbf{\Sigma}^*\right)\right)_{\mathcal{E}}\|_{\mathrm{F}}\left\|\widetilde{\mathbf{\Delta}}\right\|_{\mathrm{F}}-\|\left(\nabla f\left(\mathbf{\Sigma}^*\right)\right)\|_{\max}\left\|\widetilde{\mathbf{\Delta}}\right\|_1.\end{aligned}$$

For term $\mathbf{II}$, separating the support of $\mathbf{\Lambda}\odot\widetilde{\mathbf{\Xi}}$ and $\widetilde{\mathbf{\Delta}}$ to $\mathcal{E}$ and $\overline{\mathcal{E}}$, and then using the matrix Hölder's inequality, we obtain

$$\begin{aligned}\mathbf{II}=&\left\langle\left(\mathbf{\Lambda}\odot\widetilde{\mathbf{\Xi}}\right)_{\mathcal{E}},\widetilde{\mathbf{\Delta}}_{\mathcal{E}}\right\rangle+\left\langle\left(\mathbf{\Lambda}\odot\widetilde{\mathbf{\Xi}}\right)_{\overline{\mathcal{E}}},\widetilde{\mathbf{\Delta}}_{\overline{\mathcal{E}}}\right\rangle\\\geq&-\|\mathbf{\Lambda}_{\mathcal{E}}\|_{\max}\left\|\widetilde{\mathbf{\Delta}}_{\mathcal{E}}\right\|_1+\|\mathbf{\Lambda}_{\overline{\mathcal{E}}}\|_{\min}\left\|\widetilde{\mathbf{\Delta}}_{\overline{\mathcal{E}}}\right\|_1.\end{aligned}$$

Plugging the above results into 15 yields

$$\begin{aligned}\left\|\widetilde{\mathbf{\Delta}}_{\overline{\mathcal{E}}}\right\|_1\leq&\frac{\lambda+\|\left(\nabla f\left(\mathbf{\Sigma}^*\right)\right)\|_{\max}+\omega_{\mathbf{\Lambda}}\left(\widetilde{\mathbf{\Sigma}}\right)}{\|\mathbf{\Lambda}_{\overline{\mathcal{E}}}\|_{\min}-\left(\|\left(\nabla f\left(\mathbf{\Sigma}^*\right)\right)\|_{\max}+\omega_{\mathbf{\Lambda}}\left(\widetilde{\mathbf{\Sigma}}\right)\right)}\left\|\widetilde{\mathbf{\Delta}}_{\mathcal{E}}\right\|_1\\\underset{(i)}{\leq}&5\left\|\widetilde{\mathbf{\Delta}}_{\mathcal{E}}\right\|_1,\end{aligned}$$

where (i) is from $\|\mathbf{\Lambda}_{\overline{\mathcal{E}}}\|_{\min}\geq\frac{\lambda}{2}$, Assumption 12 and $\omega_{\mathbf{\Lambda}}\left(\widetilde{\mathbf{\Sigma}}\right)\leq\varepsilon$. □

Lemma 27. *Suppose that Assumption 9, 10, 11 and 12 hold. Consider the general problem in* (7)*. Let there exists a set $\mathcal{E}$ such that*

$$\mathcal{S}^*\subseteq\mathcal{E},|\mathcal{E}|\leq 2s^*,\text{ and }\|\mathbf{\Lambda}_{\overline{\mathcal{E}}}\|_{\min}\geq\frac{\lambda}{2}.$$

Then the ε-optimal solution $\widetilde{\mathbf{\Sigma}}$ satisfies

$$\begin{aligned}\left\|\widetilde{\mathbf{\Sigma}}-\mathbf{\Sigma}^*\right\|_{\mathrm{F}}\leq&\frac{1}{\rho^-_{2s^*+2\widetilde{s}}}\left(\|\left(\nabla f\left(\mathbf{\Sigma}^*\right)\right)_{\mathcal{E}}\|_{\mathrm{F}}+\varepsilon\sqrt{|\mathcal{E}|}+\|\mathbf{\Lambda}_{\mathcal{S}^*}\|_{\mathrm{F}}\right)\\\leq&\frac{2+\sqrt{2}}{2\rho^-_{2s^*+2\widetilde{s}}}\lambda\sqrt{s^*}.\end{aligned}$$

Proof. Given the Assumption 12, it follows that $\lambda\geq 2\left(\|\nabla f\left(\mathbf{\Sigma}^*\right)\|_{\max}+\varepsilon\right)$. We introduce an intermediate estimator $\widetilde{\mathbf{\Sigma}}^*=\mathbf{\Sigma}^*+\theta(\widetilde{\mathbf{\Sigma}}-\mathbf{\Sigma}^*)$, where θ is chosen to ensure $\|\widetilde{\mathbf{\Sigma}}^*-\mathbf{\Sigma}^*\|_{\mathrm{F}}=r$ if $\|\widetilde{\mathbf{\Sigma}}^*-\mathbf{\Sigma}^*\|_{\mathrm{F}}>r$; otherwise $\theta=1$. This construction guarantees that $\|\widetilde{\mathbf{\Sigma}}^*-\mathbf{\Sigma}^*\|_{\mathrm{F}}\leq r$. By leveraging Lemma 26, we deduce that the approximate solution is contained within the ℓ_1-cone. Given the structure of $\widetilde{\mathbf{\Sigma}}^*$, we find that $\widetilde{\mathbf{\Sigma}}^*-\mathbf{\Sigma}^*=\theta(\widetilde{\mathbf{\Sigma}}-\mathbf{\Sigma}^*)$. This leads to the following inequality

$$\left\|\left(\widetilde{\mathbf{\Sigma}}^*-\mathbf{\Sigma}^*\right)_{\mathcal{E}}\right\|_1\leq 5\left\|\left(\widetilde{\mathbf{\Sigma}}^*-\mathbf{\Sigma}^*\right)_{\overline{\mathcal{E}}}\right\|_1.$$

By integrating the aforementioned inequality with the premise that $|\mathcal{E}|\leq 2s^*$, it follows that $\widetilde{\mathbf{\Sigma}}^*$ resides within the local ℓ_1-cone. Furthermore, by synthesizing Definition 21, Proposition 22, and Lemma 24, one can deduce the presence of localized restricted strong convexity, i.e.

$$\rho^-_{2s^*+2\widetilde{s}}\left\|\widetilde{\mathbf{\Sigma}}^*-\mathbf{\Sigma}^*\right\|_{\mathrm{F}}^2\leq G_f^s\left(\widetilde{\mathbf{\Sigma}}^*,\mathbf{\Sigma}^*\right).\tag{16}$$

Using Lemma 25, we can bound the right-hand side of the above inequality, such as

$$\begin{aligned}G_f^s\left(\widetilde{\mathbf{\Sigma}}^*,\mathbf{\Sigma}^*\right)\leq&\theta G_f^s\left(\widetilde{\mathbf{\Sigma}},\mathbf{\Sigma}^*\right)\\=&\theta\left\langle\nabla f\left(\widetilde{\mathbf{\Sigma}}\right)-\nabla f\left(\mathbf{\Sigma}^*\right),\widetilde{\mathbf{\Sigma}}-\mathbf{\Sigma}^*\right\rangle.\end{aligned}\tag{17}$$

Plugging (17) back into (16) yields

$$\begin{aligned}\rho^-_{2s^*+2\widetilde{s}}\left\|\widetilde{\mathbf{\Sigma}}^*-\mathbf{\Sigma}^*\right\|_{\mathrm{F}}^2\leq&\theta\left\langle\nabla f\left(\widetilde{\mathbf{\Sigma}}\right)-\nabla f\left(\mathbf{\Sigma}^*\right),\widetilde{\mathbf{\Delta}}\right\rangle\\=&\theta\underbrace{\left\langle\nabla f\left(\widetilde{\mathbf{\Sigma}}\right)+\mathbf{\Lambda}\odot\widetilde{\mathbf{\Xi}},\widetilde{\mathbf{\Delta}}\right\rangle}_{\mathbf{I}}\\&-\theta\underbrace{\left\langle\nabla f\left(\mathbf{\Sigma}^*\right),\widetilde{\mathbf{\Delta}}\right\rangle}_{\mathbf{II}}-\theta\underbrace{\left\langle\mathbf{\Lambda}\odot\widetilde{\mathbf{\Xi}},\widetilde{\mathbf{\Delta}}\right\rangle}_{\mathbf{III}},\end{aligned}\tag{18}$$

where $\widetilde{\mathbf{\Delta}}=\widetilde{\mathbf{\Sigma}}-\mathbf{\Sigma}^*$, $\mathbf{\Xi}^{(t+1)}\in\partial\|\mathbf{\Sigma}^{(t+1)}\|_1$. Then we establish limits for terms $\mathbf{I}$, $\mathbf{II}$ and $\mathbf{III}$, respectively.

Define $\boldsymbol{\Omega}=\nabla f\left(\widetilde{\boldsymbol{\Sigma}}\right)+\boldsymbol{\Lambda}\odot\widetilde{\boldsymbol{\Xi}}$. For term **I**, we partition the support of $\boldsymbol{\Omega}$ and $\widetilde{\boldsymbol{\Delta}}$ into the set $\mathcal{E}$ and its complement $\overline{\mathcal{E}}$. By employing the matrix Hölder's inequality, we derive

$$\begin{aligned}\mathbf{I}&=\left\langle\boldsymbol{\Omega}_{\mathcal{E}},\widetilde{\boldsymbol{\Delta}}_{\mathcal{E}}\right\rangle+\left\langle\boldsymbol{\Omega}_{\overline{\mathcal{E}}},\widetilde{\boldsymbol{\Delta}}_{\overline{\mathcal{E}}}\right\rangle\\&\leq\|\boldsymbol{\Omega}_{\mathcal{E}}\|_{\mathrm{F}}\left\|\widetilde{\boldsymbol{\Delta}}_{\mathcal{E}}\right\|_{\mathrm{F}}+\|\boldsymbol{\Omega}_{\overline{\mathcal{E}}}\|_{\max}\left\|\widetilde{\boldsymbol{\Delta}}_{\overline{\mathcal{E}}}\right\|_{1}\\&\leq\sqrt{|\mathcal{E}|}\,\|\boldsymbol{\Omega}_{\mathcal{E}}\|_{\max}\left\|\widetilde{\boldsymbol{\Delta}}_{\mathcal{E}}\right\|_{\mathrm{F}}+\|\boldsymbol{\Omega}\|_{\max}\left\|\widetilde{\boldsymbol{\Delta}}_{\overline{\mathcal{E}}}\right\|_{1}\\&\underset{(i)}{\leq}\varepsilon\sqrt{|\mathcal{E}|}\left\|\widetilde{\boldsymbol{\Delta}}_{\mathcal{E}}\right\|_{\mathrm{F}}+\varepsilon\left\|\widetilde{\boldsymbol{\Delta}}_{\overline{\mathcal{E}}}\right\|_{1},\end{aligned}$$

where (i) is from $\|\boldsymbol{\Omega}\|_{\max}\leq\varepsilon$ by Definition 8. For term **II**, we divide the support of $\nabla f\left(\boldsymbol{\Sigma}^*\right)$ and $\widetilde{\boldsymbol{\Delta}}$ into the set $\mathcal{E}$ and $\overline{\mathcal{E}}$. Applying the matrix Hölder's inequality, we obtain

$$\begin{aligned}\mathbf{II}&=\left\langle\left(\nabla f\left(\boldsymbol{\Sigma}^*\right)\right)_{\mathcal{E}},\widetilde{\boldsymbol{\Delta}}_{\mathcal{E}}\right\rangle+\left\langle\left(\nabla f\left(\boldsymbol{\Sigma}^*\right)\right)_{\overline{\mathcal{E}}},\widetilde{\boldsymbol{\Delta}}_{\overline{\mathcal{E}}}\right\rangle\\&\geq-\|(\nabla f\left(\boldsymbol{\Sigma}^*\right))_{\mathcal{E}}\|_{\mathrm{F}}\left\|\widetilde{\boldsymbol{\Delta}}_{\mathcal{E}}\right\|_{\mathrm{F}}-\|(\nabla f\left(\boldsymbol{\Sigma}^*\right))_{\overline{\mathcal{E}}}\|_{\max}\left\|\widetilde{\boldsymbol{\Delta}}_{\overline{\mathcal{E}}}\right\|_{1}.\end{aligned}$$

For term **III**, we separate the support of $\boldsymbol{\Lambda}\odot\widetilde{\boldsymbol{\Xi}}$ and $\widetilde{\boldsymbol{\Delta}}$ into the set $\mathcal{S}^*$ and $\overline{\mathcal{S}^*}$. And then applying the matrix Hölder's inequality, we obtain

$$\begin{aligned}\mathbf{III}&=\left\langle\left(\boldsymbol{\Lambda}\odot\widetilde{\boldsymbol{\Xi}}\right)_{\mathcal{S}^*},\widetilde{\boldsymbol{\Delta}}_{\mathcal{S}^*}\right\rangle+\left\langle\left(\boldsymbol{\Lambda}\odot\widetilde{\boldsymbol{\Xi}}\right)_{\overline{\mathcal{S}^*}},\widetilde{\boldsymbol{\Delta}}_{\overline{\mathcal{S}^*}}\right\rangle\\&\underset{(i)}{=}\left\langle\left(\boldsymbol{\Lambda}\odot\widetilde{\boldsymbol{\Xi}}\right)_{\mathcal{S}^*},\widetilde{\boldsymbol{\Delta}}_{\mathcal{S}^*}\right\rangle+\left\langle\boldsymbol{\Lambda}_{\overline{\mathcal{S}^*}},\left|\widetilde{\boldsymbol{\Delta}}_{\overline{\mathcal{S}^*}}\right|\right\rangle\\&\geq-\|\boldsymbol{\Lambda}_{\mathcal{S}^*}\|_{\mathrm{F}}\left\|\widetilde{\boldsymbol{\Delta}}_{\mathcal{S}^*}\right\|_{\mathrm{F}}+\left\langle\boldsymbol{\Lambda}_{\overline{\mathcal{E}}},\left|\widetilde{\boldsymbol{\Delta}}_{\overline{\mathcal{E}}}\right|\right\rangle\\&\underset{(ii)}{\geq}-\|\boldsymbol{\Lambda}_{\mathcal{S}^*}\|_{\mathrm{F}}\left\|\widetilde{\boldsymbol{\Delta}}_{\mathcal{E}}\right\|_{\mathrm{F}}+\|\boldsymbol{\Lambda}_{\overline{\mathcal{E}}}\|_{\min}\left\|\widetilde{\boldsymbol{\Delta}}_{\overline{\mathcal{E}}}\right\|_{1},\end{aligned}$$

where (i) is from

$$\left\langle\left(\boldsymbol{\Lambda}\odot\widetilde{\boldsymbol{\Xi}}\right)_{\overline{\mathcal{S}^*}},\widetilde{\boldsymbol{\Delta}}_{\overline{\mathcal{S}^*}}\right\rangle=\left\langle\boldsymbol{\Lambda}_{\overline{\mathcal{S}^*}},\left|\widetilde{\boldsymbol{\Sigma}}_{\overline{\mathcal{S}^*}}\right|\right\rangle=\left\langle\boldsymbol{\Lambda}_{\overline{\mathcal{S}^*}},\left|\widetilde{\boldsymbol{\Delta}}_{\overline{\mathcal{S}^*}}\right|\right\rangle,$$

and (ii) is from

$$\begin{aligned}\left\langle\boldsymbol{\Lambda}_{\overline{\mathcal{E}}},\left|\widetilde{\boldsymbol{\Delta}}_{\overline{\mathcal{E}}}\right|\right\rangle&=\sum_{(i,j)\in\overline{\mathcal{E}}}\Lambda_{ij}\left|\widetilde{\Delta}_{ij}\right|\geq\|\boldsymbol{\Lambda}_{\overline{\mathcal{E}}}\|_{\min}\sum_{(i,j)\in\overline{\mathcal{E}}}\left|\widetilde{\Delta}_{ij}\right|\\&=\|\boldsymbol{\Lambda}_{\overline{\mathcal{E}}}\|_{\min}\left\|\widetilde{\boldsymbol{\Delta}}_{\overline{\mathcal{E}}}\right\|_{1},\end{aligned}$$

and

$$\left\|\widetilde{\boldsymbol{\Delta}}_{\mathcal{S}^*}\right\|_{\mathrm{F}}\leq\left\|\widetilde{\boldsymbol{\Delta}}_{\mathcal{E}}\right\|_{\mathrm{F}}.$$

Combining the above results into (18) yields

$$\begin{aligned}&\rho^{-}_{2s^*+2\widetilde{s}}\left\|\widetilde{\boldsymbol{\Sigma}}^*-\boldsymbol{\Sigma}^*\right\|_{\mathrm{F}}^2\\&\leq\left(\|\boldsymbol{\Lambda}_{\mathcal{S}^*}\|_{\mathrm{F}}+\|(\nabla f\left(\boldsymbol{\Sigma}^*\right))_{\mathcal{E}}\|_{\mathrm{F}}+\varepsilon\sqrt{|\mathcal{E}|}\right)\times\theta\left\|\widetilde{\boldsymbol{\Delta}}_{\mathcal{E}}\right\|_{\mathrm{F}}\\&\quad-\theta\left(\|\boldsymbol{\Lambda}_{\overline{\mathcal{E}}}\|_{\min}-\left(\|(\nabla f\left(\boldsymbol{\Sigma}^*\right))_{\overline{\mathcal{E}}}\|_{\max}+\varepsilon\right)\right)\left\|\widetilde{\boldsymbol{\Delta}}_{\overline{\mathcal{E}}}\right\|_{1}\\&\underset{(i)}{\leq}\left(\|\boldsymbol{\Lambda}_{\mathcal{S}^*}\|_{\mathrm{F}}+\|(\nabla f\left(\boldsymbol{\Sigma}^*\right))_{\mathcal{E}}\|_{\mathrm{F}}+\varepsilon\sqrt{|\mathcal{E}|}\right)\times\theta\left\|\left(\widetilde{\boldsymbol{\Sigma}}-\boldsymbol{\Sigma}^*\right)_{\mathcal{E}}\right\|_{\mathrm{F}}\\&\leq\left(\left(\|(\nabla f\left(\boldsymbol{\Sigma}^*\right))_{\mathcal{E}}\|_{\max}+\varepsilon\right)\sqrt{|\mathcal{E}|}+\|\boldsymbol{\Lambda}_{\mathcal{S}^*}\|_{\max}\sqrt{|\mathcal{S}^*|}\right)\\&\quad\times\theta\left\|\left(\widetilde{\boldsymbol{\Sigma}}-\boldsymbol{\Sigma}^*\right)_{\mathcal{E}}\right\|_{\mathrm{F}}\\&\leq\left(\left(\|(\nabla f\left(\boldsymbol{\Sigma}^*\right))_{\mathcal{E}}\|_{\max}+\varepsilon\right)\sqrt{2s^*}+\lambda\sqrt{s^*}\right)\\&\quad\times\theta\left\|\left(\widetilde{\boldsymbol{\Sigma}}-\boldsymbol{\Sigma}^*\right)_{\mathcal{E}}\right\|_{\mathrm{F}}\\&\leq\frac{2+\sqrt{2}}{2}\lambda\sqrt{s^*}\times\underbrace{\theta\left\|\left(\widetilde{\boldsymbol{\Sigma}}-\boldsymbol{\Sigma}^*\right)_{\mathcal{E}}\right\|_{\mathrm{F}}}_{\mathbf{IV}},\end{aligned}$$

where (i) is due to the fact that $\|\boldsymbol{\Lambda}_{\overline{\mathcal{E}}}\|_{\min}\geq\frac{\lambda}{2}\geq\|\nabla f\left(\boldsymbol{\Sigma}^*\right)\|_{\max}+\varepsilon$.

For **IV**, we have $\theta\|(\widetilde{\boldsymbol{\Sigma}}-\boldsymbol{\Sigma}^*)_{\mathcal{E}}\|_{\mathrm{F}}=\|(\widetilde{\boldsymbol{\Sigma}}^*-\boldsymbol{\Sigma}^*)_{\mathcal{E}}\|_{\mathrm{F}}$. Consequently, we derive

$$\rho^{-}_{2s^*+2\widetilde{s}}\left\|\widetilde{\boldsymbol{\Sigma}}^*-\boldsymbol{\Sigma}^*\right\|_{\mathrm{F}}\leq\frac{2+\sqrt{2}}{2}\lambda\sqrt{s^*},$$

which serves as a contraction in relation to the construction of $\widetilde{\boldsymbol{\Sigma}}^*$. This leads to the conclusion that $\widetilde{\boldsymbol{\Sigma}}^*=\widetilde{\boldsymbol{\Sigma}}$. Therefore, the desired bound is satisfied for $\widetilde{\boldsymbol{\Sigma}}$. ☐

Lemma 28. *Suppose that Assumptions 4 and 10 hold. Consider the problem in* (7). *Define the set* $\mathcal{E}^{(k)}=\mathcal{S}^*\cup\mathcal{S}^{(k)}$, *where* $\mathcal{S}^{(k)}=\left\{(i,j)\mid\boldsymbol{\Lambda}_{ij}^{(k-1)}\leq p'_{\lambda}\left(u\right)\right\}$ *with* $u=c\lambda$ *and* $c=\frac{2+\sqrt{2}}{2\rho^{-}_{2s^*+2\widetilde{s}}}\in(0,\alpha)$, *such that* $p'_{\lambda}\left(c\lambda\right)\geq\frac{\lambda}{2}$. *This can always hold due to Assumption 3. Then for* $k\geq1$, *we establish that:* (1) $\lambda\geq2(\|\nabla f(\boldsymbol{\Sigma}^*)\|_{\max}+\varepsilon)$, (2) $|\mathcal{E}^{(k)}|\leq2s^*$, (3) $\left\|\boldsymbol{\Lambda}^{(k-1)}_{\mathcal{E}^{(k)}}\right\|_{\min}\geq\frac{\lambda}{2}$, *and*

$$\begin{aligned}\left\|\widetilde{\boldsymbol{\Sigma}}^{(k)}-\boldsymbol{\Sigma}^*\right\|_{\mathrm{F}}\leq&\frac{1}{\rho^{-}_{2s^*+2\widetilde{s}}}\left(\|(\nabla f\left(\boldsymbol{\Sigma}^*\right))_{\mathcal{E}^{(k)}}\|_{\mathrm{F}}+\varepsilon\sqrt{\left|\mathcal{E}^{(k)}\right|}\right)\\&+\frac{1}{\rho^{-}_{2s^*+2\widetilde{s}}}\left\|\boldsymbol{\Lambda}^{(k-1)}_{\mathcal{S}^*}\right\|_{\mathrm{F}}\\\leq&\frac{2+\sqrt{2}}{2\rho^{-}_{s^*+2\widetilde{s}}}\lambda\sqrt{s^*}.\end{aligned}$$

Proof. We first show that $\left|\mathcal{E}^{(k)}\right|\leq2s^*$ for all k by induction on k.

Base Case $(k=1)$**:**

For $k=1$, we have $\Lambda_{ij}^{(0)}=\lambda$ for $i\neq j$, where λ is a threshold parameter satisfying $\lambda\geq p'_{\lambda}\left(u\right)$. It follows that $\left|\mathcal{S}^{(1)}\right|\leq s^*$. And we see $\mathcal{E}^{(1)}=\mathcal{S}^*\cup\mathcal{S}^{(1)}$, which implies $\left|\mathcal{E}^{(1)}\right|\leq2s^*$ holds.

Inductive Step $(k\geq2)$**:**

Assume for some $k \geq 2$, $\left|\mathcal{E}^{(k-1)}\right| \leq 2s^*$ holds true. We aim to show that $\left|\mathcal{E}^{(k)}\right| \leq 2s^*$. For any $(i,j) \in \mathcal{S}^{(k)}$, we obtain $\left|\widetilde{\Sigma}_{ij}^{(k-1)}\right| \geq u$ and further have

$$\begin{aligned}\sqrt{\left|\mathcal{S}^{(k)} \backslash \mathcal{S}^*\right|} &\leq \sqrt{\sum_{(i,j)\in\mathcal{S}^{(k)}\backslash\mathcal{S}^*} \left(u^{-1}\widetilde{\Sigma}_{ij}^{(k-1)}\right)^2} \\ &= u^{-1}\left\|\widetilde{\mathbf{\Sigma}}_{\mathcal{S}^{(k)}\backslash\mathcal{S}^*}^{(k-1)}\right\|_{\mathrm{F}} \\ &= u^{-1}\left\|\left(\widetilde{\mathbf{\Sigma}}^{(k-1)} - \mathbf{\Sigma}^*\right)_{\mathcal{S}^{(k)}\backslash\mathcal{S}^*}\right\|_{\mathrm{F}} \\ &\leq u^{-1}\left\|\widetilde{\mathbf{\Sigma}}^{(k-1)} - \mathbf{\Sigma}^*\right\|_{\mathrm{F}}.\end{aligned} \tag{19}$$

For any $(i,j) \in \overline{\mathcal{S}^{(k-1)}}$, it follows that $\Lambda_{ij}^{(k-2)} = p'_\lambda\left(\widetilde{\Sigma}_{ij}^{(k-2)}\right) \geq p'_\lambda(u) \geq \frac{\lambda}{2}$. This chain of inequalities indicates that

$$\left\|\mathbf{\Lambda}_{\overline{\mathcal{E}^{(k-1)}}}^{(k-2)}\right\|_{\min} \geq \left\|\mathbf{\Lambda}_{\overline{\mathcal{S}^{(k-1)}}}^{(k-2)}\right\|_{\min} \geq p'_\lambda(u) \geq \frac{\lambda}{2}.$$

Furthermore, it is established that $\left|\mathcal{E}^{(k-1)}\right| \leq 2s^*$ and $\mathcal{S}^* \subseteq \mathcal{E}^{(k-1)}$. By applying Lemma 27 with $\widetilde{\mathbf{\Sigma}} = \widetilde{\mathbf{\Sigma}}^{(k-1)}$, $\mathcal{E} = \mathcal{E}^{(k-1)}$, and $\mathbf{\Lambda}_{\mathcal{S}^*} = \mathbf{\Lambda}_{\mathcal{S}^*}^{(k-2)}$ yields

$$\left\|\widetilde{\mathbf{\Sigma}}^{(k-1)} - \mathbf{\Sigma}^*\right\|_{\mathrm{F}} \leq \frac{2+\sqrt{2}}{2\rho^-_{2s^*+2\widetilde{s}}}\lambda\sqrt{s^*}.$$

Substituting this result into the inequality (19) yields

$$\sqrt{\left|\mathcal{S}^{(k)}\backslash\mathcal{S}^*\right|} \leq \frac{2+\sqrt{2}}{2u\rho^-_{2s^*+2\widetilde{s}}}\lambda\sqrt{s^*} = \sqrt{s^*}.$$

Thus, we have

$$\left|\mathcal{E}^{(k)}\right| = \left|\mathcal{S}^* \cup \left(\mathcal{S}^{(k)}\backslash\mathcal{S}^*\right)\right| = |\mathcal{S}^*| + \left|\mathcal{S}^{(k)}\backslash\mathcal{S}^*\right| \leq 2s^*,$$

completing the induction process.

Then according to the definition of $\mathcal{E}^{(k)}$ and $\mathcal{S}^{(k)}$, it follows that

$$\left\|\mathbf{\Lambda}_{\overline{\mathcal{E}^{(k)}}}^{(k-1)}\right\|_{\min} \geq \left\|\mathbf{\Lambda}_{\overline{\mathcal{S}^{(k)}}}^{(k-1)}\right\|_{\min} \geq p'_\lambda(u) \geq \frac{\lambda}{2}.$$

By employing Lemma 27 with $\widetilde{\mathbf{\Sigma}} = \widetilde{\mathbf{\Sigma}}^{(k)}$, $\mathcal{E} = \mathcal{E}^{(k)}$, and $\mathbf{\Lambda}_{\mathcal{S}^*} = \mathbf{\Lambda}_{\mathcal{S}^*}^{(k-1)}$, the ε-optimal solution $\widetilde{\mathbf{\Sigma}}^{(k)}$ to (7) satisfies

$$\begin{aligned}&\left\|\widetilde{\mathbf{\Sigma}}^{(k)} - \mathbf{\Sigma}^*\right\|_{\mathrm{F}} \\ \leq &\frac{1}{\rho^-_{2s^*+2\widetilde{s}}}\left(\left\|\left(\nabla f\left(\mathbf{\Sigma}^*\right)\right)_{\mathcal{E}^{(k)}}\right\|_{\mathrm{F}} + \varepsilon\sqrt{\left|\mathcal{E}^{(k)}\right|} + \left\|\mathbf{\Lambda}_{\mathcal{S}^*}^{(k-1)}\right\|_{\mathrm{F}}\right) \\ \leq &\frac{2+\sqrt{2}}{2\rho^-_{2s^*+2\widetilde{s}}}\lambda\sqrt{s^*}.\end{aligned}$$

□

B. Proof of Theorem 14

Proof. Based on Lemma 28, we have

$$\begin{aligned}\left\|\widetilde{\mathbf{\Sigma}}^{(k)} - \mathbf{\Sigma}^*\right\|_{\mathrm{F}} \leq &\frac{1}{\rho^-_{2s^*+2\widetilde{s}}}\underbrace{\left(\left\|\left(\nabla f\left(\mathbf{\Sigma}^*\right)\right)_{\mathcal{E}^{(k)}}\right\|_{\mathrm{F}} + \varepsilon\sqrt{\left|\mathcal{E}^{(k)}\right|}\right)}_{\mathbf{I}} \\ &+ \frac{1}{\rho^-_{2s^*+2\widetilde{s}}}\underbrace{\left\|\mathbf{\Lambda}_{\mathcal{S}^*}^{(k-1)}\right\|_{\mathrm{F}}}_{\mathbf{II}}.\end{aligned} \tag{20}$$

Then, we proceed to establish bounds for the term $\mathbf{I}$ and $\mathbf{II}$, respectively.

For term $\mathbf{I}$, dividing the support set into $\mathcal{S}^*$ and $\mathcal{E}^{(q)}\backslash\mathcal{S}^*$, we obtain

$$\begin{aligned}\mathbf{I} \leq &\left\|\left(\nabla f\left(\mathbf{\Sigma}^*\right)\right)_{\mathcal{S}^*}\right\|_{\mathrm{F}} + \varepsilon\sqrt{s^*} \\ &+ \left(\left\|\nabla f\left(\mathbf{\Sigma}^*\right)\right\|_{\max} + \varepsilon\right)\sqrt{\left|\mathcal{E}^{(k)}\backslash\mathcal{S}^*\right|} \\ \leq &\left\|\left(\nabla f\left(\mathbf{\Sigma}^*\right)\right)_{\mathcal{S}^*}\right\|_{\mathrm{F}} + \varepsilon\sqrt{s^*} + \frac{\lambda}{2u}\left\|\widetilde{\mathbf{\Sigma}}^{(k-1)} - \mathbf{\Sigma}^*\right\|_{\mathrm{F}},\end{aligned}$$

where the second inequality is due to

$$\sqrt{\left|\mathcal{E}^{(k)}\backslash\mathcal{S}^*\right|} = \sqrt{\left|\mathcal{S}^{(k)}\backslash\mathcal{S}^*\right|} \leq u^{-1}\left\|\widetilde{\mathbf{\Sigma}}^{(k-1)} - \mathbf{\Sigma}^*\right\|_{\mathrm{F}},$$

which follows from the inequality (19).

By Assumptions 3 and 11, for any $\mathbf{\Sigma}$, if $\left|\Sigma_{ij} - \Sigma_{ij}^*\right| \geq u$, then $p'_\lambda(\Sigma_{ij}) \leq \lambda \leq \lambda u^{-1}\left|\Sigma_{ij} - \Sigma_{ij}^*\right|$; otherwise, $p'_\lambda(\Sigma_{ij}) \leq p'_\lambda\left(\left|\Sigma_{ij}^*\right| - u\right) = 0$. Therefore, for term $\mathbf{II}$, we have

$$\mathbf{II} \leq \lambda u^{-1}\left\|\widetilde{\mathbf{\Sigma}}_{\mathcal{S}^*}^{(k-1)} - \mathbf{\Sigma}_{\mathcal{S}^*}^*\right\|_{\mathrm{F}} \leq \lambda u^{-1}\left\|\widetilde{\mathbf{\Sigma}}^{(k-1)} - \mathbf{\Sigma}^*\right\|_{\mathrm{F}}.$$

Substituting the above results into (20) yields

$$\begin{aligned}&\left\|\widetilde{\mathbf{\Sigma}}^{(k)} - \mathbf{\Sigma}^*\right\|_{\mathrm{F}} \\ \leq &\frac{1}{\rho^-_{2s^*+2\widetilde{s}}}\left(\left\|\left(\nabla f\left(\mathbf{\Sigma}^*\right)\right)_{\mathcal{S}^*}\right\|_{\mathrm{F}} + \varepsilon\sqrt{s^*}\right) + \delta\left\|\widetilde{\mathbf{\Sigma}}^{(k-1)} - \mathbf{\Sigma}^*\right\|_{\mathrm{F}},\end{aligned}$$

where $\delta = \frac{3\lambda}{2u\rho^-_{2s^*+2\widetilde{s}}} = \frac{3}{2+\sqrt{2}} \in (0,1)$. □

C. Concentration Inequality

Lemma 29 (Hanson-Wright Inequality)**.** *Let $\mathbf{x} = (x_1, \ldots, x_n) \in \mathbb{R}^n$ be a random vector with independent components x_i which satisfy $\mathbb{E}(x_i) = 0$ and $\|x_i\|_{\psi_2} \leq L$. Let $\mathbf{B}$ be an $n \times n$ matrix. Then for any $t > 0$,*

$$\begin{aligned}&\mathbb{P}\left\{\left|\mathbf{x}^\top\mathbf{B}\mathbf{x} - \mathbb{E}\left(\mathbf{x}^\top\mathbf{B}\mathbf{x}\right)\right| > t\right\} \\ \leq &2\exp\left(-c'\min\left(\frac{t^2}{L^4\|\mathbf{B}\|_{\mathrm{F}}^2}, \frac{t}{L^2\|\mathbf{B}\|}\right)\right).\end{aligned} \tag{21}$$

Lemma 30 (Lemma D.1 in [43])**.** *Let $\mathbf{x}$ be a sub-Gaussian random vector with zero mean and covariance $\mathbf{\Sigma}^*$ and $\{\mathbf{x}_i\}_{i=1}^n$ be a collection of i.i.d. samples from $\mathbf{x}$. There exists some constant c_1, c_2, and t_0 such that for all t with $0 < t < t_0$, the sample covariance matrix $\mathbf{S}$ satisfies the following tail-bound*

$$\mathbb{P}\left(\left|\Sigma_{ij}^* - S_{ij}\right| > t\right) \leq c_1\exp\left(-c_2nt^2\right).$$

Lemma 31. *Under Assumptions 11 and the same conditions in Lemma 30, the columns of $\widetilde{\mathbf{A}}$ are normalized such that $\max_l \left\|\widetilde{\mathbf{A}}_{\cdot l}\right\|_2 \leq \sqrt{m}$, there exists some constant c_1 such that*

$$\mathbb{P}\left(\left\|\frac{1}{m}\mathcal{A}^*\left(\boldsymbol{y}-\mathcal{A}\left(\boldsymbol{\Sigma}^*\right)\right)\right\|_{\max} \leq \lambda\right) \geq 1-\frac{c_1}{d}.$$

Proof. We begin by establishing a bound on the probability that $\left\|\frac{1}{m}\mathcal{A}^*\left(\boldsymbol{y}-\mathcal{A}\left(\boldsymbol{\Sigma}^*\right)\right)\right\|_{\max} > t$ using a union bound approach.

First, consider the bound $\mathbb{P}\left(\left\|\frac{1}{m}\sum_{i=1}^m \eta_i' \boldsymbol{a}_i\boldsymbol{a}_i^\top\right\|_{\max} > t\right)$, where $\eta_i' = \boldsymbol{a}_i^\top\left(\boldsymbol{\Sigma}_N-\boldsymbol{\Sigma}^*\right)\boldsymbol{a}_i + \eta_i$. By applying the Hanson-Wright inequality (21), we establish

$$\begin{aligned}
&\mathbb{P}\left(\boldsymbol{a}_i^\top\left(\boldsymbol{\Sigma}_N-\boldsymbol{\Sigma}^*\right)\boldsymbol{a}_i > t\right)\\
=&\mathbb{P}\left(\left|\left\langle \boldsymbol{a}_i\boldsymbol{a}_i^\top, \left(\boldsymbol{\Sigma}_N-\boldsymbol{\Sigma}^*\right)\right\rangle\right| > t\right)\\
\leq& 2\exp\left(-c'\min\left(\frac{t^2}{L^4\left\|\boldsymbol{\Sigma}_N-\boldsymbol{\Sigma}^*\right\|_{\mathrm{F}}^2}, \frac{t}{L^2\left\|\boldsymbol{\Sigma}_N-\boldsymbol{\Sigma}^*\right\|}\right)\right),
\end{aligned}$$

where c' is a positive constant, and L is a scaling factor. This indicates that $\left\langle \boldsymbol{a}_i\boldsymbol{a}_i^\top, \boldsymbol{\Sigma}_N-\boldsymbol{\Sigma}^*\right\rangle$ is a sub-exponential random variable. Additionally, given that η_i follows a sub-exponential distribution, η_i' is also a sub-exponential random variable. By summing over individual probabilities, we obtain

$$\begin{aligned}
&\mathbb{P}\left(\left\|\frac{1}{m}\mathcal{A}^*\left(\boldsymbol{y}-\mathcal{A}\left(\boldsymbol{\Sigma}^*\right)\right)\right\|_{\max} > t\right)\\
=&\mathbb{P}\left(\left\|\frac{1}{m}\sum_{i=1}^m \eta_i'\boldsymbol{a}_i\boldsymbol{a}_i^\top\right\|_{\max} > t\right)\\
\leq&\sum_{l=1}^{d^2}\mathbb{P}\left(\frac{1}{m}\left|\widetilde{\mathbf{A}}_{\cdot l}^\top\cdot\boldsymbol{\eta}'\right| > t\right),
\end{aligned} \tag{22}$$

Define $\zeta_l = \widetilde{\mathbf{A}}_{\cdot l}^\top\cdot\boldsymbol{\eta}'$. Given that η_i' is sub-exponential$\left(0,\xi^2\right)$ for $i=1,\ldots,m$, we apply Lemma 30 and employ concentration inequalities

$$\mathbb{E}\left(\exp\left\{t_0\zeta_l\right\}+\exp\left\{-t_0\zeta_l\right\}\right) \leq 2\exp\left\{nm^{-2}\left\|\widetilde{\mathbf{A}}_{\cdot l}\right\|^2\xi^2t_0^2/2\right\},$$

which implies

$$\mathbb{P}\left(\left|\zeta_l\right| \geq t\right)\exp\left\{t_0t\right\} \leq 2\exp\left\{nm^{-2}\left\|\widetilde{\mathbf{A}}_{\cdot l}\right\|^2\xi^2t_0^2/2\right\}.$$

Selecting $t_0 = t\left(nm^{-2}\left\|\widetilde{\mathbf{A}}_{\cdot l}\right\|_2^2\xi^2\right)^{-1}$ yields that

$$\mathbb{P}\left(\left|\zeta_l\right| \geq t\right) \leq 2\exp\left\{-nt^2\cdot\left(2\xi^2\left\|\widetilde{\mathbf{A}}_{\cdot l}\right\|_2^2/m^2\right)^{-1}\right\}.$$

Plugging it into (22) results

$$\begin{aligned}
&\mathbb{P}\left(\left\|\frac{1}{m}\mathcal{A}^*\left(\boldsymbol{y}-\mathcal{A}\left(\boldsymbol{\Sigma}^*\right)\right)\right\|_{\max} > t\right)\\
\underset{(i)}{\leq}& 2d^2\exp\left\{-nmt^2\cdot\left(2\xi^2\max_l\left\{\left\|\widetilde{\mathbf{A}}_{\cdot l}\right\|_2^2/m\right\}\right)^{-1}\right\}\\
=&c_1\exp\left\{-c_2nmt^2+2\log d\right\},
\end{aligned}$$

where (i) is from the column of $\widetilde{\mathbf{A}}$ are normalized such that $\max_l\left\|\widetilde{\mathbf{A}}_{\cdot l}\right\|_2 \leq \sqrt{m}$, and c_1, c_2 are constants. Then taking $\lambda = \sqrt{\frac{3\log d}{c_3mn}} \asymp \sqrt{\frac{\log d}{mn}}$, we obtain

$$\begin{aligned}
&\mathbb{P}\left(\left\|\frac{1}{m}\mathcal{A}^*\left(\boldsymbol{y}-\mathcal{A}\left(\boldsymbol{\Sigma}^*\right)\right)\right\|_{\max} \leq \lambda\right)\\
\geq& 1-c_1\exp\left(-c_2nm\lambda^2+2\log d\right)\\
=&1-\frac{c_1}{d}.
\end{aligned}$$

$\square$

Lemma 32. *Under the same conditions as in Lemma 31, the following result holds*

$$\left\|\frac{1}{m}\mathcal{A}^*\left(\boldsymbol{y}-\mathcal{A}\left(\boldsymbol{\Sigma}^*\right)\right)_{\mathcal{S}^*}\right\|_{\mathrm{F}} = \mathcal{O}_p\left(\sqrt{\frac{s^*}{mn}}\right).$$

Proof. Following the analysis presented in Lemma 31 analysis, consider any M such that $0 < M\sqrt{\frac{1}{mn}}$, we have

$$\begin{aligned}
&\mathbb{P}\left(\left\|\frac{1}{m}\mathcal{A}^*\left(\boldsymbol{y}-\mathcal{A}\left(\boldsymbol{\Sigma}^*\right)\right)_{\mathcal{S}^*}\right\|_{\max} > M\sqrt{\frac{1}{mn}}\right)\\
\leq& c_1s^*\exp\left(-c_2M\right)\\
=&c_1\exp\left(-c_2M+\log s^*\right).
\end{aligned}$$

Setting M such that $\sqrt{\frac{2\log s^*}{c_3}} < M$ and letting $M\to\infty$, we obtain

$$\lim_{M\to\infty}\sup_m\mathbb{P}\left(\left\|\frac{1}{m}\mathcal{A}^*\left(\boldsymbol{y}-\mathcal{A}\left(\boldsymbol{\Sigma}^*\right)\right)_{\mathcal{S}^*}\right\|_{\max} > M\sqrt{\frac{1}{mn}}\right) = 0.$$

The proof is completed by applying

$$\begin{aligned}
&\left\|\frac{1}{m}\mathcal{A}^*\left(\boldsymbol{y}-\mathcal{A}\left(\boldsymbol{\Sigma}^*\right)\right)_{\mathcal{S}^*}\right\|_{\mathrm{F}}\\
\leq&\sqrt{s^*}\left\|\frac{1}{m}\mathcal{A}^*\left(\boldsymbol{y}-\mathcal{A}\left(\boldsymbol{\Sigma}^*\right)\right)_{\mathcal{S}^*}\right\|_{\max}.
\end{aligned}$$

$\square$

D. Proof of Corollary 16

Proof. We begin by bound the gradient norm of $f\left(\boldsymbol{\Sigma}\right)$ at $\boldsymbol{\Sigma}^*$:

$$\begin{aligned}
\left\|\nabla f\left(\boldsymbol{\Sigma}^*\right)\right\|_{\max} \leq& \left\|\frac{1}{m}\mathcal{A}^*\left(\boldsymbol{y}-\mathcal{A}\left(\boldsymbol{\Sigma}^*\right)\right)\right\|_{\max}\\
&+\tau\left\|\left(\boldsymbol{\Sigma}^*\right)^{-1}\right\|_{\max}.
\end{aligned}$$

Assuming $\lambda \asymp \sqrt{\frac{\log d}{mn}}$ and $\tau \lesssim \sqrt{\frac{1}{mn}}\left\|\left(\boldsymbol{\Sigma}^*\right)^{-1}\right\|_{\max}^{-1}$, and invoking Lemma 31, we establish that $\lambda \geq 2\left(\left\|\nabla f\left(\boldsymbol{\Sigma}^*\right)\right\|_{\max}+\varepsilon\right)$ holds w.h.p.

Applying Lemma 28 with $k=1$, we obtain

$$\left\|\widetilde{\boldsymbol{\Sigma}}^{(1)}-\boldsymbol{\Sigma}^*\right\|_{\mathrm{F}} \leq \frac{2+\sqrt{2}}{2\rho^-_{2s^*+2\widetilde{s}}}\lambda\sqrt{s^*}.$$

Under the condition $\lambda \asymp \sqrt{\frac{\log d}{mn}}$, the Frobenius norm difference is asymptotically bounded as $\left\|\widetilde{\boldsymbol{\Sigma}}^{(1)}-\boldsymbol{\Sigma}^*\right\|_{\mathrm{F}} \lesssim \sqrt{\frac{s^*\log d}{mn}}$ w.h.p. $\square$

E. Proof of Corollary 17

Proof. We begin by bound the gradient norm of $f(\boldsymbol{\Sigma})$ at $\boldsymbol{\Sigma}^*$:

$$\|\nabla f(\boldsymbol{\Sigma}^*)\|_{\max} \leq \left\|\frac{1}{m}\mathcal{A}^*(\boldsymbol{y} - \mathcal{A}(\boldsymbol{\Sigma}^*))\right\|_{\max} + \tau\left\|(\boldsymbol{\Sigma}^*)^{-1}\right\|_{\max}.$$

Under Assumptions 9 and 11 condition, and with parameters λ and τ scaled as $\lambda \asymp \sqrt{\frac{\log d}{mn}}$ and $\tau \lesssim \sqrt{\frac{1}{mn}}\left\|(\boldsymbol{\Sigma}^*)^{-1}\right\|_{\max}^{-1}$, then by Lemma 31, $\lambda \geq 2(\|\nabla f(\boldsymbol{\Sigma}^*)\|_{\max} + \varepsilon)$ holds w.h.p.

Next, applying Theorem 14, we derive

$$\begin{aligned}
&\left\|\widetilde{\boldsymbol{\Sigma}}^{(1)} - \boldsymbol{\Sigma}^*\right\|_{\mathrm{F}} \\
&\leq \frac{1}{\rho^-_{2s^*+2\widetilde{s}}}\left(\|(\nabla f(\boldsymbol{\Sigma}^*))_{\mathcal{S}^*}\|_{\mathrm{F}} + \varepsilon\sqrt{s^*}\right) + \delta\left\|\widetilde{\boldsymbol{\Sigma}}^{(k-1)} - \boldsymbol{\Sigma}^*\right\|_{\mathrm{F}} \\
&\leq \frac{1}{\rho^-_{2s^*+2\widetilde{s}}}\left(\|(\nabla f(\boldsymbol{\Sigma}^*))_{\mathcal{S}^*}\|_{\mathrm{F}} + \varepsilon\sqrt{s^*}\right) + \delta^{k-1}\left\|\widetilde{\boldsymbol{\Sigma}}^{(1)} - \boldsymbol{\Sigma}^*\right\|_{\mathrm{F}} \\
&\leq \frac{1}{\rho^-_{2s^*+2\widetilde{s}}}\left(\|(\nabla f(\boldsymbol{\Sigma}^*))_{\mathcal{S}^*}\|_{\mathrm{F}} + \varepsilon\sqrt{s^*}\right) + \delta^{k-1}\frac{2+\sqrt{2}}{2\rho^-_{2s^*+2\widetilde{s}}}\lambda\sqrt{s^*},
\end{aligned}$$

where the last inequality is due to $\|\widetilde{\boldsymbol{\Sigma}}^{(1)} - \boldsymbol{\Sigma}^*\|_{\mathrm{F}} \leq \frac{2+\sqrt{2}}{2\rho^-_{2s^*+2\widetilde{s}}}\lambda\sqrt{s^*}$, which follows from Lemma 28 with $k=1$.

Finally, considering the norm of the projected gradient within the support set $\mathcal{S}^*$, we obtain

$$\begin{aligned}
&\|(\nabla f(\boldsymbol{\Sigma}^*))_{\mathcal{S}^*}\|_{\mathrm{F}} \\
=&\left\|\left(-\frac{1}{m}\mathcal{A}^*(\boldsymbol{y} - \mathcal{A}(\boldsymbol{\Sigma}^*)) - \tau(\boldsymbol{\Sigma}^*)^{-1}\right)_{\mathcal{S}^*}\right\|_{\mathrm{F}} \\
\leq&\left\|\frac{1}{m}\mathcal{A}^*(\boldsymbol{y} - \mathcal{A}(\boldsymbol{\Sigma}^*))_{\mathcal{S}^*}\right\|_{\mathrm{F}} + \tau\left\|\left((\boldsymbol{\Sigma}^*)^{-1}\right)_{\mathcal{S}^*}\right\|_{\mathrm{F}} \\
\leq&\left\|\frac{1}{m}\mathcal{A}^*(\boldsymbol{y} - \mathcal{A}(\boldsymbol{\Sigma}^*))_{\mathcal{S}^*}\right\|_{\mathrm{F}} + \tau\left\|(\boldsymbol{\Sigma}^*)^{-1}\right\|_{\mathrm{F}}
\end{aligned}$$

By Lemma 32, $\left\|\frac{1}{m}\mathcal{A}^*(\boldsymbol{y} - \mathcal{A}(\boldsymbol{\Sigma}^*))_{\mathcal{S}^*}\right\|_{\mathrm{F}} = \mathcal{O}_p\left(\sqrt{\frac{s^*}{mn}}\right)$. If $\tau \lesssim \sqrt{\frac{1}{mn}}\left\|(\boldsymbol{\Sigma}^*)^{-1}\right\|_{\max}^{-1}$, then $\|(\nabla f(\boldsymbol{\Sigma}^*))_{\mathcal{S}^*}\|_{\mathrm{F}} = \mathcal{O}_p\left(\sqrt{\frac{s^*}{mn}}\right)$.

Furthermore, if $K \geq 1 + \frac{\log(\lambda\sqrt{mn})}{\log\delta^{-1}} \gtrsim \log(\lambda\sqrt{mn}) \gtrsim \log\log d$, we derive

$$\delta^{K-1}\lambda\sqrt{s^*} \leq \frac{1}{\lambda\sqrt{mn}}\lambda\sqrt{s^*} \leq \sqrt{\frac{s^*}{mn}}.$$

By integrating all findings with the condition $\varepsilon \lesssim \sqrt{\frac{1}{mn}}$, we conclude that $\left\|\widetilde{\boldsymbol{\Sigma}}^{(K)} - \boldsymbol{\Sigma}^*\right\|_{\mathrm{F}} = \mathcal{O}_p\left(\sqrt{\frac{s^*}{mn}}\right)$. $\square$

APPENDIX D
PROOF OF COMPUTATIONAL THEORY 20

In this appendix, we first show $f(\boldsymbol{\Sigma})$ exhibits properties of the local strong convexity, local strong smoothness and local restricted Hessian smoothness. Then we demonstrate the sparsity of the solution is preserved within the vicinity of the true model parameter $\boldsymbol{\Sigma}^*$, provided that the starting point is a sparse solution. Following this, it is established that proximal Newton updates achieve a quadratic rate of convergence towards a local minimizer at each step within each stage, assuming the initialization occurs within a suitably refined sparse region. Subsequently, Lemmas 40 and 37 confirm that each subproblem experiences a substantial reduction. Furthermore, we outline the necessary number of iterations required at each convex relaxation stage to satisfy the approximate KKT conditions.

A. Proof of Lemma 18

Proof. We first verify the convexity of the set $\mathcal{B}\left(\boldsymbol{\Sigma}^*, \frac{\rho^-_{2s^*+2\widetilde{s}}}{4\tau\kappa}\right)$. For any $\boldsymbol{\Sigma}_1, \boldsymbol{\Sigma}_2 \in \mathcal{B}\left(\boldsymbol{\Sigma}^*, \frac{\rho^-_{2s^*+2\widetilde{s}}}{4\tau\kappa}\right)$, the linearity of the space implies that any convex combination of these points also resides within the set due to the properties of the norm and the definition of the set.

To establish the sparse strong convexity of $f(\boldsymbol{\Sigma})$ within this set, consider two arbitrary points $\boldsymbol{\Sigma}_3, \boldsymbol{\Sigma}_4$ and convex combination $\boldsymbol{\Sigma}_\rho = \rho\boldsymbol{\Sigma}_3 + (1-\rho)\boldsymbol{\Sigma}_4$ for $\rho \in [0,1]$. The function $f(\boldsymbol{\Sigma})$ can be expanded using a Taylor series approximation around $\boldsymbol{\Sigma}_4$:

$$\begin{aligned}
f(\boldsymbol{\Sigma}_3) =& f(\boldsymbol{\Sigma}_4) + \langle\nabla f(\boldsymbol{\Sigma}_4), \boldsymbol{\Sigma}_3 - \boldsymbol{\Sigma}_4\rangle \\
&+ \frac{1}{2}\mathrm{vec}^\top(\boldsymbol{\Sigma}_3 - \boldsymbol{\Sigma}_4)\nabla^2 f(\boldsymbol{\Sigma}_\rho)\mathrm{vec}(\boldsymbol{\Sigma}_3 - \boldsymbol{\Sigma}_4).
\end{aligned}$$

The next step is to estimate the lower bound of $\lambda_{\min}\left(\nabla^2 f(\boldsymbol{\Sigma}_\rho)\right)$. Using the fact that $\lambda_{\min}\left(\mathbf{X}^{-1}\right) = \|\mathbf{X}\|_2^{-1}$ and $\|\mathbf{X}\otimes\mathbf{Y}\|_2 = \|\mathbf{X}\|_2\|\mathbf{Y}\|_2$ for any matrices $\mathbf{X}$, $\mathbf{Y}$, we deduce

$$\begin{aligned}
&\lambda_{\min}\left(\nabla^2 f(\boldsymbol{\Sigma}_\rho)\right) \\
=&\lambda_{\min}\left(\sum_{i=1}^m \mathbf{A}_i\otimes\mathbf{A}_i^\top + \tau\boldsymbol{\Sigma}_\rho^{-1}\otimes\boldsymbol{\Sigma}_\rho^{-1}\right) \\
\geq&\lambda_{\min}\left(\sum_{i=1}^m \mathbf{A}_i\otimes\mathbf{A}_i^\top\right) + \tau\lambda_{\min}\left(\boldsymbol{\Sigma}_\rho^{-1}\otimes\boldsymbol{\Sigma}_\rho^{-1}\right) \\
\underset{(i)}{\geq}&\rho^-_{2s^*+2\widetilde{s}} + \tau\|\boldsymbol{\Sigma}_\rho\|_2^{-2},
\end{aligned}$$

where (i) is from Assumption 7. One has

$$\begin{aligned}
\|\boldsymbol{\Sigma}_\rho\|_2 &= \|\boldsymbol{\Sigma}^* - \boldsymbol{\Sigma}^* + \rho\boldsymbol{\Sigma}_1 + (1-\rho)\boldsymbol{\Sigma}_2\|_2 \\
&\leq \|\boldsymbol{\Sigma}^*\|_2 + \rho\|\boldsymbol{\Sigma}_1 - \boldsymbol{\Sigma}^*\|_2 + (1-\rho)\|\boldsymbol{\Sigma}_2 - \boldsymbol{\Sigma}^*\|_2 \\
&\leq \|\boldsymbol{\Sigma}^*\|_2 + \rho\|\boldsymbol{\Sigma}_1 - \boldsymbol{\Sigma}^*\|_{\mathrm{F}} + (1-\rho)\|\boldsymbol{\Sigma}_2 - \boldsymbol{\Sigma}^*\|_{\mathrm{F}} \\
&\underset{(i)}{\leq} \kappa + \frac{\rho^-_{2s^*+2\widetilde{s}}}{4\tau\kappa} \leq \kappa + \frac{\kappa\rho^-_{2s^*+2\widetilde{s}}}{4\tau},
\end{aligned}$$

where (i) holds due to $\|\boldsymbol{\Sigma}^*\|_2 \leq \kappa$ and $\boldsymbol{\Sigma}_1, \boldsymbol{\Sigma}_2 \in \mathcal{B}\left(\boldsymbol{\Sigma}^*, \frac{\rho^-_{2s^*+2\widetilde{s}}}{4\tau\kappa}\right)$, the last inequality holds due to $\kappa \geq 1$. Thus, $f(\boldsymbol{\Sigma})$ is shown to be strongly convex over $\mathcal{B}\left(\boldsymbol{\Sigma}^*, \frac{\rho^-_{2s^*+2\widetilde{s}}}{4\tau\kappa}\right)$ with a convexity parameter of $\left(\rho^-_{2s^*+2\widetilde{s}} + \frac{16\tau^3}{\kappa^2\left(4\tau+\rho^-_{2s^*+2\widetilde{s}}\right)^2}\right)$.

Next, we establish the strong smoothness of $f(\boldsymbol{\Sigma})$, which involves setting an upper bound for $\lambda_{\max}\left(\nabla^2 f(\boldsymbol{\Sigma}_\rho)\right)$. By the fact that $\lambda_{\max}\left(\mathbf{X}^{-1}\right)=\lambda_{\min}^{-1}(\mathbf{X})$ and $\lambda_{\min}(\mathbf{X}\otimes\mathbf{Y})$ for any matrices $\mathbf{X}$, $\mathbf{Y}$. Therefore,

$$\begin{aligned}&\lambda_{\max}\left(\nabla^2 f\left(\boldsymbol{\Sigma}_\rho\right)\right)\\=&\lambda_{\max}\left(\sum_{i=1}^m \mathbf{A}_i\otimes\mathbf{A}_i^\top+\tau\boldsymbol{\Sigma}_\rho^{-1}\otimes\boldsymbol{\Sigma}_\rho^{-1}\right)\\\leq&\lambda_{\max}\left(\sum_{i=1}^m \mathbf{A}_i\otimes\mathbf{A}_i^\top\right)+\tau\lambda_{\max}\left(\boldsymbol{\Sigma}_\rho^{-1}\otimes\boldsymbol{\Sigma}_\rho^{-1}\right)\\\leq&\rho_{2s^*+2\widetilde{s}}^{+}+\tau\lambda_{\min}^{-2}\left(\boldsymbol{\Sigma}_\rho\right).\end{aligned}$$

One has

$$\begin{aligned}&\lambda_{\min}\left(\boldsymbol{\Sigma}_\rho\right)\\=&\lambda_{\min}\left(\boldsymbol{\Sigma}^*-\boldsymbol{\Sigma}^*+\rho\boldsymbol{\Sigma}_1+(1-\rho)\boldsymbol{\Sigma}_2\right)\\\geq&\lambda_{\min}\left(\boldsymbol{\Sigma}^*\right)+\rho\lambda_{\min}\left(\boldsymbol{\Sigma}_1-\boldsymbol{\Sigma}^*\right)+(1-\rho)\lambda_{\min}\left(\boldsymbol{\Sigma}_2-\boldsymbol{\Sigma}^*\right)\\\geq&\lambda_{\min}\left(\boldsymbol{\Sigma}^*\right)-\rho\lambda_{\max}\left(\boldsymbol{\Sigma}_1-\boldsymbol{\Sigma}^*\right)-(1-\rho)\lambda_{\max}\left(\boldsymbol{\Sigma}_2-\boldsymbol{\Sigma}^*\right)\\\geq&\lambda_{\min}\left(\boldsymbol{\Sigma}^*\right)-\rho\left\|\boldsymbol{\Sigma}_1-\boldsymbol{\Sigma}^*\right\|_{\mathrm{F}}-(1-\rho)\left\|\boldsymbol{\Sigma}_2-\boldsymbol{\Sigma}^*\right\|_{\mathrm{F}}\\\geq&\frac{1}{\kappa}-\frac{\rho_{2s^*+2\widetilde{s}}^{-}}{4\tau\kappa},\end{aligned}$$

where the last inequality holds due to $\lambda_{\min}\left(\boldsymbol{\Sigma}^*\right)\geq\frac{1}{\kappa}$ and $\boldsymbol{\Sigma}_1,\boldsymbol{\Sigma}_2\in\mathcal{B}\left(\boldsymbol{\Sigma}^*,\frac{\rho_{2s^*+2\widetilde{s}}^{-}}{4\tau\kappa}\right)$. Thus, $f(\boldsymbol{\Sigma})$ is proven to be $\left(\rho_{2s^*+2\widetilde{s}}^{+}+\frac{16\tau^3\kappa^2}{\left(4\tau-\rho_{2s^*+2\widetilde{s}}^{-}\right)^2}\right)$-smooth over $\mathcal{B}\left(\boldsymbol{\Sigma}^*,\frac{\rho_{2s^*+2\widetilde{s}}^{-}}{4\tau\kappa}\right)$. □

B. Preliminary Lemmas

Lemma 33 (Local Restricted Hessian Smoothness)**.** *Recall $\widetilde{s}$ introduced in Assumption 7. We assert the existence of constant $L_{2s^*+2\widetilde{s}}$ and r such that for any $\boldsymbol{\Sigma},\boldsymbol{\Sigma}'\in\mathcal{B}\left(\boldsymbol{\Sigma}^*,r\right)$ with $\left\|\boldsymbol{\Sigma}_{\overline{\mathcal{S}}}\right\|_0\leq\widetilde{s}$ and $\left\|\boldsymbol{\Sigma}'_{\overline{\mathcal{S}}}\right\|_0\leq\widetilde{s}$, the following inequality is satisfied:*

$$\sup_{\mathbf{U}\in\boldsymbol{\Omega},\|\mathbf{U}\|_{\mathrm{F}}=1}\|\mathbf{U}\|_{\nabla^2 f(\boldsymbol{\Sigma}')-\nabla^2 f(\boldsymbol{\Sigma})}^2\leq L_{2s^*+2\widetilde{s}}\left\|\boldsymbol{\Sigma}-\boldsymbol{\Sigma}'\right\|_{\mathrm{F}}^2,$$

where $\boldsymbol{\Omega}=\left\{\mathbf{U}\mid\operatorname{supp}(\mathbf{U})\subseteq\left(\operatorname{supp}(\boldsymbol{\Sigma})\cup\operatorname{supp}\left(\boldsymbol{\Sigma}'\right)\right)\right\}$.

Proof. To establish the Lipschitz continuity of $\nabla^2 f(\boldsymbol{\Sigma})$, consider the third derivative $\nabla^3 f(\boldsymbol{\Sigma})$, which is expressed as $-2\tau\boldsymbol{\Sigma}^{-1}\otimes\boldsymbol{\Sigma}^{-1}\otimes\boldsymbol{\Sigma}^{-1}$. Employing a proof technique analogous to that utilized in Lemma 18, we have

$$\lambda_{\max}\left(\left|\nabla^3 f(\boldsymbol{\Sigma})\right|\right)\leq 2\tau\lambda_{\min}^{-3}(\boldsymbol{\Sigma})\leq 2\tau\kappa^3.$$

This finding allows us to conveniently define the Lipschitz constant for the Hessian matrix, represented as $L_{2s^*+2\widetilde{s}}$, to be $2\tau\kappa^3$. □

Lemma 34. *Under the same condition of Lemma 28, we have the following basic inequality*

$$\left\langle\nabla f\left(\widetilde{\boldsymbol{\Sigma}}^{(1)}\right)-\nabla f\left(\boldsymbol{\Sigma}^*\right),\widetilde{\boldsymbol{\Sigma}}^{(1)}-\boldsymbol{\Sigma}^*\right\rangle\leq\frac{c_1\lambda^2 s^*}{\rho_{2s^*+2\widetilde{s}}^{-}}.$$

Proof. Applying Lemma 28 with $k=1$, we obtain

$$\left\|\widetilde{\boldsymbol{\Sigma}}^{(1)}-\boldsymbol{\Sigma}^*\right\|_{\mathrm{F}}\leq\frac{2+\sqrt{2}}{2\rho_{2s^*+2\widetilde{s}}^{-}}\lambda\sqrt{s^*}.\tag{23}$$

Moreover, by invoking Lemma 26 yields that

$$\begin{aligned}\left\|\widetilde{\boldsymbol{\Sigma}}^{(1)}-\boldsymbol{\Sigma}^*\right\|_1&\leq\left\|\left(\widetilde{\boldsymbol{\Sigma}}^{(1)}-\boldsymbol{\Sigma}^*\right)_{\overline{\mathcal{E}^{(1)}}}\right\|_1+\left\|\left(\widetilde{\boldsymbol{\Sigma}}^{(1)}-\boldsymbol{\Sigma}^*\right)_{\mathcal{E}^{(1)}}\right\|_1\\&\leq 6\left\|\left(\widetilde{\boldsymbol{\Sigma}}^{(1)}-\boldsymbol{\Sigma}^*\right)_{\mathcal{E}^{(1)}}\right\|_1,\end{aligned}$$

where $\mathcal{E}^{(1)}$ can be taken as $\mathcal{S}^*$. Combining these results with (23), we conclude

$$\left\|\left(\widetilde{\boldsymbol{\Sigma}}^{(1)}-\boldsymbol{\Sigma}^*\right)_{\mathcal{S}^*}\right\|_1\leq\sqrt{s^*}\left\|\widetilde{\boldsymbol{\Sigma}}^{(1)}-\boldsymbol{\Sigma}^*\right\|_{\mathrm{F}}\leq\frac{2+\sqrt{2}}{2\rho_{2s^*+2\widetilde{s}}^{-}}\lambda s^*.$$

Therefore, we obtain

$$\left\|\widetilde{\boldsymbol{\Sigma}}^{(1)}-\boldsymbol{\Sigma}^*\right\|_1\leq 6\left\|\left(\widetilde{\boldsymbol{\Sigma}}^{(1)}-\boldsymbol{\Sigma}^*\right)_{\mathcal{S}^*}\right\|_1\leq\frac{3\left(2+\sqrt{2}\right)}{\rho_{2s^*+2\widetilde{s}}^{-}}\lambda s^*.$$

Since $\widetilde{\boldsymbol{\Sigma}}^{(1)}$ is a ε-optimal solution, we obtain

$$\begin{aligned}&\left\langle\nabla f\left(\widetilde{\boldsymbol{\Sigma}}^{(1)}\right)-\nabla f\left(\boldsymbol{\Sigma}^*\right),\widetilde{\boldsymbol{\Sigma}}^{(1)}-\boldsymbol{\Sigma}^*\right\rangle\\\leq&\left\|\nabla f\left(\widetilde{\boldsymbol{\Sigma}}^{(1)}\right)+\boldsymbol{\Lambda}\odot\widetilde{\boldsymbol{\Xi}}^{(1)}\right\|_{\max}\left\|\widetilde{\boldsymbol{\Sigma}}^{(1)}-\boldsymbol{\Sigma}^*\right\|_1\\&+\left\|\boldsymbol{\Lambda}\odot\widetilde{\boldsymbol{\Xi}}^{(1)}+\nabla f\left(\boldsymbol{\Sigma}^*\right)\right\|_{\max}\left\|\widetilde{\boldsymbol{\Sigma}}^{(1)}-\boldsymbol{\Sigma}^*\right\|_1\\\leq&c_0\lambda\left\|\widetilde{\boldsymbol{\Sigma}}^{(1)}-\boldsymbol{\Sigma}^*\right\|_1\leq\frac{c_1\lambda^2 s^*}{\rho_{2s^*+2\widetilde{s}}^{-}},\end{aligned}$$

for some constants c_0 and c_1. □

Lemma 35. *Given $\omega_{\boldsymbol{\Lambda}^{(k-1)}}\left(\widetilde{\boldsymbol{\Sigma}}^{(k)}\right)\leq\frac{\lambda}{8}$, we have that for all $t\geq 1$ at the $(k+1)$-th stage, $\omega_{\boldsymbol{\Lambda}^{(k)}}\left(\boldsymbol{\Sigma}_t^{(k+1)}\right)\leq\frac{\lambda}{4}$ and*

$$F_{\boldsymbol{\Lambda}^{(k)}}\left(\boldsymbol{\Sigma}_t^{(k+1)}\right)\leq F_{\boldsymbol{\Lambda}^{(k)}}\left(\boldsymbol{\Sigma}^*\right)+\frac{\lambda}{4}\left\|\boldsymbol{\Sigma}_t^{(k)}-\boldsymbol{\Sigma}^*\right\|_1.$$

Proof. We begin by establishing the foundational inequality at the $(k+1)$-th stage. Note that $\boldsymbol{\Sigma}_0^{(k+1)}=\widetilde{\boldsymbol{\Sigma}}^{(k)}$. By definition, we have

$$\omega_{\boldsymbol{\Lambda}^{(k)}}\left(\boldsymbol{\Sigma}_0\right)=\min_{\boldsymbol{\Xi}\in\partial\|\boldsymbol{\Sigma}_0\|_1}\left\|\nabla f\left(\boldsymbol{\Sigma}_0\right)+\boldsymbol{\Lambda}^{(k)}\odot\boldsymbol{\Xi}\right\|_{\max}.$$

Using the triangle inequality, we can expand the above expression as follows:

$$\begin{aligned}\omega_{\boldsymbol{\Lambda}^{(k)}}\left(\boldsymbol{\Sigma}_0\right)&\leq\min_{\boldsymbol{\Xi}\in\partial\|\boldsymbol{\Sigma}_0\|_1}\left\|\nabla f\left(\boldsymbol{\Sigma}_0\right)+\boldsymbol{\Lambda}^{(k-1)}\odot\boldsymbol{\Xi}\right\|_{\max}\\&\quad+\left\|\left(\boldsymbol{\Lambda}^{(k)}-\boldsymbol{\Lambda}^{(k-1)}\right)\odot\boldsymbol{\Xi}\right\|_{\max}\\&\underset{(i)}{\leq}\omega_{\boldsymbol{\Lambda}^{(k-1)}}\left(\boldsymbol{\Sigma}_0\right)+\left\|\boldsymbol{\Lambda}^{(k)}-\boldsymbol{\Lambda}^{(k-1)}\right\|_{\max}\\&\underset{(ii)}{\leq}\frac{\lambda}{8}+\frac{\lambda}{8}\leq\frac{\lambda}{4},\end{aligned}$$

where (i) is from the definition of the approximate KKT condition and $\boldsymbol{\Xi}$, and (ii) is from $\omega_{\boldsymbol{\Lambda}^{(k-1)}}\left(\boldsymbol{\Sigma}_0\right)=\omega_{\boldsymbol{\Lambda}^{(k-1)}}\left(\widetilde{\boldsymbol{\Sigma}}^{(k)}\right)\leq\frac{\lambda}{8}$ and $\left\|\boldsymbol{\Lambda}^{(k)}-\boldsymbol{\Lambda}^{(k-1)}\right\|_{\max}\leq\frac{\lambda}{8}$.

Continuing, for any $t\geq 0$, consider

$$\boldsymbol{\Xi}_t=\underset{\boldsymbol{\Xi}\in\partial\|\boldsymbol{\Sigma}_t\|_1}{\arg\min}\left\|\nabla f\left(\boldsymbol{\Sigma}_t\right)+\boldsymbol{\Lambda}^{(k)}\odot\boldsymbol{\Xi}\right\|_{\max}.$$

$$
\begin{aligned}
&\eta_t \left\langle \nabla f\left(\boldsymbol{\Sigma}_t\right), \Delta\boldsymbol{\Sigma}_t \right\rangle + \frac{\eta_t^2}{2}\left\|\Delta\boldsymbol{\Sigma}_t\right\|^2_{\nabla^2 f(\boldsymbol{\Sigma}_t)} + \left\|\boldsymbol{\Lambda}^{(k-1)} \odot \left(\boldsymbol{\Sigma}_t + \eta_t \Delta\boldsymbol{\Sigma}_t\right)\right\|_1 \\
\geq &\left\langle \nabla f\left(\boldsymbol{\Sigma}_t\right), \Delta\boldsymbol{\Sigma}_t \right\rangle + \frac{1}{2}\left\|\Delta\boldsymbol{\Sigma}_t\right\|^2_{\nabla^2 f(\boldsymbol{\Sigma}_t)} + \left\|\boldsymbol{\Lambda}^{(k-1)} \odot \left(\boldsymbol{\Sigma}_t + \Delta\boldsymbol{\Sigma}_t\right)\right\|_1
\end{aligned} \tag{24}
$$

$$
\begin{aligned}
&\eta_t \left\langle \nabla f\left(\boldsymbol{\Sigma}_t\right), \Delta\boldsymbol{\Sigma}_t \right\rangle + \frac{\eta_t^2}{2}\left\|\Delta\boldsymbol{\Sigma}_t\right\|^2_{\nabla^2 f(\boldsymbol{\Sigma}_t)} + \eta_t\left\|\boldsymbol{\Lambda}^{(k-1)} \odot \left(\boldsymbol{\Sigma}_t + \eta_t \Delta\boldsymbol{\Sigma}_t\right)\right\|_1 + (1-\eta_t)\left\|\boldsymbol{\Lambda}^{(k-1)} \odot \boldsymbol{\Sigma}_t\right\|_1 \\
\geq &\left\langle \nabla f\left(\boldsymbol{\Sigma}_t\right), \Delta\boldsymbol{\Sigma}_t \right\rangle + \frac{1}{2}\left\|\Delta\boldsymbol{\Sigma}_t\right\|^2_{\nabla^2 f(\boldsymbol{\Sigma}_t)} + \left\|\boldsymbol{\Lambda}^{(k-1)} \odot \left(\boldsymbol{\Sigma}_t + \Delta\boldsymbol{\Sigma}_t\right)\right\|_1
\end{aligned} \tag{25}
$$

By the convexity of F, we have

$$
\begin{aligned}
&F_{\boldsymbol{\Lambda}^{(k)}}\left(\boldsymbol{\Sigma}^*\right) \\
\geq &F_{\boldsymbol{\Lambda}^{(k)}}\left(\boldsymbol{\Sigma}_t\right) - \left\langle \nabla f\left(\boldsymbol{\Sigma}_t\right) + \boldsymbol{\Lambda}^{(k)} \odot \boldsymbol{\Xi}_t, \boldsymbol{\Sigma}_t - \boldsymbol{\Sigma}^* \right\rangle \\
\geq &F_{\boldsymbol{\Lambda}^{(k)}}\left(\boldsymbol{\Sigma}_t\right) - \left\|\nabla f\left(\boldsymbol{\Sigma}_t\right) + \boldsymbol{\Lambda}^{(k)} \odot \boldsymbol{\Xi}_t\right\|_{\max} \left\|\boldsymbol{\Sigma}_t - \boldsymbol{\Sigma}^*\right\|_1 \\
\underset{(i)}{\geq} &F_{\boldsymbol{\Lambda}^{(k)}}\left(\boldsymbol{\Sigma}_t\right) - \frac{\lambda}{4}\left\|\boldsymbol{\Sigma}_t - \boldsymbol{\Sigma}^*\right\|_1,
\end{aligned}
$$

where (ii) is from that $\left\|\nabla f\left(\boldsymbol{\Sigma}_t\right) + \boldsymbol{\Lambda}^{(k)} \odot \boldsymbol{\Xi}_t\right\|_{\max} \leq \frac{\lambda}{4}$. ☐

Lemma 36. *Suppose* $\|(\boldsymbol{\Sigma}_t)_{\overline{\mathcal{S}}}\|_0 \leq \widetilde{s}$ *and* $\omega_{\boldsymbol{\Lambda}^{(k)}}\left(\boldsymbol{\Sigma}_t\right) \leq \frac{\lambda}{4}$. *Then, there exists a generic constant* c_1 *such that*

$$
\left\|\boldsymbol{\Sigma}_t - \boldsymbol{\Sigma}^*\right\|_{\mathrm{F}} \leq \frac{c_1 \lambda \sqrt{s^*}}{\rho^-_{2s^*+2\widetilde{s}}}.
$$

Proof. The proof is analogous to that of Lemma 28. Therefore, we omit it for brevity. For more details, we refer readers to [58]. ☐

Lemma 37. *Recall* $\boldsymbol{\Sigma}_{t+\frac{1}{2}}$ *defined in* (9) *and* δ_t *as in* (10). *Denote* $\Delta\boldsymbol{\Sigma}_t = \boldsymbol{\Sigma}_{t+\frac{1}{2}} - \boldsymbol{\Sigma}_t$. *Then we have*

$$
\delta_t \leq -\left\|\Delta\boldsymbol{\Sigma}_t\right\|^2_{\nabla^2 f(\boldsymbol{\Sigma}_t)}.
$$

Proof. We note $\Delta\boldsymbol{\Sigma}_t$ is the solution for $\overline{F}\left(\boldsymbol{\Sigma} \mid \boldsymbol{\Sigma}_t, \boldsymbol{\Lambda}^{(k-1)}\right)$. For any $\eta_t \in (0,1]$, we have formula (24). By the convexity of ℓ_1-norm, we have formula (25). Rearranging the terms, canceling the $(1-\eta_t)$ factor from both sides and let $\eta_t \to 1$, we obtain the desired inequality $\delta_t \leq -\left\|\Delta\boldsymbol{\Sigma}_t\right\|^2_{\nabla^2 f(\boldsymbol{\Sigma}_t)}$. ☐

C. Proof of Theorem 20

Lemma 38 (Sparsity Preserving Lemma)**.** *Suppose that Assumptions 10 and 11 hold. Given* $\boldsymbol{\Sigma}_t^{(k)} \in \mathcal{B}\left(\boldsymbol{\Sigma}^*, r\right)$ *and* $\left\|\left(\boldsymbol{\Sigma}_t^{(k)}\right)_{\overline{\mathcal{S}}}\right\|_0 \leq \widetilde{s}$, *there exists a generic constant* c_1 *such that*

$$
\left\|\left(\boldsymbol{\Sigma}_{t+1}^{(k)}\right)_{\overline{\mathcal{S}}}\right\|_0 \leq \widetilde{s} \quad \textit{and} \quad \left\|\boldsymbol{\Sigma}_{t+1}^{(k)} - \boldsymbol{\Sigma}^*\right\|_{\mathrm{F}} \leq \frac{c_1 \lambda \sqrt{s^*}}{\rho^-_{2s^*+2\widetilde{s}}}.
$$

Proof. To simplify notation, we omit the stage index (k). Considering $\boldsymbol{\Sigma}_{t+1}$ as the proximal Newton update, the system satisfies the following equation:

$$
\operatorname{mat}\left(\nabla^2 f\left(\boldsymbol{\Sigma}_t\right) \operatorname{vec}\left(\boldsymbol{\Sigma}_{t+1} - \boldsymbol{\Sigma}_t\right)\right) + \nabla f\left(\boldsymbol{\Sigma}_t\right) + \boldsymbol{\Lambda} \odot \boldsymbol{\Xi}_{t+1} = \mathbf{0},
$$

where $\boldsymbol{\Xi}_{t+1} \in \partial\left\|\boldsymbol{\Sigma}_{t+1}\right\|_1$.

From Lemma 27, under identical conditions, it holds that $\left\|\boldsymbol{\Lambda}_{\overline{\mathcal{E}}}\right\|_{\min} \geq \frac{\lambda}{2}$ and $|\mathcal{E}| \leq 2s^*$ for some set $\mathcal{E} \supseteq \mathcal{S}^*$. In the following, we decompose

$$
\operatorname{mat}\left(\nabla^2 f\left(\boldsymbol{\Sigma}_t\right) \operatorname{vec}\left(\boldsymbol{\Sigma}_{t+1} - \boldsymbol{\Sigma}_t\right)\right) + \nabla f\left(\boldsymbol{\Sigma}_t\right)
$$

into four parts:

- $\mathbf{V}_1 := \operatorname{mat}\left(\nabla^2 f\left(\boldsymbol{\Sigma}_t\right) \operatorname{vec}\left(\boldsymbol{\Sigma}_{t+1} - \boldsymbol{\Sigma}^*\right)\right)$,
- $\mathbf{V}_2 := \operatorname{mat}\left(\nabla^2 f\left(\boldsymbol{\Sigma}_t\right) \operatorname{vec}\left(\boldsymbol{\Sigma}^* - \boldsymbol{\Sigma}_t\right)\right)$,
- $\mathbf{V}_3 := \nabla f\left(\boldsymbol{\Sigma}_t\right) - \nabla f\left(\boldsymbol{\Sigma}^*\right)$,
- $\mathbf{V}_4 := \nabla f\left(\boldsymbol{\Sigma}^*\right)$.

(1) For $\mathbf{V}_1$, we define set

$$
\mathcal{H}_1 = \left\{(i,j) \in \overline{\mathcal{E}} \mid \left|\left(\mathbf{V}_1\right)_{ij}\right| \geq \frac{\lambda}{4}\right\}.
$$

Referring to Lemma 35, we establish that

$$
F_{\boldsymbol{\Lambda}}\left(\boldsymbol{\Sigma}_{t+1}\right) \leq F_{\boldsymbol{\Lambda}}\left(\boldsymbol{\Sigma}^*\right) + \frac{\lambda}{4}\left\|\boldsymbol{\Sigma}_{t+1} - \boldsymbol{\Sigma}^*\right\|_1.
$$

This relationship leads to the following inequality:

$$
\begin{aligned}
&f\left(\boldsymbol{\Sigma}_{t+1}\right) - f\left(\boldsymbol{\Sigma}^*\right) \\
\leq &\lambda\left(\left\|\boldsymbol{\Sigma}^*\right\|_1 - \left\|\boldsymbol{\Sigma}_{t+1}\right\|_1\right) + \frac{\lambda}{4}\left\|\boldsymbol{\Sigma}_{t+1} - \boldsymbol{\Sigma}^*\right\|_1 \\
= &\lambda\left(\left\|\boldsymbol{\Sigma}^*_{\mathcal{E}}\right\|_1 - \left\|\left(\boldsymbol{\Sigma}_{t+1}\right)_{\mathcal{E}}\right\|_1 - \left\|\left(\boldsymbol{\Sigma}_{t+1}\right)_{\overline{\mathcal{E}}}\right\|_1\right) + \frac{\lambda}{4}\left\|\boldsymbol{\Sigma}_{t+1} - \boldsymbol{\Sigma}^*\right\|_1 \\
\leq &\frac{5\lambda}{4}\left\|\left(\boldsymbol{\Sigma}_{t+1}\right)_{\mathcal{E}} - \boldsymbol{\Sigma}^*_{\mathcal{E}}\right\|_1 - \frac{3\lambda}{4}\left\|\left(\boldsymbol{\Sigma}_{t+1}\right)_{\overline{\mathcal{E}}} - \boldsymbol{\Sigma}^*_{\overline{\mathcal{E}}}\right\|_1,
\end{aligned} \tag{26}
$$

where the equality holds since $\boldsymbol{\Sigma}^*_{\overline{\mathcal{E}}} = 0$. On the other hand, we have

$$
\begin{aligned}
&f\left(\boldsymbol{\Sigma}_{t+1}\right) - f\left(\boldsymbol{\Sigma}^*\right) \\
\underset{(i)}{\geq} &\left\langle \nabla f\left(\boldsymbol{\Sigma}^*\right), \boldsymbol{\Sigma}_{t+1} - \boldsymbol{\Sigma}^* \right\rangle \\
\geq &-\left\|\nabla f\left(\boldsymbol{\Sigma}^*\right)\right\|_{\max}\left\|\boldsymbol{\Sigma}_{t+1} - \boldsymbol{\Sigma}^*\right\|_1 \\
\underset{(ii)}{\geq} &-\frac{\lambda}{4}\left\|\boldsymbol{\Sigma}_{t+1} - \boldsymbol{\Sigma}^*\right\|_1 \\
= &-\frac{\lambda}{4}\left\|\left(\boldsymbol{\Sigma}_{t+1}\right)_{\mathcal{E}} - \boldsymbol{\Sigma}^*_{\mathcal{E}}\right\|_1 - \frac{\lambda}{4}\left\|\left(\boldsymbol{\Sigma}_{t+1}\right)_{\overline{\mathcal{E}}} - \boldsymbol{\Sigma}^*_{\overline{\mathcal{E}}}\right\|_1,
\end{aligned} \tag{27}
$$

where (i) is from the convexity of f and (ii) is from Assumption 11. Combining (26) and (27), we conclude

$$
\left\|\left(\boldsymbol{\Sigma}_{t+1}\right)_{\overline{\mathcal{E}}} - \boldsymbol{\Sigma}^*_{\overline{\mathcal{E}}}\right\|_1 \leq 3\left\|\left(\boldsymbol{\Sigma}_{t+1}\right)_{\mathcal{E}} - \boldsymbol{\Sigma}^*_{\mathcal{E}}\right\|_1.
$$

Consider a subset $\mathcal{S}' \subset \mathcal{H}_1$ with $|\mathcal{S}'| = s' \leq \widetilde{s}$. Select a vector $\mathbf{v} \in \mathbb{R}^{d^2}$ such that $\|\mathbf{v}\|_{\max} = 1$ and $\|\mathbf{v}\|_0 = s'$, satisfying the condition $s'\frac{\lambda}{4} \leq \mathbf{v}^\top \nabla^2 f(\boldsymbol{\Sigma}_t) \operatorname{vec}(\boldsymbol{\Sigma}_{t+1} - \boldsymbol{\Sigma}^*)$. Then, we have

$$\begin{aligned}
s'\frac{\lambda}{4} &\leq \mathbf{v}^\top \nabla^2 f(\boldsymbol{\Sigma}_t) \operatorname{vec}(\boldsymbol{\Sigma}_{t+1} - \boldsymbol{\Sigma}^*) \\
&\leq \left\|\mathbf{v}^\top \left(\nabla^2 f(\boldsymbol{\Sigma}_t)\right)^{\frac{1}{2}}\right\|_2 \left\|\left(\nabla^2 f(\boldsymbol{\Sigma}_t)\right)^{\frac{1}{2}} \operatorname{vec}(\boldsymbol{\Sigma}_{t+1} - \boldsymbol{\Sigma}^*)\right\|_2 \\
&\underset{(i)}{\leq} c_1 \sqrt{\rho^+_{2s^*+2\widetilde{s}}\rho^+_{s'}}\, \|\mathbf{v}\|_2 \|\boldsymbol{\Sigma}^* - \boldsymbol{\Sigma}_{t+1}\|_{\mathrm{F}} \\
&\underset{(ii)}{\leq} c_1 \sqrt{s'\rho^+_{2s^*+2\widetilde{s}}\rho^+_{s'}}\, \|\boldsymbol{\Sigma}^* - \boldsymbol{\Sigma}_{t+1}\|_{\mathrm{F}} \\
&\underset{(iii)}{\leq} \frac{c_1 \sqrt{s'\rho^+_{2s^*+2\widetilde{s}}\rho^+_{s'}}}{\rho^-_{2s^*+2\widetilde{s}}} \lambda \sqrt{s^*},
\end{aligned}$$

where (i) is from SE condition, (ii) is from the definition of $\mathbf{v}$, and (iii) is from Lemma 36. From the above inequalities, it follows that

$$s' \leq \frac{c_1 \rho^+_{2s^*+2\widetilde{s}}\rho^+_{s'} s^*}{\left(\rho^-_{2s^*+2\widetilde{s}}\right)^2} \leq c_1 \varpi^2_{2s^*+2\widetilde{s}} s^*,$$

where $\varpi_{2s^*+2\widetilde{s}} = \rho^+_{2s^*+2\widetilde{s}}/\rho^-_{2s^*+2\widetilde{s}}$ is the condition number. Given that $s' = |\mathcal{S}'|$ achieves the maximum possible value such that $s' \leq \widetilde{s}$ for any subset $\mathcal{S}'$ of $\mathcal{H}_1$. We conclude $\mathcal{S}' = \mathcal{H}_1$ to attain the maximum and thus $|\mathcal{H}_1| = s' \leq c_1 \varpi^2_{2s^*+2\widetilde{s}} s^*$.

(2) For $\mathbf{V}_2$, we define set

$$\mathcal{H}_2 = \left\{(i,j) \in \overline{\mathcal{E}} \mid \left|(\mathbf{V}_2)_{ij}\right| \geq \frac{\lambda}{4}\right\}.$$

Similar to the argument of $\mathcal{H}_1$, we have $|\mathcal{H}_2| \leq c_2 \varpi^2_{2s^*+2\widetilde{s}} s^*$.

(3) For $\mathbf{V}_3$, we define set

$$\mathcal{H}_3 = \left\{(i,j) \in \overline{\mathcal{E}} \mid \left|(\mathbf{V}_3)_{ij}\right| \geq \frac{\lambda}{4}\right\}.$$

Consider a vector $\mathbf{V} \in \mathbb{R}^{d\times d}$ such that $\|\mathbf{V}\|_{\max} = 1$, $\|\mathbf{V}\|_0 = |\mathcal{H}_3|$, and

$$\begin{aligned}
&\langle \mathbf{V}, \nabla f(\boldsymbol{\Sigma}_t) - \nabla f(\boldsymbol{\Sigma}^*)\rangle \\
=& \sum_{(i',j')\in\mathcal{H}_3} \mathbf{V}_{i'j'} \left(\nabla f(\boldsymbol{\Sigma}_t) - \nabla f(\boldsymbol{\Sigma}^*)\right)_{i'j'} \\
=& \sum_{(i',j')\in\mathcal{H}_3} \left(\nabla f(\boldsymbol{\Sigma}_t) - \nabla f(\boldsymbol{\Sigma}^*)\right)_{ij} \\
\geq& \frac{\lambda}{4}.
\end{aligned} \tag{30}$$

Then we have

$$\begin{aligned}
&\langle \mathbf{V}, \nabla f(\boldsymbol{\Sigma}_t) - \nabla f(\boldsymbol{\Sigma}^*)\rangle \\
&\leq \|\mathbf{V}\|_{\mathrm{F}} \|\nabla f(\boldsymbol{\Sigma}_t) - \nabla f(\boldsymbol{\Sigma}^*)\|_{\mathrm{F}} \\
&\underset{(i)}{\leq} \sqrt{|\mathcal{H}_3|}\, \|\nabla f(\boldsymbol{\Sigma}_t) - \nabla f(\boldsymbol{\Sigma}^*)\|_{\mathrm{F}} \\
&\underset{(ii)}{\leq} \rho^+_{2s^*+2\widetilde{s}} \sqrt{|\mathcal{H}_3|}\, \|\boldsymbol{\Sigma}_t - \boldsymbol{\Sigma}^*\|_{\mathrm{F}},
\end{aligned} \tag{31}$$

where (i) is from the definition of $\mathbf{V}$, and (ii) is from the mean value theorem and analogous argument for $\mathcal{H}_3$.

Combining (30) and (31), we have

$$\begin{aligned}
\lambda |\mathcal{H}_3| &\leq 4\rho^+_{2s^*+2\widetilde{s}} \sqrt{|\mathcal{H}_3|}\, \|\boldsymbol{\Sigma}_t - \boldsymbol{\Sigma}^*\|_{\mathrm{F}} \\
&\underset{(i)}{\leq} 8\lambda \varpi_{2s^*+2\widetilde{s}} \sqrt{3s^* |\mathcal{H}_3|},
\end{aligned}$$

where (i) is from Lemma 36. This implies $|\mathcal{H}_3| \leq c_3 \varpi^2_{2s^*+2\widetilde{s}} s^*$.

(4) For $\mathbf{V}_4$, we define set

$$\mathcal{H}_4 = \left\{(i,j) \in \overline{\mathcal{E}} \mid \left|(\boldsymbol{V}_4)_{ij}\right| \geq \frac{\lambda}{4}\right\}.$$

By Assumption 11, we have

$$\begin{aligned}
&0 \leq |\mathcal{H}_4| \leq \sum_{(i,j)\in\overline{\mathcal{E}}} \frac{4}{\lambda} \left|(\nabla f(\boldsymbol{\Sigma}^*))_{ij}\right| \cdot \mathbb{I}\left(\left|(\nabla f(\boldsymbol{\Sigma}^*))_{ij}\right| > \frac{\lambda}{4}\right) \\
&= \sum_{(i,j)\in\overline{\mathcal{E}}} \frac{4}{\lambda} \left|(\nabla f(\boldsymbol{\Sigma}^*))_{ij}\right| \cdot 0 = 0.
\end{aligned}$$

Combining the results for Set $\mathcal{H}_1 \sim \mathcal{H}_4$, we have that there exists some constant c_0 such that

$$\left\|\left(\boldsymbol{\Sigma}_{t+\frac{1}{2}}\right)_{\overline{\mathcal{S}^*}}\right\|_0 \leq c_0 \kappa^2_{2s^*+2\widetilde{s}} s^* \leq \widetilde{s}.$$

□

Lemma 39. *Suppose Assumptions 10 and 11 hold. If $\boldsymbol{\Sigma}^{(t)} \in \mathcal{B}(\boldsymbol{\Sigma}^*, r)$ and $\|\boldsymbol{\Sigma}^{(t+1)}_{\overline{\mathcal{S}^*}}\|_0 \leq \widetilde{s}$, then for each stage $k \geq 2$, we have*

$$\left\|\boldsymbol{\Sigma}^{(k)}_{t+1} - \widehat{\boldsymbol{\Sigma}}^{(k)}\right\|_{\mathrm{F}} \leq \frac{\tau\kappa^3}{\rho^-_{2s^*+2\widetilde{s}}} \left\|\boldsymbol{\Sigma}^{(k)}_t - \widehat{\boldsymbol{\Sigma}}^{(k)}\right\|^2_{\mathrm{F}}.$$

Proof. First, we reformulate the prox-Newton update (9) as formula (28). By the Lemma 38, we have $\left\|\left(\boldsymbol{\Sigma}^{(k)}_{t+1}\right)_{\overline{\mathcal{S}^*}}\right\|_0 \leq \widetilde{s}$. And by the KKT condition, we have formula (29). By the strictly non-expansive property of the proximal operator, we obtain the formula (32).

$$\boldsymbol{\Sigma}^{(k)}_{t+1} = \underset{\boldsymbol{\Sigma}}{\operatorname{argmin}} \left\langle \nabla f\left(\boldsymbol{\Sigma}^{(k)}_t\right), \boldsymbol{\Sigma} - \boldsymbol{\Sigma}^{(k)}_t \right\rangle + \frac{1}{2}\left\|\boldsymbol{\Sigma} - \boldsymbol{\Sigma}^{(k)}_t\right\|^2_{\nabla^2 f\left(\boldsymbol{\Sigma}^{(k)}_t\right)} + \left\|\boldsymbol{\Lambda}^{(k-1)} \odot \boldsymbol{\Sigma}\right\|_1 \tag{28}$$

$$\widehat{\boldsymbol{\Sigma}}^{(k)} = \underset{\boldsymbol{\Sigma}}{\operatorname{argmin}} \left\langle \nabla f\left(\widehat{\boldsymbol{\Sigma}}^{(k)}\right), \boldsymbol{\Sigma} - \widehat{\boldsymbol{\Sigma}}^{(k)} \right\rangle + \frac{1}{2}\left\|\boldsymbol{\Sigma} - \widehat{\boldsymbol{\Sigma}}^{(k)}\right\|^2_{\nabla^2 f\left(\boldsymbol{\Sigma}^{(k)}_t\right)} + \left\|\boldsymbol{\Lambda}^{(k-1)} \odot \widehat{\boldsymbol{\Sigma}}^{(k)}\right\|_1 \tag{29}$$

$$\left\|\boldsymbol{\Sigma}_{t+1}^{(k)}-\widehat{\boldsymbol{\Sigma}}^{(k)}\right\|_{\nabla^2 f\left(\widehat{\boldsymbol{\Sigma}}^{(k)}\right)}^2 \leq \operatorname{vec}^{\top}\left(\boldsymbol{\Sigma}_{t+1}^{(k)}-\widehat{\boldsymbol{\Sigma}}^{(k)}\right)\left[\nabla^2 f\left(\boldsymbol{\Sigma}_t^{(k)}\right) \operatorname{vec}\left(\boldsymbol{\Sigma}_t^{(k)}-\widehat{\boldsymbol{\Sigma}}^{(k)}\right)+\left(\nabla f\left(\widehat{\boldsymbol{\Sigma}}^{(k)}\right)-\nabla f\left(\boldsymbol{\Sigma}_t^{(k)}\right)\right)\right]$$
$$\leq\left\|\boldsymbol{\Sigma}_{t+1}^{(k)}-\widehat{\boldsymbol{\Sigma}}^{(k)}\right\|_{\mathrm{F}}\left\|\nabla^2 f\left(\boldsymbol{\Sigma}_t^{(k)}\right) \operatorname{vec}\left(\boldsymbol{\Sigma}_t^{(k)}-\widehat{\boldsymbol{\Sigma}}^{(k)}\right)+\left(\nabla f\left(\widehat{\boldsymbol{\Sigma}}^{(k)}\right)-\nabla f\left(\boldsymbol{\Sigma}_t^{(k)}\right)\right)\right\|_2 \tag{32}$$

$$\begin{aligned}
\left\|\boldsymbol{\Sigma}_{t+1}^{(k)}-\widehat{\boldsymbol{\Sigma}}^{(k)}\right\|_{\mathrm{F}} &\leq \frac{1}{\rho_{2s^*+2\widetilde{s}}^{-}}\left\|\nabla^2 f\left(\boldsymbol{\Sigma}_t^{(k)}\right) \operatorname{vec}\left(\boldsymbol{\Sigma}_t^{(k)}-\widehat{\boldsymbol{\Sigma}}^{(k)}\right)+\left(\nabla f\left(\widehat{\boldsymbol{\Sigma}}^{(k)}\right)-\nabla f\left(\boldsymbol{\Sigma}_t^{(k)}\right)\right)\right\|_2 \\
&=\frac{1}{\rho_{2s^*+2\widetilde{s}}^{-}}\left\|\int_0^1\left[\nabla^2 f\left(\boldsymbol{\Sigma}_t^{(k)}+\theta\left(\widehat{\boldsymbol{\Sigma}}^{(k)}-\boldsymbol{\Sigma}_t^{(k)}\right)\right)-\nabla^2 f\left(\boldsymbol{\Sigma}_t^{(k)}\right)\right] \cdot \operatorname{vec}\left(\widehat{\boldsymbol{\Sigma}}^{(k)}-\boldsymbol{\Sigma}_t^{(k)}\right) d\theta\right\|_2 \\
&\leq \frac{1}{\rho_{2s^*+2\widetilde{s}}^{-}} \int_0^1\left\|\left[\nabla^2 f\left(\boldsymbol{\Sigma}_t^{(k)}+\theta\left(\widehat{\boldsymbol{\Sigma}}^{(k)}-\boldsymbol{\Sigma}_t^{(k)}\right)\right)-\nabla^2 f\left(\boldsymbol{\Sigma}_t^{(k)}\right)\right] \cdot \operatorname{vec}\left(\widehat{\boldsymbol{\Sigma}}^{(k)}-\boldsymbol{\Sigma}_t^{(k)}\right)\right\|_2 d\theta \\
&\underset{(i)}{\leq} \frac{\tau\kappa^3}{\rho_{2s^*+2\widetilde{s}}^{-}}\left\|\boldsymbol{\Sigma}_t^{(k)}-\widehat{\boldsymbol{\Sigma}}^{(k)}\right\|_{\mathrm{F}}^2
\end{aligned} \tag{34}$$

Note that both $\|\boldsymbol{\Sigma}_{t+1}\|_0 \leq \widetilde{s}$ and $\|\widehat{\boldsymbol{\Sigma}}^{(k)}\|_0 \leq \widetilde{s}$. On the other hand, from the SE properties, we have

$$\left\|\boldsymbol{\Sigma}_{t+1}^{(k)}-\widehat{\boldsymbol{\Sigma}}^{(k)}\right\|_{\nabla^2 f\left(\widehat{\boldsymbol{\Sigma}}^{(k)}\right)}^2 \geq \rho_{2s^*+2\widetilde{s}}^{-}\left\|\boldsymbol{\Sigma}_{t+1}^{(k)}-\widehat{\boldsymbol{\Sigma}}^{(k)}\right\|_{\mathrm{F}}^2. \tag{33}$$

Combining (32) and (33), we have formula (34), where (i) is due to the local restricted Hessian smoothness of f. Then the proof is finished. ☐

In the subsequent analysis, it is essential to utilize the property that the iterates $\boldsymbol{\Sigma}_t \in \mathcal{B}(\widehat{\boldsymbol{\Sigma}}^{(k)}, 2r)$ rather than $\boldsymbol{\Sigma}_t \in \mathcal{B}\left(\boldsymbol{\Sigma}^*, r\right)$ for the convergence analysis of the proximal Newton method. This property is valid due to the simultaneous inclusion of $\boldsymbol{\Sigma}_t \in \mathcal{B}\left(\boldsymbol{\Sigma}^*, r\right)$ and $\widehat{\boldsymbol{\Sigma}}^{(k)} \in \mathcal{B}\left(\boldsymbol{\Sigma}^*, r\right)$. Consequently, it follows that $\boldsymbol{\Sigma}_t \in \mathcal{B}(\widehat{\boldsymbol{\Sigma}}^{(k)}, 2r)$, where $2r=\frac{\rho_{2s^*+2\widetilde{s}}^{-}}{2\tau\kappa}$ represents the radius of the quadratic convergence region for the proximal Newton algorithm.

Lemma 40. *Suppose that Assumptions 4, 7, 9, and 10 hold. If $\boldsymbol{\Sigma}_t \in \mathcal{B}(\widehat{\boldsymbol{\Sigma}}^{(k)}, 2r)$ and $\|(\boldsymbol{\Sigma}_t)_{\overline{\mathcal{S}^*}}\|_0 \leq \widetilde{s}$ at each stage $k \geq 2$ with $\alpha=0.3$, then $\eta_t=1$. Further, we have*

$$F\left(\boldsymbol{\Sigma}_{t+1}^{(k)}\right) \leq F\left(\boldsymbol{\Sigma}_t^{(k)}\right)+\frac{1}{4}\delta_t.$$

Proof. Recall $\boldsymbol{\Sigma}_{t+\frac{1}{2}}$ defined in (9) and denote $\Delta\boldsymbol{\Sigma}_t=\boldsymbol{\Sigma}_{t+\frac{1}{2}}-\boldsymbol{\Sigma}_t$. Then we have

$$\begin{aligned}
&\left\|\Delta\boldsymbol{\Sigma}_t^{(k)}\right\|_{\mathrm{F}}^2 \\
&\underset{(i)}{\leq}\left\|\boldsymbol{\Sigma}_t^{(k)}-\widehat{\boldsymbol{\Sigma}}^{(k)}\right\|_{\mathrm{F}}+\left\|\boldsymbol{\Sigma}_{t+\frac{1}{2}}^{(k)}-\widehat{\boldsymbol{\Sigma}}^{(k)}\right\|_{\mathrm{F}} \\
&\underset{(ii)}{\leq}\left\|\boldsymbol{\Sigma}_t^{(k)}-\widehat{\boldsymbol{\Sigma}}^{(k)}\right\|_{\mathrm{F}}+\frac{\tau\kappa^3}{\rho_{2s^*+2\widetilde{s}}^{-}}\left\|\boldsymbol{\Sigma}_t^{(k)}-\widehat{\boldsymbol{\Sigma}}^{(k)}\right\|_{\mathrm{F}}^2 \\
&\underset{(iii)}{\leq} \frac{3}{2}\left\|\boldsymbol{\Sigma}_t^{(k)}-\widehat{\boldsymbol{\Sigma}}^{(k)}\right\|_{\mathrm{F}},
\end{aligned} \tag{35}$$

where (i) is from triangle inequality, (ii) is from Lemma 39, and (iii) is from $\left\|\boldsymbol{\Sigma}_t^{(k)}-\widehat{\boldsymbol{\Sigma}}^{(k)}\right\|_{\mathrm{F}} \leq R \leq \frac{\rho_{2s^*+2\widetilde{s}}^{-}}{2\tau\kappa^3}$.

By Lemma 38, we have $\|(\boldsymbol{\Sigma}_t)_{\overline{\mathcal{S}^*}}\|_0 \leq 2\widetilde{s}$. Then by expanding F, we have formula (36), where (i) is from the restricted Hessian smooth condition, (ii) and (iv) are from Lemma 37, (iii) is from the same argument of (33), and (v) is from (35), $\delta_t<0$, and $\left\|\boldsymbol{\Sigma}_t^{(k)}-\widehat{\boldsymbol{\Sigma}}^{(k)}\right\|_{\mathrm{F}} \leq r \leq \frac{\rho_{2s^*+2\widetilde{s}}^{-}}{2\tau\kappa^3}$. The $\eta_t=1$ can be demonstrated analogously from the analysis in [75] and [76], thus we omit it. ☐

Lemma 41. *Suppose that Assumptions 4, 7, 9, 10, and 11 hold. To achieve the approximate KKT condition $\omega\left(\boldsymbol{\Sigma}_t\right) \leq \varepsilon$ for any $\varepsilon>0$ at each stage $k \geq 2$, the number of iterations for proximal Newton updates is at most*

$$\log\log\left(3\rho_{2s^*+2\widetilde{s}}^{+}/\varepsilon\right).$$

Proof. Based on the solution $\boldsymbol{\Sigma}_{t-1}$ obtained from the $(t-1)$-th iteration, we consider the optimal solution at t-th iteration. By the KKT condition, we have

$$\operatorname{mat}\left(\nabla^2 f\left(\boldsymbol{\Sigma}_{t-1}\right) \operatorname{vec}\left(\boldsymbol{\Sigma}_t-\boldsymbol{\Sigma}_{t-1}\right)\right)+\nabla f\left(\boldsymbol{\Sigma}_{t-1}\right)+\boldsymbol{\Lambda}\odot\boldsymbol{\Xi}_t=\mathbf{0},$$

where $\boldsymbol{\Xi}_t \in \partial\left\|\boldsymbol{\Sigma}_t\right\|_1$. Given any $\mathbf{V} \in \mathbb{R}^{d\times d}$ satisfying $\|\mathbf{V}\|_2 \leq \|\mathbf{V}\|_1=1$ and $\|\mathbf{V}\|_0 \leq 2s^*+2\widetilde{s}$, we derive formula (37), where (i) is from mean value theorem with some $\check{\boldsymbol{\Sigma}}=(1-\theta)\boldsymbol{\Sigma}_{t-1}+\theta\boldsymbol{\Sigma}_t$ for some $\theta\in[0,1]$ and Cauchy-Schwarz inequality, and (ii) is from the SE properties. By taking the supremum of the L.H.S of (37) with respect to $\mathbf{V}$, we have

$$\left\|\nabla f\left(\boldsymbol{\Sigma}_t\right)+\boldsymbol{\Lambda}\odot\boldsymbol{\Xi}_t\right\|_{\max} \leq 2\rho_{2s^*+2\widetilde{s}}^{+}\left\|\boldsymbol{\Sigma}_t-\boldsymbol{\Sigma}_{t-1}\right\|_{\mathrm{F}}. \tag{38}$$

$$
\begin{aligned}
& F\left(\boldsymbol{\Sigma}_t^{(k)}+\Delta \boldsymbol{\Sigma}_t^{(k)}\right)-F\left(\boldsymbol{\Sigma}_t^{(k)}\right) \\
= & f\left(\boldsymbol{\Sigma}_t^{(k)}+\Delta \boldsymbol{\Sigma}_t^{(k)}\right)-f\left(\boldsymbol{\Sigma}_t^{(k)}\right)+\left\|\boldsymbol{\Lambda}^{(k-1)} \odot\left(\boldsymbol{\Sigma}_t^{(k)}+\Delta \boldsymbol{\Sigma}_t^{(k)}\right)\right\|_1-\left\|\boldsymbol{\Lambda}^{(k-1)} \odot \boldsymbol{\Sigma}_t^{(k)}\right\|_1 \\
\underset{(i)}{\leq} & \left\langle\nabla f\left(\boldsymbol{\Sigma}_t^{(k)}\right), \Delta \boldsymbol{\Sigma}_t^{(k)}\right\rangle+\frac{1}{2}\left\|\Delta \boldsymbol{\Sigma}_t^{(k)}\right\|_{\nabla^2 f\left(\boldsymbol{\Sigma}_t^{(k)}\right)}^2+\frac{\tau \kappa^3}{3}\left\|\Delta \boldsymbol{\Sigma}_t^{(k)}\right\|_{\mathrm{F}}^3+\left\|\boldsymbol{\Lambda}^{(k-1)} \odot\left(\boldsymbol{\Sigma}_t^{(k)}+\Delta \boldsymbol{\Sigma}_t^{(k)}\right)\right\|_1-\left\|\boldsymbol{\Lambda}^{(k-1)} \odot \boldsymbol{\Sigma}_t^{(k)}\right\|_1 \\
\underset{(ii)}{\leq} & \delta_t-\frac{1}{2} \delta_t+\frac{\tau \kappa^3}{3}\left\|\Delta \boldsymbol{\Sigma}_t^{(k)}\right\|_{\mathrm{F}}^3 \underset{(iii)}{\leq} \frac{1}{2} \delta_t+\frac{\tau \kappa^3}{3 \rho_{2 s^*+2 \widetilde{s}}^{-}}\left\|\Delta \boldsymbol{\Sigma}_t^{(k)}\right\|_{\nabla^2 f(\boldsymbol{\Sigma})}^2\left\|\Delta \boldsymbol{\Sigma}_t^{(k)}\right\|_{\mathrm{F}} \\
\underset{(iv)}{\leq} & \left(\frac{1}{2}-\frac{\tau \kappa^3}{3 \rho_{2 s^*+2 \widetilde{s}}^{-}}\left\|\Delta \boldsymbol{\Sigma}_t^{(k)}\right\|_{\mathrm{F}}\right) \delta_t \underset{(v)}{\leq} \frac{1}{4} \delta_t
\end{aligned} \tag{36}
$$

$$
\begin{aligned}
\left\langle\nabla f\left(\boldsymbol{\Sigma}_t\right)+\boldsymbol{\Lambda} \odot \boldsymbol{\Xi}_t, \mathbf{V}\right\rangle & =\left\langle\nabla f\left(\boldsymbol{\Sigma}_t\right)-\nabla f\left(\boldsymbol{\Sigma}_{t-1}\right)-\operatorname{mat}\left(\nabla^2 f\left(\boldsymbol{\Sigma}_{t-1}\right) \operatorname{vec}\left(\boldsymbol{\Sigma}_t-\boldsymbol{\Sigma}_{t-1}\right)\right), \mathbf{V}\right\rangle \\
& =\left\langle\nabla f\left(\boldsymbol{\Sigma}_t\right)-\nabla f\left(\boldsymbol{\Sigma}_{t-1}\right), \mathbf{V}\right\rangle-\left(\nabla^2 f\left(\boldsymbol{\Sigma}_{t-1}\right) \operatorname{vec}\left(\boldsymbol{\Sigma}_t-\boldsymbol{\Sigma}_{t-1}\right)\right)^{\top} \operatorname{vec}(\mathbf{V}) \\
& \underset{(i)}{\leq}\left\|\left(\nabla^2 f\left(\breve{\boldsymbol{\Sigma}}\right)\right)^{\frac{1}{2}} \operatorname{vec}\left(\boldsymbol{\Sigma}_t-\boldsymbol{\Sigma}_{t-1}\right)\right\|_2 \cdot\left\|\operatorname{vec}^{\top}(\mathbf{V})\left(\nabla^2 f\left(\breve{\boldsymbol{\Sigma}}\right)\right)^{\frac{1}{2}}\right\|_2 \\
& \quad+\left\|\left(\nabla^2 f\left(\boldsymbol{\Sigma}_{t-1}\right)\right)^{\frac{1}{2}} \operatorname{vec}\left(\boldsymbol{\Sigma}_t-\boldsymbol{\Sigma}_{t-1}\right)\right\|_2 \cdot\left\|\operatorname{vec}^{\top}(\mathbf{V})\left(\nabla^2 f\left(\boldsymbol{\Sigma}_{t-1}\right)\right)^{\frac{1}{2}}\right\|_2 \\
& \underset{(ii)}{\leq} 2 \rho_{2 s^*+2 \widetilde{s}}^{+}\left\|\boldsymbol{\Sigma}_t-\boldsymbol{\Sigma}_{t-1}\right\|_{\mathrm{F}},
\end{aligned} \tag{37}
$$

Then from Lemma (39), we have

$$
\begin{aligned}
& \left\|\boldsymbol{\Sigma}_{t+1}^{(k)}-\widehat{\boldsymbol{\Sigma}}^{(k)}\right\|_{\mathrm{F}} \\
\leq & \left(\tau \kappa^3 / \rho_{2 s^*+2 \widetilde{s}}^{-}\right)^{1+2+4+\ldots+2^{t-1}}\left\|\boldsymbol{\Sigma}_0^{(k)}-\widehat{\boldsymbol{\Sigma}}^{(k)}\right\|_{\mathrm{F}}^2 \\
\leq & \left(\left(\tau \kappa^3 / \rho_{2 s^*+2 \widetilde{s}}^{-}\right)\left\|\boldsymbol{\Sigma}_0^{(k)}-\widehat{\boldsymbol{\Sigma}}^{(k)}\right\|_{\mathrm{F}}^2\right)^{2^t} .
\end{aligned}
$$

By (37) and (35) by taking $\Delta \boldsymbol{\Sigma}_{t-1}=\boldsymbol{\Sigma}_t-\boldsymbol{\Sigma}_{t-1}$, we obtain

$$
\begin{aligned}
& \omega_{\boldsymbol{\Lambda}^{(k-1)}}\left(\boldsymbol{\Sigma}_t^{(k)}\right) \\
\leq & 2 \rho_{2 s^*+2 \widetilde{s}}^{+}\left\|\boldsymbol{\Sigma}_t^{(k)}-\boldsymbol{\Sigma}_{t-1}^{(k)}\right\|_{\mathrm{F}} \\
\leq & 3 \rho_{2 s^*+2 \widetilde{s}}^{+}\left\|\boldsymbol{\Sigma}_{t-1}^{(k)}-\widehat{\boldsymbol{\Sigma}}^{(k)}\right\|_{\mathrm{F}} \\
\leq & 3 \rho_{2 s^*+2 \widetilde{s}}^{+}\left(\left(\tau \kappa^3 / \rho_{2 s^*+2 \widetilde{s}}^{-}\right)\left\|\boldsymbol{\Sigma}_0^{(k)}-\widehat{\boldsymbol{\Sigma}}^{(k)}\right\|_{\mathrm{F}}\right)^{2^t} .
\end{aligned}
$$

By requiring the R.H.S equal to ε we obtain

$$
\begin{aligned}
t & =\log \left(\log \left(\frac{3 \rho_{2 s^*+2 \widetilde{s}}^{+}}{\varepsilon}\right) / \log \left(\frac{\rho_{2 s^*+2 \widetilde{s}}^{-}}{\tau \kappa^3\left\|\boldsymbol{\Sigma}_0^{(k)}-\widehat{\boldsymbol{\Sigma}}^{(k)}\right\|_{\mathrm{F}}}\right)\right) \\
& =\log \log \left(\frac{3 \rho_{2 s^*+2 \widetilde{s}}^{+}}{\varepsilon}\right)-\log \log \left(\frac{\rho_{2 s^*+2 \widetilde{s}}^{-}}{\tau \kappa^3\left\|\boldsymbol{\Sigma}_0^{(k)}-\widehat{\boldsymbol{\Sigma}}^{(k)}\right\|_{\mathrm{F}}}\right) \\
& \underset{(i)}{\leq} \log \log \left(\frac{3 \rho_{2 s^*+2 \widetilde{s}}^{+}}{\varepsilon}\right)-\log \log 4 \\
& \leq \log \log \left(\frac{3 \rho_{2 s^*+2 \widetilde{s}}^{+}}{\varepsilon}\right),
\end{aligned}
$$

where (i) is from the fact that $\left\|\boldsymbol{\Sigma}_0^{(k)}-\widehat{\boldsymbol{\Sigma}}^{(k)}\right\|_{\mathrm{F}} \leq r=\frac{\rho_{2 s^*+2 \widetilde{s}}^{-}}{4 \tau \kappa^3}$. □

D. *Proof of Theorem 19*

Lemma 42. *Suppose that Assumptions 4, 7, 9, 10, and 11 hold. After sufficiently many iterations $T<\infty$, the following results hold for all $t \geq T$:*

$$
\left\|\left(\boldsymbol{\Sigma}_t\right)_{\overline{\mathcal{S}^*}}\right\|_0 \leq \widetilde{s} \text{ and } F_{\boldsymbol{\Lambda}^{(0)}}\left(\boldsymbol{\Sigma}_t^{(1)}\right) \leq F_{\boldsymbol{\Lambda}^{(0)}}\left(\boldsymbol{\Sigma}^*\right)+\frac{15 \lambda^2 s^*}{4 \rho_{2 s^*+2 \widetilde{s}}^{-}} .
$$

Proof. Building upon the framework established in Lemma 37, Lemma 40, it can be concluded that the objective $F_{\boldsymbol{\Lambda}^{(0)}}$ exhibits a sufficient decrease in each iteration of the proximal

Newton step, as also discussed in [77]. Consequently, there exists a constant T such that for all $t \geq T$, we have

$$F_{\mathbf{\Lambda}^{(0)}}\left(\mathbf{\Sigma}_t^{(1)}\right) \leq F_{\mathbf{\Lambda}^{(0)}}\left(\mathbf{\Sigma}^*\right) + \frac{\lambda}{4}\left\|\mathbf{\Sigma}_t^{(1)} - \mathbf{\Sigma}^*\right\|_1,$$

where $\left\|\mathbf{\Sigma}_t^{(1)} - \mathbf{\Sigma}^*\right\|_1 \leq \frac{c_1\lambda\sqrt{s^*}}{\rho^-_{2s^*+2\widetilde{s}}}$ derived from a similar argument presented in the proof of Lemma 27. The subsequent analysis follows a pattern analogous to that of Lemma 38. Let $S_n = \left\{(i,j) \mid \left|\nabla f\left(\widetilde{\mathbf{\Sigma}}\right)_{ij}\right| = \lambda\right\}$ and consider the set $\left\{(i,j) \mid \left(\widetilde{\mathbf{\Sigma}}\right)_{ij} \neq 0\right\} \subseteq S_n$. It is required to show $|S_n| \leq s^* + \widetilde{s}$. To demonstrate this, we decompose S_n into two distinct parts:

- $S_n^1 : \left\{(i,j) \notin \overline{\mathcal{S}^*} \mid \left|\left(\nabla f\left(\widetilde{\mathbf{\Sigma}}\right) - \nabla f\left(\mathbf{\Sigma}^*\right)\right)_{ij}\right| \geq \frac{\lambda}{2}\right\}$,
- $S_n^2 : \left\{(i,j) \notin \overline{\mathcal{S}^*} \mid \left|\left(\nabla f\left(\mathbf{\Sigma}^*\right)\right)_{ij}\right| > \frac{\lambda}{2}\right\}$.

We have $S_n \subseteq \mathcal{S}^* \cup S_n^1 \cup S_n^2$ and establish upper bounds for each.

(1) For S_n^1, consider S' with maximum size $s' = |S'| \leq \widetilde{s}$ such that $S' \subseteq S_n^1$. Then there exists a matrix $\mathbf{V}$ such that $\|\mathbf{V}\|_{\max} = 1$ and $\|\mathbf{V}\|_0 = s'$, satisfying

$$\frac{\lambda s'}{2} \leq \left\langle \nabla f\left(\widetilde{\mathbf{\Sigma}}\right) - \nabla f\left(\mathbf{\Sigma}^*\right), \mathbf{V}\right\rangle.$$

By the mean value theorem, there exists some $\theta \in [0,1]$ such that

$$\begin{aligned}&\nabla f\left(\widetilde{\mathbf{\Sigma}}\right) - \nabla f\left(\mathbf{\Sigma}^*\right)\\ &= \operatorname{mat}\left(\nabla^2 f\left(\theta\widetilde{\mathbf{\Sigma}} + (1-\theta)\mathbf{\Sigma}^*\right)\operatorname{vec}\left(\widetilde{\mathbf{\Sigma}} - \mathbf{\Sigma}^*\right)\right).\end{aligned}$$

For simplicity, we define $\mathbf{H} = \nabla^2 f\left(\rho\widetilde{\mathbf{\Sigma}} + (1-\rho)\mathbf{\Sigma}^*\right)$. Applying the Cauchy-Schwartz inequality, we have

$$\begin{aligned}\frac{\lambda s'}{2} &\leq \operatorname{vec}^\top(\mathbf{V})\ \mathbf{H}\operatorname{vec}\left(\widetilde{\mathbf{\Sigma}} - \mathbf{\Sigma}^*\right)\\ &\leq \underbrace{\left\|\operatorname{vec}^\top(\mathbf{V})(\mathbf{H})^{\frac{1}{2}}\right\|_2}_{\mathbf{I}}\underbrace{\left\|(\mathbf{H})^{\frac{1}{2}}\operatorname{vec}\left(\widetilde{\mathbf{\Sigma}} - \mathbf{\Sigma}^*\right)\right\|_2}_{\mathbf{II}}.\end{aligned} \tag{39}$$

We now establish upper bounds for terms $\mathbf{I}$ and $\mathbf{II}$ respectively. Let $\widetilde{\mathbf{\Sigma}}, \mathbf{\Sigma}^* \in \mathcal{B}(\mathbf{\Sigma}^*, r)$, any convex combination of $\widetilde{\mathbf{\Sigma}}$ and $\mathbf{\Sigma}^*$ also falls in $\mathcal{B}(\mathbf{\Sigma}^*, r)$. The localized sparse eigenvalue condition can be applied to $\mathbf{H}$.

i) For term $\mathbf{I}$, based on Definition (6) and Assumption 7, we have

$$\begin{aligned}\left\|\operatorname{vec}^\top(\mathbf{U})(\mathbf{H})^{\frac{1}{2}}\right\|_2 &\leq \sqrt{\rho^+_{2s^*+2\widetilde{s}}}\,\|\mathbf{U}\|_{\mathrm{F}}^2\\ &\leq \sqrt{\rho^+_{2s^*+2\widetilde{s}}}\left(\|\mathbf{U}\|_1\|\mathbf{U}\|_{\max}\right)^{\frac{1}{2}}\\ &\leq \sqrt{\rho^+_{2s^*+2\widetilde{s}}}\sqrt{s'}.\end{aligned}$$

ii) For $\mathbf{II}$, using Lemma 34, we have

$$\begin{aligned}&\left\|(\mathbf{H})^{\frac{1}{2}}\operatorname{vec}\left(\widetilde{\mathbf{\Sigma}} - \mathbf{\Sigma}^*\right)\right\|_2\\ \leq&\left\langle \nabla f\left(\widetilde{\mathbf{\Sigma}}\right) - \nabla f\left(\mathbf{\Sigma}^*\right), \widetilde{\mathbf{\Sigma}} - \mathbf{\Sigma}^*\right\rangle\\ \leq&\frac{c_1\lambda^2 s^*}{\rho^-_{2s^*+2\widetilde{s}}}.\end{aligned}$$

By substituting the bounds for $\mathbf{I}$ and $\mathbf{II}$ back into (39), we obtain

$$\frac{\lambda s'}{2} \leq \sqrt{\rho^+_{2s^*+2\widetilde{s}}}\sqrt{s'} \times \frac{c_1\lambda\sqrt{s^*}}{\rho^-_{2s^*+2\widetilde{s}}}.$$

Multiplying both sides of this inequality by $\left(\frac{\lambda}{2}\right)^{\frac{1}{2}}$ and squaring gives us

$$s' \leq \frac{4\rho^+_{2s^*+2\widetilde{s}}c_2 s^*}{\left(\rho^-_{2s^*+2\widetilde{s}}\right)^2} < \widetilde{s}, \tag{40}$$

where the last inequality follows from our assumption. Since $s' = |S'|$ achieves the maximum possible value such that $s' \leq \widetilde{s}$ for any subset S' of S_n^1 and 40 shows that $s' < \widetilde{s}$, we conclude that $S' = S_n^1$, and thus

$$\left|S_n^1\right| = s' \leq \left\lfloor \frac{4\rho^+_{2s^*+2\widetilde{s}}c_2 s^*}{\left(\rho^-_{2s^*+2\widetilde{s}}\right)^2}\right\rfloor < \widetilde{s}.$$

(2) For S_n^2, Given $\|\nabla f(\mathbf{\Sigma}^*)\|_{\max} + \varepsilon \leq \frac{\lambda}{4}$, it follows that $S_n^2 = \emptyset$ and thus $\left|S_n^2\right| = 0$.

Combining all the results, we have $\left\|\mathbf{\Sigma}^{(t)}_{\overline{\mathcal{S}^*}}\right\|_0 \leq \widetilde{s}$. ☐

Lemma 43. *Suppose that Assumptions 7, 10 and 11 hold. If* $\left\|\left(\mathbf{\Sigma}_t^{(1)}\right)_{\overline{\mathcal{S}^*}}\right\|_0 \leq \widetilde{s}$, *and* $F_{\mathbf{\Lambda}^{(0)}}\left(\mathbf{\Sigma}_t^{(1)}\right) \leq F_{\mathbf{\Lambda}^{(0)}}\left(\mathbf{\Sigma}^*\right) + \frac{15\lambda^2 s^*}{4\rho^-_{2s^*+2\widetilde{s}}}$, *we have*

$$\left\|\mathbf{\Sigma}_t^{(1)} - \mathbf{\Sigma}^*\right\|_2 \leq \frac{c_1\lambda\sqrt{s^*}}{\rho^-_{2s^*+2\widetilde{s}}}$$

for some constant c_1 *and*

$$\left\|\mathbf{\Sigma}_{t+1}^{(1)} - \widehat{\mathbf{\Sigma}}^{(1)}\right\|_2 \leq \frac{\tau\kappa^3}{\rho^-_{2s^*+2\widetilde{s}}}\left\|\mathbf{\Sigma}_t^{(1)} - \widehat{\mathbf{\Sigma}}^{(1)}\right\|_2^2.$$

Proof. The estimation error is derived analogously from [58] and [76]; thus, we omit it here. The claim of quadratic convergence follows directly from Lemma (39) given sparse solutions. ☐

Lemma 44. *Suppose that Assumptions 7, 10, and 11 hold. If* $\left\|\left(\mathbf{\Sigma}_T^{(1)}\right)_{\overline{\mathcal{S}^*}}\right\|_0 \leq \widetilde{s}$, *and* $F_{\mathbf{\Lambda}^{(0)}}\left(\mathbf{\Sigma}_T^{(1)}\right) \leq F_{\mathbf{\Lambda}^{(1)}}\left(\mathbf{\Sigma}^*\right) + \frac{15\lambda^2 s^*}{4\rho^-_{2s^*+2\widetilde{s}}}$ *at some iteration* T, *we need at most*

$$T_1 \leq \log\log\left(\frac{3\rho^+_{2s^*+2\widetilde{s}}}{\varepsilon}\right)$$

extra iterations of the proximal Newton updates such that $\omega_{\mathbf{\Lambda}^{(0)}}\left(\mathbf{\Sigma}_{T+T_1}^{(1)}\right) \leq \frac{\lambda}{8}$.

Proof. The maximum number of iterations for the proximal Newton update is determined by integrating Lemma (42) and Lemma (41). Note that

$$T_1 \leq \log\frac{\log\left(\frac{3\rho^+_{2s^*+2\widetilde{s}}}{\varepsilon}\right)}{\log\left(\frac{\rho^-_{2s^*+2\widetilde{s}}}{\tau\kappa^3\left\|\mathbf{\Sigma}^{(T+1)} - \widehat{\mathbf{\Sigma}}^{\{1\}}\right\|_{\mathrm{F}}}\right)}.$$

Then we obtain the result from $\left\|\mathbf{\Sigma}_{T+1}^{(1)} - \widehat{\mathbf{\Sigma}}^{(1)}\right\|_{\mathrm{F}} \leq r = \frac{\rho^-_{2s^*+2\widetilde{s}}}{4\tau\kappa}$. ☐

TABLE I
QUANTITATIVE COMPARISON AMONG DIFFERENT PENALTIES FOR THE BANDED MATRIX

	ℓ_1				MCP				ℓ_1				MCP			
	$m=100$	$m=500$	$m=1500$	$m=3000$	$m=100$	$m=500$	$m=1500$	$m=3000$	$m=100$	$m=500$	$m=1500$	$m=3000$	$m=100$	$m=500$	$m=1500$	$m=3000$
	$d=80, n=50$								$d=80, n=100$							
FAE	27.7825	24.1565	22.1561	19.9998	24.3152	20.1542	19.4295	18.6415	26.5923	23.8546	21.4165	19.4620	23.7515	19.5612	18.5164	17.8962
	(0.2616)	(0.1648)	(0.1246)	(0.1051)	(0.2551)	(0.1566)	(0.1047)	(0.1015)	(0.2436)	(0.1534)	(0.1132)	(0.1052)	(0.2435)	(0.1530)	(0.1106)	(0.0945)
FRE	1.3876	1.2065	1.1066	0.9989	1.2144	1.0066	0.9704	0.9311	1.3282	1.1914	1.0697	0.9720	1.1863	0.9770	0.9248	0.8938
	(0.0546)	(0.0591)	(0.0198)	(0.0095)	(0.0357)	(0.0296)	(0.0210)	(0.0090)	(0.0973)	(0.0275)	(0.1016)	(0.0099)	(0.0773)	(0.0176)	(0.0012)	(0.0009)
FPR	0.1615	0.1528	0.1137	0.1053	0.1256	0.1038	0.0681	0.0311	0.1572	0.1315	0.1152	0.0873	0.1137	0.0852	0.0489	0.0241
	(0.0702)	(0.0765)	(0.0686)	(0.0552)	(0.0699)	(0.0438)	(0.0423)	(0.0217)	(0.0425)	(0.0407)	(0.0319)	(0.0287)	(0.0366)	(0.0287)	(0.0209)	(0.0174)
TPR	0.8072	0.8491	0.8774	0.8955	0.8413	0.8618	0.8957	0.9214	0.8117	0.8546	0.8810	0.9016	0.8455	0.8821	0.9016	0.9425
	(0.0368)	(0.0382)	(0.0397)	(0.0399)	(0.0397)	(0.0407)	(0.0412)	(0.0420)	(0.0371)	(0.0392)	(0.0401)	(0.0408)	(0.0400)	(0.0412)	(0.0418)	(0.0431)
	$d=100, n=50$								$d=100, n=100$							
FAE	28.1276	26.5312	23.1562	20.1532	24.8612	21.0017	20.1562	18.8964	27.8613	24.2361	22.2613	19.9131	24.2315	21.1156	19.8216	18.1566
	(0.2492)	(0.2151)	(0.1813)	(0.1513)	(0.2263)	(0.2052)	(0.1673)	(0.1233)	(0.2161)	(0.2111)	(0.1562)	(0.1162)	(0.2023)	(0.1762)	(0.1230)	(0.1055)
FRE	1.1012	1.0387	0.9066	0.7890	0.9734	0.8223	0.7892	0.7398	1.0908	0.9489	0.8716	0.7796	0.9487	0.8267	0.7761	0.7109
	(0.0435)	(0.0051)	(0.0736)	(0.0435)	(0.0315)	(0.0184)	(0.0163)	(0.0056)	(0.0251)	(0.0156)	(0.0013)	(0.0012)	(0.0130)	(0.0104)	(0.0061)	(0.0023)
FPR	0.1511	0.1341	0.1056	0.0915	0.1105	0.1004	0.0557	0.0303	0.1462	0.1227	0.1005	0.0905	0.1126	0.0756	0.0448	0.0131
	(0.0851)	(0.0793)	(0.0712)	(0.0663)	(0.0714)	(0.0627)	(0.0487)	(0.0226)	(0.0487)	(0.0415)	(0.0326)	(0.0291)	(0.0392)	(0.0311)	(0.0265)	(0.0059)
TPR	0.8157	0.8581	0.9167	0.9294	0.8780	0.8860	0.9070	0.9421	0.8307	0.8706	0.8913	0.9212	0.8605	0.8915	0.9172	0.9512
	(0.0337)	(0.0391)	(0.0401)	(0.0413)	(0.0402)	(0.0412)	(0.0422)	(0.0430)	(0.0380)	(0.0401)	(0.0409)	(0.0411)	(0.0416)	(0.0423)	(0.0427)	(0.0436)

TABLE II
QUANTITATIVE COMPARISON AMONG DIFFERENT PENALTIES FOR THE BLOCK MATRIX

	ℓ_1				MCP				ℓ_1				MCP			
	$m=100$	$m=500$	$m=1500$	$m=3000$	$m=100$	$m=500$	$m=1500$	$m=3000$	$m=100$	$m=500$	$m=1500$	$m=3000$	$m=100$	$m=500$	$m=1500$	$m=3000$
	$d=80, n=50$								$d=80, n=100$							
FAE	14.2562	12.5602	11.5627	10.1561	13.5615	11.4833	10.1566	9.1562	13.9231	12.1569	10.9158	9.7123	12.8873	11.8153	10.0862	9.0561
	(0.0172)	(0.0107)	(0.0095)	(0.0073)	(0.0147)	(0.0081)	(0.0072)	(0.0056)	(0.0133)	(0.0110)	(0.0089)	(0.0071)	(0.0109)	(0.0102)	(0.0074)	(0.0062)
FRE	0.9612	0.8468	0.7796	0.6847	0.9143	0.7742	0.6848	0.6173	0.9387	0.8196	0.7359	0.6548	0.8689	0.7966	0.6800	0.6106
	(0.0183)	(0.0113)	(0.0099)	(0.0081)	(0.0176)	(0.0106)	(0.0094)	(0.0075)	(0.0164)	(0.0161)	(0.0088)	(0.0067)	(0.0154)	(0.0106)	(0.0079)	(0.0058)
FPR	0.1771	0.1662	0.1322	0.1162	0.1607	0.1430	0.1017	0.0924	0.1682	0.1531	0.1285	0.1056	0.1502	0.1428	0.0977	0.0887
	(0.0848)	(0.0821)	(0.0792)	(0.0627)	(0.0873)	(0.0791)	(0.0628)	(0.0597)	(0.0692)	(0.0665)	(0.0594)	(0.0556)	(0.0671)	(0.0598)	(0.0541)	(0.0426)
TPR	0.8462	0.8716	0.9051	0.9243	0.8513	0.8819	0.9158	0.9356	0.8532	0.8762	0.9132	0.9287	0.8762	0.8816	0.9266	0.9532
	(0.0817)	(0.0794)	(0.0762)	(0.0711)	(0.0808)	(0.0791)	(0.0743)	(0.0706)	(0.0809)	(0.0786)	(0.0758)	(0.0707)	(0.0802)	(0.0775)	(0.0752)	(0.0701)
	$d=100, n=50$								$d=100, n=100$							
FAE	15.9891	15.0652	14.8767	13.9917	14.9291	14.6727	13.9806	13.2177	15.4827	14.6944	14.4958	13.5837	14.4938	13.9848	13.1766	12.5748
	(0.0551)	(0.0301)	(0.0314)	(0.0277)	(0.0165)	(0.0124)	(0.0182)	(0.0206)	(0.0369)	(0.0504)	(0.0227)	(0.0187)	(0.0178)	(0.0112)	(0.0143)	(0.0126)
FRE	0.8869	0.8357	0.8252	0.7761	0.8281	0.8139	0.7755	0.7332	0.8588	0.8151	0.8041	0.7535	0.8040	0.7757	0.7309	0.6975
	(0.0244)	(0.0196)	(0.0127)	(0.0101)	(0.0186)	(0.0153)	(0.0121)	(0.0098)	(0.0199)	(0.0185)	(0.0182)	(0.0113)	(0.0178)	(0.0140)	(0.0099)	(0.0067)
FPR	0.1672	0.1531	0.1310	0.1011	0.1584	0.1386	0.0996	0.0854	0.1593	0.1488	0.1247	0.1006	0.1477	0.1323	0.1109	0.0909
	(0.1024)	(0.0868)	(0.0773)	(0.0545)	(0.0997)	(0.0798)	(0.0661)	(0.0602)	(0.0924)	(0.0803)	(0.0722)	(0.0521)	(0.0911)	(0.0754)	(0.0612)	(0.0509)
TPR	0.8554	0.8817	0.9111	0.9306	0.8606	0.8996	0.9224	0.9434	0.8595	0.8906	0.9323	0.9344	0.8874	0.9001	0.9389	0.9562
	(0.0993)	(0.0813)	(0.0661)	(0.0523)	(0.0985)	(0.0751)	(0.0561)	(0.0413)	(0.0862)	(0.0615)	(0.0515)	(0.0356)	(0.0731)	(0.0515)	(0.0417)	(0.0258)

TABLE III
QUANTITATIVE COMPARISON AMONG DIFFERENT PENALTIES FOR THE TOEPLITZ MATRIX

	ℓ_1				MCP				ℓ_1				MCP			
	$m=100$	$m=500$	$m=1500$	$m=3000$	$m=100$	$m=500$	$m=1500$	$m=3000$	$m=100$	$m=500$	$m=1500$	$m=3000$	$m=100$	$m=500$	$m=1500$	$m=3000$
	$d=80, n=50$								$d=80, n=100$							
FAE	20.3550	16.7333	15.3850	11.6680	17.7370	15.2278	14.1555	11.1678	18.7959	15.8283	15.0287	11.3017	16.5025	14.5135	12.9327	9.8112
	(0.1918)	(0.1217)	(0.0933)	(0.0421)	(0.1065)	(0.0803)	(0.0622)	(0.0531)	(0.1241)	(0.0887)	(0.0450)	(0.0326)	(0.0915)	(0.0754)	(0.0301)	(0.0222)
FRE	1.2168	1.0003	0.9197	0.6975	1.0603	0.9103	0.8462	0.6676	1.1236	0.9462	0.8984	0.6756	0.9865	0.8676	0.7731	0.5865
	(0.0218)	(0.0051)	(0.0062)	(0.0007)	(0.0045)	(0.0003)	(0.0011)	(0.0002)	(0.0062)	(0.0047)	(0.0010)	(0.0005)	(0.0071)	(0.0033)	(0.0009)	(0.0003)
	$d=100, n=50$								$d=100, n=100$							
FAE	24.1613	19.2561	17.6243	14.9918	20.3272	17.3691	15.6213	12.1561	20.3576	18.0317	16.7666	13.5243	18.0767	16.6280	14.1802	10.6530
	(1.0406)	(0.7621)	(0.0773)	(0.0712)	(0.7348)	(0.0804)	(0.0674)	(0.0526)	(1.0006)	(0.6862)	(0.2165)	(0.0465)	(0.8611)	(0.5626)	(0.0560)	(0.0447)
FRE	1.2891	1.0274	0.9404	0.7999	1.0846	0.9267	0.8335	0.6486	1.0862	0.9621	0.8946	0.7216	0.9645	0.8872	0.7566	0.5684
	(0.0982)	(0.0461)	(0.0100)	(0.0071)	(0.0021)	(0.0002)	(0.0011)	(0.0003)	(0.0861)	(0.0355)	(0.0086)	(0.0045)	(0.0633)	(0.0072)	(0.0041)	(0.0009)

FPR and TPR are not reported due to the fact that it is not sparse.

APPENDIX E
SUPPLEMENTARY TABLES

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TABLE IV
QUANTITATIVE COMPARISON AMONG DIFFERENT PENALTIES FOR THE PROBABILITY MATRIX WITH $\mathbf{P} = 0.97$

	ℓ_1				MCP				ℓ_1				MCP			
	$m=100$	$m=500$	$m=1500$	$m=3000$	$m=100$	$m=500$	$m=1500$	$m=3000$	$m=100$	$m=500$	$m=1500$	$m=3000$	$m=100$	$m=500$	$m=1500$	$m=3000$
	$d=80, n=50$								$d=80, n=100$							
FAE	38.9147	26.4884	14.8860	14.2507	33.0702	18.7424	13.3777	9.6156	36.6029	24.7278	14.5060	14.0390	29.3833	17.0730	12.2319	8.3190
	(0.2249)	(0.1971)	(0.1666)	(0.1215)	(0.2355)	(0.1998)	(0.1572)	(0.1147)	(0.2201)	(0.1615)	(0.1322)	(0.1162)	(0.1981)	(0.1455)	(0.1106)	(0.0893)
FRE	1.3416	0.9132	0.5132	0.4913	1.1401	0.6462	0.4612	0.3315	1.2619	0.8525	0.5001	0.4840	1.0130	0.5886	0.4217	0.2868
	(0.0107)	(0.0095)	(0.0074)	(0.0052)	(0.0100)	(0.0083)	(0.0065)	(0.0048)	(0.0112)	(0.0089)	(0.0083)	(0.0062)	(0.0099)	(0.0075)	(0.0030)	(0.0024)
FPR	0.1216	0.1027	0.0892	0.0774	0.1163	0.0994	0.0796	0.0652	0.1186	0.1003	0.0844	0.0698	0.1026	0.0952	0.0774	0.0588
	(0.1106)	(0.0917)	(0.0432)	(0.0327)	(0.1023)	(0.0755)	(0.0523)	(0.0348)	(0.1027)	(0.0856)	(0.0561)	(0.0322)	(0.0953)	(0.0413)	(0.0232)	(0.0103)
TPR	0.8314	0.8612	0.8913	0.9128	0.8510	0.8864	0.9176	0.9355	0.8531	0.8751	0.9065	0.9261	0.8644	0.8923	0.9216	0.9462
	(0.0562)	(0.0231)	(0.0213)	(0.0127)	(0.0856)	(0.0335)	(0.0245)	(0.0156)	(0.0756)	(0.0565)	(0.0216)	(0.0163)	(0.0565)	(0.0326)	(0.0261)	(0.1052)
	$d=100, n=50$								$d=100, n=100$							
FAE	55.2209	36.1787	28.7683	20.2725	48.5060	25.6995	17.0870	12.3318	47.8352	31.5048	27.4390	17.3699	37.5328	21.2660	15.8426	10.4085
	(0.2115)	(0.1956)	(0.1561)	(0.1193)	(0.1915)	(0.1566)	(0.1325)	(0.1033)	(0.1986)	(0.1568)	(0.1262)	(0.1091)	(0.1535)	(0.1217)	(0.1066)	(0.0997)
FRE	1.5619	1.0233	0.8137	0.5734	1.3720	0.7269	0.4833	0.3488	1.3530	0.8911	0.7761	0.4913	1.0616	0.6015	0.4481	0.2944
	(0.0096)	(0.0081)	(0.0063)	(0.0047)	(0.0089)	(0.0066)	(0.0038)	(0.0010)	(0.0101)	(0.0076)	(0.0058)	(0.0036)	(0.0084)	(0.0061)	(0.0029)	(0.0011)
FPR	0.1162	0.0986	0.0848	0.0725	0.1056	0.0955	0.0769	0.0643	0.1098	0.0926	0.0813	0.0699	0.0999	0.0896	0.0753	0.0619
	(0.1097)	(0.0896)	(0.0653)	(0.0313)	(0.0995)	(0.0766)	(0.0575)	(0.0243)	(0.0995)	(0.0766)	(0.0589)	(0.0304)	(0.0865)	(0.0561)	(0.0237)	(0.0111)
TPR	0.8426	0.8847	0.9017	0.9273	0.8553	0.8962	0.9172	0.9326	0.8543	0.8914	0.9116	0.9306	0.8668	0.9006	0.9233	0.9447
	(0.0155)	(0.0274)	(0.0136)	(0.0115)	(0.0117)	(0.0206)	(0.0184)	(0.0095)	(0.0143)	(0.0103)	(0.0085)	(0.0062)	(0.0097)	(0.0016)	(0.0008)	(0.0004)

TABLE V
QUANTITATIVE COMPARISON AMONG DIFFERENT PENALTIES FOR THE PROBABILITY MATRIX WITH $\mathbf{P} = 0.85$

	ℓ_1				MCP				ℓ_1				MCP			
	$m=100$	$m=500$	$m=1500$	$m=3000$	$m=100$	$m=500$	$m=1500$	$m=3000$	$m=100$	$m=500$	$m=1500$	$m=3000$	$m=100$	$m=500$	$m=1500$	$m=3000$
	$d=80, n=50$								$d=80, n=100$							
FAE	74.5959	49.0515	41.2813	24.7726	57.8763	41.2046	29.7818	19.9504	71.2069	46.3288	37.8156	23.3010	52.7664	36.9864	25.7169	16.5902
	(0.2752)	(0.1538)	(0.1210)	(0.1006)	(0.2647)	(0.1433)	(0.1157)	(0.0948)	(0.2556)	(0.1315)	(0.1057)	(0.0839)	(0.2123)	(0.1625)	(0.1100)	(0.0872)
FRE	1.5562	1.0233	0.8612	0.5168	1.2074	0.8596	0.6213	0.4162	1.4855	0.9665	0.7889	0.4861	1.1008	0.7716	0.5365	0.3461
	(0.0261)	(0.0175)	(0.0099)	(0.0043)	(0.0197)	(0.0088)	(0.0074)	(0.0016)	(0.0085)	(0.0078)	(0.0064)	(0.0051)	(0.0080)	(0.0063)	(0.0035)	(0.0017)
FPR	0.1398	0.1245	0.1135	0.1027	0.1316	0.1160	0.0983	0.0937	0.1264	0.1152	0.1008	0.0983	0.1163	0.1134	0.0934	0.0863
	(0.0258)	(0.0284)	(0.0443)	(0.0209)	(0.0411)	(0.0169)	(0.0237)	(0.0652)	(0.0436)	(0.0674)	(0.0156)	(0.0175)	(0.0178)	(0.0454)	(0.0137)	(0.0112)
TPR	0.8453	0.8642	0.8894	0.9204	0.8637	0.9092	0.9464	0.9513	0.8755	0.8972	0.8981	0.9257	0.9111	0.9268	0.9415	0.9574
	(0.0381)	(0.0183)	(0.0241)	(0.0239)	(0.0216)	(0.0128)	(0.0337)	(0.0278)	(0.0297)	(0.0240)	(0.0281)	(0.0226)	(0.0331)	(0.0205)	(0.0184)	(0.0121)
	$d=100, n=50$								$d=100, n=100$							
FAE	91.4426	61.4585	47.5946	35.9809	79.8289	48.5239	38.6520	24.7354	85.0191	57.6535	43.5675	33.6955	73.4463	41.8549	34.2157	18.4405
	(0.2482)	(0.2688)	(0.1369)	(0.1346)	(0.1405)	(0.1269)	(0.1343)	(0.1086)	(0.1232)	(0.1666)	(0.1349)	(0.1282)	(0.1246)	(0.0958)	(0.1174)	(0.1039)
FRE	1.5645	1.0515	0.8143	0.6156	1.3658	0.8302	0.6613	0.4232	1.4546	0.9864	0.7454	0.5765	1.2566	0.7161	0.5854	0.3155
	(0.0234)	(0.0209)	(0.0301)	(0.0146)	(0.0105)	(0.0149)	(0.0134)	(0.0129)	(0.0197)	(0.0166)	(0.0108)	(0.0114)	(0.0121)	(0.0097)	(0.0132)	(0.0086)
FPR	0.1263	0.1122	0.1061	0.1009	0.1038	0.0988	0.0852	0.0787	0.1188	0.1059	0.0905	0.0878	0.1037	0.0963	0.0843	0.0639
	(0.0342)	(0.0228)	(0.0139)	(0.0114)	(0.0143)	(0.0231)	(0.0127)	(0.0113)	(0.0175)	(0.0202)	(0.0163)	(0.0145)	(0.0128)	(0.0106)	(0.0086)	(0.0079)
TPR	0.8163	0.8207	0.8427	0.8452	0.8309	0.8596	0.8978	0.9071	0.8288	0.8364	0.8553	0.8774	0.8538	0.8739	0.9126	0.9232
	(0.0349)	(0.0224)	(0.0131)	(0.0101)	(0.0133)	(0.0116)	(0.0147)	(0.0122)	(0.0113)	(0.0157)	(0.0118)	(0.0107)	(0.0184)	(0.0197)	(0.0108)	(0.0134)

TABLE VI
QUANTITATIVE COMPARISON AMONG DIFFERENT PENALTIES FOR THE SPRANDSYM MATRIX

	ℓ_1				MCP				ℓ_1				MCP			
	$m=100$	$m=500$	$m=1500$	$m=3000$	$m=100$	$m=500$	$m=1500$	$m=3000$	$m=100$	$m=500$	$m=1500$	$m=3000$	$m=100$	$m=500$	$m=1500$	$m=3000$
	$d=80, n=50$								$d=80, n=100$							
FAE	5.6633	4.0051	2.5691	1.5326	4.7250	2.5927	1.3261	0.9231	5.1681	3.7130	2.3787	1.3010	4.7534	2.3718	1.1866	0.7533
	(0.2219)	(0.1039)	(0.0751)	(0.0222)	(0.1949)	(0.0416)	(0.0285)	(0.0143)	(0.2234)	(0.1037)	(0.0453)	(0.0222)	(0.1669)	(0.0949)	(0.0307)	(0.0144)
FRE	1.1540	0.8161	0.5233	0.3123	0.9628	0.5283	0.2702	0.1881	1.0531	0.7566	0.4847	0.2651	0.9686	0.4833	0.2418	0.1535
	(0.0077)	(0.0058)	(0.0049)	(0.0014)	(0.0081)	(0.0017)	(0.0012)	(0.0007)	(0.0072)	(0.0035)	(0.0027)	(0.0033)	(0.0061)	(0.0028)	(0.0014)	(0.0009)
FPR	0.1368	0.1346	0.0805	0.0649	0.1271	0.1131	0.0761	0.0546	0.1322	0.1295	0.0769	0.0608	0.1238	0.1117	0.0643	0.0445
	(0.1124)	(0.0965)	(0.0223)	(0.0118)	(0.1009)	(0.0897)	(0.0134)	(0.0106)	(0.1107)	(0.0901)	(0.0139)	(0.0113)	(0.0983)	(0.0741)	(0.0139)	(0.0085)
TPR	0.8597	0.8887	0.9068	0.9107	0.8641	0.8946	0.9176	0.9282	0.8611	0.8913	0.9139	0.9276	0.8748	0.9055	0.9226	0.9374
	(0.0224)	(0.0131)	(0.0101)	(0.0173)	(0.0158)	(0.0118)	(0.0103)	(0.0131)	(0.0147)	(0.0107)	(0.0124)	(0.0105)	(0.0110)	(0.0098)	(0.0091)	(0.00101)
	$d=100, n=50$								$d=100, n=100$							
FAE	6.3763	4.6956	3.0768	1.7689	5.8840	3.4189	1.7551	1.0624	5.7895	4.1438	2.8588	1.6846	5.4194	3.0847	1.5530	0.7971
	(0.2741)	(0.1452)	(0.1088)	(0.0820)	(0.2610)	(0.0620)	(0.0454)	(0.0218)	(0.2110)	(0.1351)	(0.0921)	(0.0742)	(0.1374)	(0.1143)	(0.0971)	(0.0669)
FRE	1.1632	0.8566	0.5613	0.3227	1.0734	0.6237	0.3202	0.1938	1.0561	0.7559	0.5215	0.3073	0.9886	0.5627	0.2833	0.1454
	(0.0057)	(0.0047)	(0.0031)	(0.0015)	(0.0087)	(0.0021)	(0.0016)	(0.0006)	(0.0033)	(0.0027)	(0.0019)	(0.0011)	(0.0032)	(0.0018)	(0.0009)	(0.0006)
FPR	0.1226	0.1245	0.0837	0.0513	0.1148	0.1045	0.0616	0.0506	0.1133	0.1076	0.0711	0.0439	0.0981	0.8861	0.0646	0.0337
	(0.0119)	(0.0105)	(0.0096)	(0.0083)	(0.0105)	(0.0104)	(0.0091)	(0.0071)	(0.0095)	(0.0076)	(0.0063)	(0.0043)	(0.0085)	(0.0061)	(0.0054)	(0.0038)
TPR	0.8645	0.8971	0.9147	0.9225	0.8861	0.9091	0.9159	0.9286	0.8748	0.8996	0.9195	0.9278	0.8849	0.9108	0.9218	0.9337
	(0.0317)	(0.0139)	(0.0116)	(0.0108)	(0.0132)	(0.0130)	(0.0104)	(0.0092)	(0.0186)	(0.0126)	(0.0109)	(0.0099)	(0.0176)	(0.0112)	(0.0081)	(0.0072)

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